Analysis IV (for EL, GM, MX)

Based on Georgios Moschidis's Lectures

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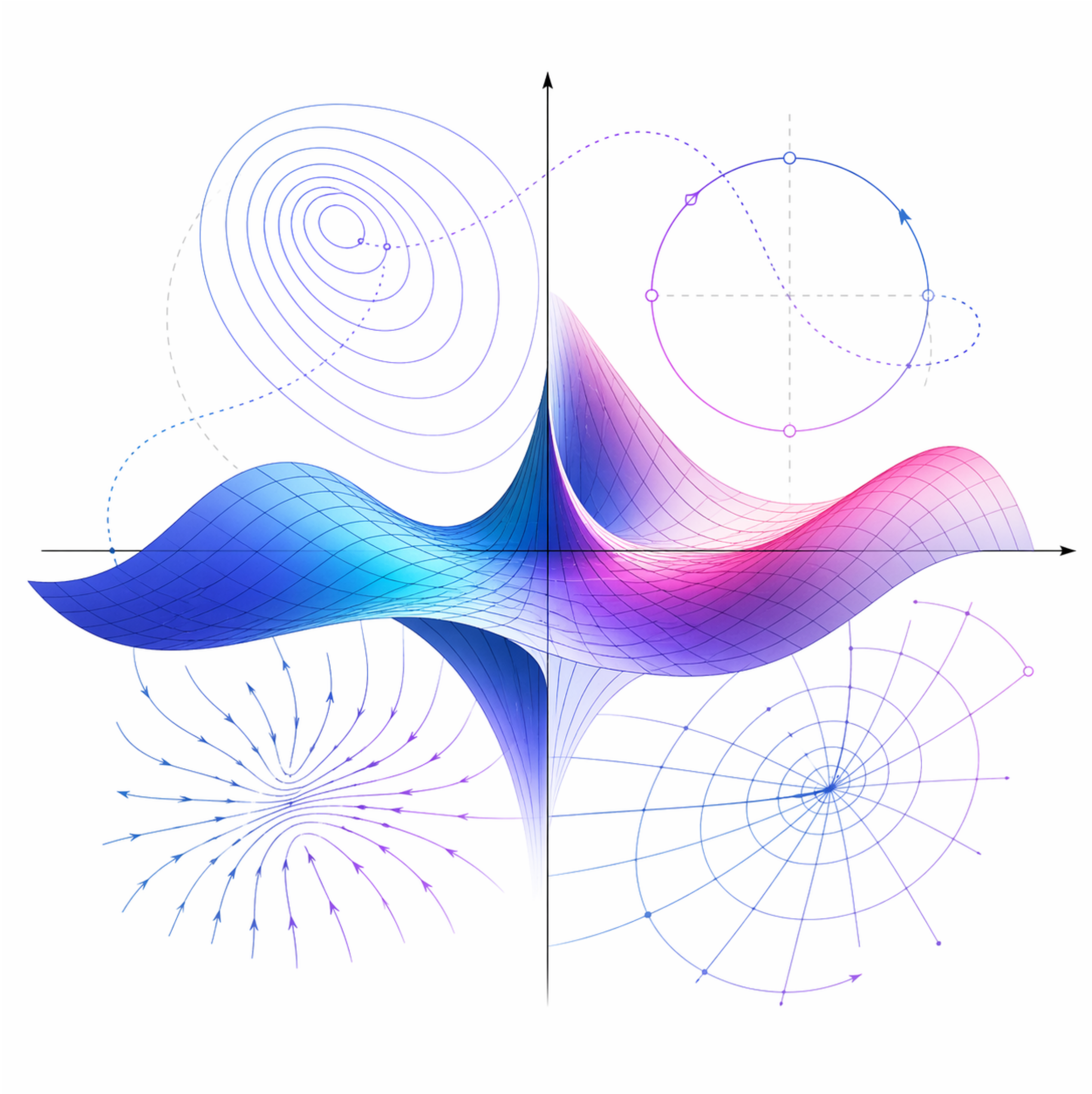

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Contents

1 Week 1 – 16.02.26 - Introduction and Complex Numbers

Overview of the course (~ 13 weeks):

- Complex analysis: Complex differentiation, Complex integration, Laurent series, Residue theorem.
- Review of Fourier transform / series.
- Laplace transform (well-suited for functions defined on the half-line $[0, +\infty)$).
- Applications to ODEs / PDEs (Initial/boundary value problems).

1.1 Review of complex numbers $\mathbb{C}$

Definition: Imaginary Unit & Complex Numbers

Imaginary unit: i
Def: $i^2 = -1$
For $z = x + iy \in \mathbb{C}$ where $x, y \in \mathbb{R}$:

- $x = \text{Re}(z)$ (Real part)
- $y = \text{Im}(z)$ (Imaginary part)

Geometric representation: Identify with $\mathbb{R}^2$.

- $z = x + iy \implies \overline{z} = x - iy$ (Complex conjugate. Reflection across the real axis).
- $|z| = \sqrt{x^2 + y^2}$ (Modulus / absolute value). Geometrically: $|z|$ is the distance of (x, y) from $(0, 0)$.
- $|z_1 - z_0|$: Distance between z_0 and z_1.

E.g.: $\{z \in \mathbb{C} : |z - 1| < 1\}$ is the open disk centered at $z_0 = 1$ of radius 1.

Algebraic operations:
If $z_1 = x_1 + iy_1$ and $z_2 = x_2 + iy_2$:

- $z_1 + z_2 = (x_1 + x_2) + i(y_1 + y_2)$
- $z_1 - z_2 = (x_1 - x_2) + i(y_1 - y_2)$
- $z_1 \cdot z_2 = (x_1 + iy_1) \cdot (x_2 + iy_2) = (x_1x_2 - y_1y_2) + i(x_1y_2 + x_2y_1)$
- $z \cdot \overline{z} = (x + iy)(x - iy) = x^2 + y^2 = |z|^2$
- $\dfrac{z_1}{z_2} = \dfrac{z_1 \cdot \overline{z_2}}{z_2 \cdot \overline{z_2}} = \dfrac{z_1 \cdot \overline{z_2}}{|z_2|^2}$ (if $z_2 \neq 0$)

Complex exponential:
If $z = x + iy$, we define $e^z = e^{x+iy} = e^x \cdot e^{iy}$, where for $y \in \mathbb{R}$, e^{iy} is defined by:

Theorem: Euler's formula

For $\theta \in \mathbb{R}$, $e^{i\theta} = \cos(\theta) + i\sin(\theta)$. Since $|e^{i\theta}| = \sqrt{\cos^2\theta + \sin^2\theta} = 1 \implies e^{i\theta}$ lies on the unit circle S_1. Also, $\overline{e^{i\theta}} = \cos(\theta) - i\sin(\theta) = e^{-i\theta}$.

Remark: An equivalent definition of e^z for $z \in \mathbb{C}$ is by assuming that e^z has the same power series expansion as in the case $\mathbb{R}$, i.e.:

$$e^z = \sum_{k=0}^{+\infty} \frac{z^k}{k!}$$

Then one can "derive" Euler's formula.

Polar expression:
For $z = x + iy \neq 0$, we can always write $z = |z| \cdot e^{i\theta}$, $\theta \in \mathbb{R}$.

- The argument θ is unique up to $2\pi\mathbb{Z}$.
- **Principal argument:** The argument with $-\pi < \theta \leq \pi$.

If $z_1 = |z_1| \cdot e^{i\theta_1}$ and $z_2 = |z_2| \cdot e^{i\theta_2}$ for some $\theta_1, \theta_2 \in \mathbb{R}$, then $z_1 = z_2 \iff |z_1| = |z_2|$ and $\theta_1 - \theta_2 \in 2\pi\mathbb{Z}$.

Operations in polar form:

- $z_1 \cdot z_2 = |z_1| \cdot |z_2| \cdot e^{i(\theta_1+\theta_2)}$
- $\dfrac{z_1}{z_2} = \dfrac{|z_1| \cdot e^{i\theta_1}}{|z_2| \cdot e^{i\theta_2}} = \dfrac{|z_1|}{|z_2|} \cdot e^{i(\theta_1-\theta_2)}$
- In particular: If $z = |z|e^{i\theta}$, then $z^k = |z|^k \cdot e^{ik\theta}$ for $k \in \mathbb{Z}$.

Remark: Multiplication by $e^{i\alpha}$ is equivalent to a rotation by α.

How to find principal argument:
For $z = x + iy$:

- If $x = 0$: $\theta = \frac{\pi}{2}$ if $y > 0$, and $\theta = -\frac{\pi}{2}$ if $y < 0$.
- If $x > 0$: $\tan(\theta) = \frac{y}{x} \implies \theta = \arctan\left(\frac{y}{x}\right)$.

Ex: If $z = 1 + i$, then $\mathrm{Arg}(z) = \arctan\left(\frac{1}{1}\right) = \frac{\pi}{4} \implies z = \sqrt{2} \cdot e^{i\frac{\pi}{4}}$ (polar expression).

Example: Solving algebraic equations

Sometimes, the polar expression is useful in solving algebraic equations.
E.g.: Solve $z^5 = 2$.
Sol: Let $z = |z| \cdot e^{i\theta}$ where θ is the principal argument. Then: $z^5 = |z|^5 \cdot e^{i5\theta} = 2$.

$$\implies \begin{cases} |z|^5 = 2 \\ 5\theta \in 2\pi\mathbb{Z} \end{cases} \implies \begin{cases} |z| = 2^{1/5} \\ \theta \in \frac{2\pi}{5}\mathbb{Z} \end{cases}$$

But θ is the principal argument: $-\pi < \theta \leq \pi$.

$$\theta \in \left\{ -\frac{4\pi}{5}, -\frac{2\pi}{5}, 0, \frac{2\pi}{5}, \frac{4\pi}{5} \right\}$$

5 solutions. (In general: a k-th order equation has k solutions, counted with multiplicity).

1.2 Complex functions

We will consider functions $f : \mathbb{C} \to \mathbb{C}$,

$$z \mapsto f(z) = f(x + iy) = u(x, y) + iv(x, y)$$

where $u, v : \mathbb{R}^2 \to \mathbb{R}$ are the Real part and Imaginary part, respectively.

Examples:

- $f(z) = z = x + iy$
- $f(z) = \overline{z} = x - iy$
- $f(z) = z^2 = (x + iy)^2 = x^2 - y^2 + 2xyi$
- $f(z) = \frac{1}{z} = \frac{\overline{z}}{|z|^2} = \frac{x-yi}{x^2+y^2} = \frac{x}{x^2+y^2} - i\frac{y}{x^2+y^2}$ (Defined on $\mathbb{C}^* = \mathbb{C} \setminus \{0\}$)
- $f(z) = e^z = e^{x+iy} = e^x \cdot e^{iy} = e^x \cos(y) + ie^x \sin(y)$

Trigonometric and hyperbolic trigonometric functions:

- $\cos(z) = \frac{e^{iz}+e^{-iz}}{2}$
- $\sin(z) = \frac{e^{iz}-e^{-iz}}{2i}$
- $\cosh(z) = \frac{e^{z}+e^{-z}}{2}$
- $\sinh(z) = \frac{e^{z}-e^{-z}}{2}$

Calculating the real and imaginary parts of $\cos(z)$:

$$\begin{aligned}
\cos(z) &= \frac{e^{iz}+e^{-iz}}{2} = \frac{e^{i(x+iy)}+e^{-i(x+iy)}}{2} \\
&= \frac{1}{2}\left(e^{-y+ix}+e^{y-ix}\right) \\
&= \frac{1}{2}\left(e^{-y}(\cos x + i\sin x) + e^{y}(\cos(-x)+i\sin(-x))\right) \\
&= \frac{1}{2}\left(e^{-y}+e^{y}\right)\cos x + i\frac{1}{2}\left(e^{-y}-e^{y}\right)\sin x \\
&= \cosh(y)\cos(x) - i\sinh(y)\sin(x)
\end{aligned}$$

This agrees with the usual definition when $z \in \mathbb{R}$ (i.e., $y = 0 \implies \cosh(0)\cos(x) = \cos(x)$).

1.3 Limits and continuity

Definition: Continuity

We will say that $f : \mathbb{C} \to \mathbb{C}$ is continuous at a point $z_0 \in \mathbb{C}$ if for all $\epsilon > 0$, there exists $\delta > 0$ such that for all z with $|z - z_0| < \delta$ we have $|f(z) - f(z_0)| < \epsilon$.
Equivalently,

$$\lim_{z \to z_0} f(z) = f(z_0)$$

Remarks: 1) Let $f(z) = f(x+iy) = u(x,y) + iv(x,y)$. Then: f is continuous at $z_0 = x_0 + iy_0 \iff u$ & v are continuous at (x_0, y_0) (as functions from $\mathbb{R}^2$ to $\mathbb{R}$). In particular: If $z_0 = x_0 + iy_0$,

$$\lim_{z \to z_0} f(z) = \lim_{(x,y)\to(x_0,y_0)} u(x,y) + i \lim_{(x,y)\to(x_0,y_0)} v(x,y)$$

2) If f, g are continuous, then the same is true for $f \pm g$, $f \cdot g$, f/g (where $g \neq 0$), etc.

1.4 Complex derivative

Definition: Complex Derivative

We say that a function $f : \mathbb{C} \to \mathbb{C}$ is (complex) differentiable at z_0 if the limit

$$\lim_{z \to z_0} \frac{f(z) - f(z_0)}{z - z_0}$$

exists and is finite. We will denote the limit by $f'(z_0)$.

- If f is differentiable at every point of an open set $W \subseteq \mathbb{C}$, f is **holomorphic** over W.
- If f is holomorphic over $\mathbb{C}$, f is **entire**.

Remark: The same rules for derivatives of products, compositions, etc. apply as in the real case.

Recall: $W \subseteq \mathbb{C}$ is open if $\forall z_0 \in W$, $\exists \epsilon > 0$ such that the disc $D_{z_0,\epsilon} = \{z \in \mathbb{C} : |z - z_0| < \epsilon\}$ is contained inside W.

Examples:

- $\mathbb{C}$ is open.
- $\mathbb{C}^* = \mathbb{C} \setminus \{0\}$ is open, since for $z_0 \in \mathbb{C}^*$ the disk of radius $\epsilon = \frac{|z_0|}{2}$ does not contain 0, so it lies inside $\mathbb{C}^*$.
- $\{z \in \mathbb{C} : \text{Re}(z) \geq 0\}$ is not open, since for z_0 on the boundary ($\text{Re}(z_0) = 0$), any open disc contains points with $\text{Re}(z) < 0$, i.e., outside the set.

- If $f(z) = z$, $\forall z_0 \in \mathbb{C}$, $\lim_{z \to z_0} \frac{z - z_0}{z - z_0} = 1$. So f is entire.
- If $f(z) = \frac{1}{z}$, $f'(z) = -\frac{1}{z^2}$. So f is holomorphic on $\mathbb{C}^*$.

2 Week 2 – 23.02.26 Cauchy's Riemann and Differentiation

Recall: A function $f : \mathbb{C} \to \mathbb{C}$ is complex differentiable at z_0 if the limit

$$f'(z_0) = \lim_{z \to z_0} \frac{f(z) - f(z_0)}{z - z_0}$$

exists and is finite. Analogous to the definition of the derivative of a real function in Analysis I & II.

2.1 Cauchy-Riemann equations

Theorem: Cauchy-Riemann equations

Let W be an open set and $f : W \to \mathbb{C}$, $z = x + iy \mapsto f(x + iy) = u(x, y) + iv(x, y)$ be a holomorphic function. Then, $u, v \in C^1(W)$ (differentiable with continuous derivatives) and:

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}$$

$$\frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$$

Conversely, if these are satisfied (and $u, v \in C^1$), then f is holomorphic.

We will prove the direct implication (the converse is more difficult).

Proof: If f is holomorphic on W, then for $z_0 \in W$:

$$f'(z_0) = \lim_{z \to z_0} \frac{f(z) - f(z_0)}{z - z_0}$$

We will calculate the limit in two ways: $z = x + iy$, $z_0 = x_0 + iy_0$.

1. Approaching z_0 from the x-direction ($y = y_0 \implies z = x + iy_0$):

$$\begin{aligned}
f'(z_0) &= \lim_{x \to x_0} \frac{f(x + iy_0) - f(x_0 + iy_0)}{x + iy_0 - (x_0 + iy_0)} \\
&= \lim_{x \to x_0} \frac{u(x, y_0) + iv(x, y_0) - u(x_0, y_0) - iv(x_0, y_0)}{x - x_0} \\
&= \lim_{x \to x_0} \frac{u(x, y_0) - u(x_0, y_0)}{x - x_0} + i \lim_{x \to x_0} \frac{v(x, y_0) - v(x_0, y_0)}{x - x_0} \\
&= \frac{\partial u}{\partial x} + i \frac{\partial v}{\partial x} \quad \text{— (1)}
\end{aligned}$$

2. Approaching z_0 from the y-direction ($x = x_0 \implies z = x_0 + iy$):

$$\begin{aligned}
f'(z_0) &= \lim_{y \to y_0} \frac{f(x_0 + iy) - f(x_0 + iy_0)}{x_0 + iy - (x_0 + iy_0)} \\
&= \lim_{y \to y_0} \frac{u(x_0, y) + iv(x_0, y) - u(x_0, y_0) - iv(x_0, y_0)}{i(y - y_0)} \\
&= \frac{1}{i} \lim_{y \to y_0} \left(\frac{u(x_0, y) - u(x_0, y_0)}{y - y_0} + i \frac{v(x_0, y) - v(x_0, y_0)}{y - y_0} \right) \\
&= \frac{1}{i} \left(\frac{\partial u}{\partial y} + i \frac{\partial v}{\partial y} \right) \\
&= \frac{\partial v}{\partial y} - i \frac{\partial u}{\partial y} \quad \text{— (2)}
\end{aligned}$$

Since $(1) = (2)$, we have:

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \quad \text{and} \quad \frac{\partial v}{\partial x} = -\frac{\partial u}{\partial y}$$

Also from this proof, expressions for $f'(z_0)$:

$$f' = \frac{\partial u}{\partial x} + i \frac{\partial v}{\partial x} = \frac{\partial v}{\partial y} - i \frac{\partial u}{\partial y}$$

Examples: Cauchy-Riemann

Ex 1. $f(z) = z^2 = (x+iy)^2 = x^2 - y^2 + i(2xy)$ $u(x,y) = x^2 - y^2$, $v(x,y) = 2xy$. Is it holomorphic over $\mathbb{C}$? We check the Cauchy-Riemann eqns:

$$\frac{\partial u}{\partial x} = 2x, \quad \frac{\partial v}{\partial x} = 2y$$
$$\frac{\partial u}{\partial y} = -2y, \quad \frac{\partial v}{\partial y} = 2x$$

C-R satisfied! And $f'(z) = \frac{\partial u}{\partial x} + i\frac{\partial v}{\partial x} = 2x + i(2y) = 2z$.
Ex 2. The complex conjugate: $f(z) = \overline{z} = x - iy$. $u = x, v = -y$.

$$\frac{\partial u}{\partial x} = 1, \quad \frac{\partial v}{\partial x} = 0$$
$$\frac{\partial u}{\partial y} = 0, \quad \frac{\partial v}{\partial y} = -1$$

Here $\partial_x u \neq \partial_y v$. Not holomorphic! Another way of seeing the same fact: If we try to compute the limit $\lim_{z \to z_0} \frac{\overline{z} - \overline{z_0}}{z - z_0}$:

- For $z = x + iy_0$, $z_0 = x_0 + iy_0$: $\lim_{x \to x_0} \frac{x - x_0}{x - x_0} = 1$.
- For $z = x_0 + iy$, $z_0 = x_0 + iy_0$: $\lim_{y \to y_0} \frac{-iy + iy_0}{iy - iy_0} = -1$.

Ex 3. $f : \mathbb{C} \setminus \{0\} \to \mathbb{C}$, $f(z) = \frac{1}{z}$. We have: $f(x+iy) = \frac{\overline{z}}{|z|^2} = \frac{x}{x^2+y^2} - i\frac{y}{x^2+y^2}$. So:

$$\frac{\partial u}{\partial x} = \frac{x^2 + y^2 - 2x^2}{(x^2+y^2)^2} = \frac{y^2 - x^2}{(x^2+y^2)^2}$$
$$\frac{\partial v}{\partial x} = \frac{2xy}{(x^2+y^2)^2}$$
$$\frac{\partial u}{\partial y} = -\frac{2xy}{(x^2+y^2)^2}$$
$$\frac{\partial v}{\partial y} = \frac{-(x^2+y^2) + 2y^2}{(x^2+y^2)^2} = \frac{y^2 - x^2}{(x^2+y^2)^2}$$

So C-R eqns are satisfied. And $f' = \frac{\partial u}{\partial x} + i\frac{\partial v}{\partial x} = \frac{y^2 - x^2}{(x^2+y^2)^2} + i\frac{2xy}{(x^2+y^2)^2}$. This is $= -\frac{1}{z^2}$. Indeed: $-\frac{1}{z^2} = -\frac{\overline{z}^2}{|z|^4} = -\frac{(x-iy)^2}{(x^2+y^2)^2} = \dots$
Ex 4. $f(z) = e^z = e^{x+iy} = e^x \cos(y) + ie^x \sin(y)$. So:

$$\frac{\partial u}{\partial x} = e^x \cos(y), \quad \frac{\partial v}{\partial x} = e^x \sin(y)$$
$$\frac{\partial u}{\partial y} = -e^x \sin(y), \quad \frac{\partial v}{\partial y} = e^x \cos(y)$$

So C-R eqns are satisfied. And $f' = \frac{\partial u}{\partial x} + i\frac{\partial v}{\partial x} = e^x \cos(y) + ie^x \sin(y) = e^z$.

2.2 Rules of differentiation for complex derivatives

Let $f, g : W \to \mathbb{C}$ be holomorphic. Then:

- $(f+g)' = f' + g'$
- $(f \cdot g)' = f' \cdot g + f \cdot g'$
- $\left(\frac{f}{g}\right)' = \frac{f' \cdot g - f \cdot g'}{g^2}$ (when $g \neq 0$)
- If the image of g is inside W: $(f \circ g)'(z) = f'(g(z)) \cdot g'(z)$.

These relations can be proved in exactly the same way as in the real case. E.g.

$$
\begin{aligned}
(f \cdot g)'(z_0) &= \lim_{z \to z_0} \frac{f(z) \cdot g(z) - f(z_0) g(z_0)}{z - z_0} \\
&= \lim_{z \to z_0} \frac{f(z)g(z) - f(z_0)g(z) + f(z_0)g(z) - f(z_0)g(z_0)}{z - z_0} \\
&= \lim_{z \to z_0} \left(\frac{f(z) - f(z_0)}{z - z_0} g(z) + f(z_0) \frac{g(z) - g(z_0)}{z - z_0} \right) \\
&= f'(z_0) g(z_0) + f(z_0) g'(z_0)
\end{aligned}
$$

With these rules and the examples we computed earlier, it follows that the usual formulas for polynomials, trigonometric functions, etc. hold. E.g. $(z^k)' = k \cdot z^{k-1}$ ($k \in \mathbb{Z}$), $(\cos(z))' = -\sin(z)$.

2.3 Aside: A geometric property of holomorphic functions

If $f : W \to \mathbb{C}$ is holomorphic, when viewed as a map $f : \mathbb{R}^2 \to \mathbb{R}^2$, the Jacobian matrix

$$
Df = \begin{pmatrix} \partial_x u & \partial_y u \\ \partial_x v & \partial_y v \end{pmatrix}
$$

tells us how f acts on infinitesimal rectangles. By the Cauchy-Riemann equations, Df takes the form of $\lambda \cdot$ Rotation matrix. So an infinitesimal square is mapped to a square (no uneven stretching, preserving angles – this is called a *conformal mapping*).

2.4 The complex logarithm

Definition: Complex Logarithm

Recall: The principal argument of a complex number $z \neq 0$: $-\pi < \text{Arg}(z) \leq \pi$. It is continuous except on the negative real axis (where it jumps from π to $-\pi$).
How would we define the complex logarithm? We want $e^{\log(z)} = z$. If $\log(z) = a + ib$, then $z = e^{a+ib}$. $|z| \cdot e^{i\text{Arg}(z)} = e^a \cdot e^{ib}$. It makes sense to set $|z| = e^a$ and $\text{Arg}(z) = b$ (up to $2\pi k$).
So: We define $\log(z) = \log|z| + i\text{Arg}(z)$.

- $\log(z)$ is not defined for $z = 0$.
- $\log(z)$ is discontinuous on the negative real axis.
- But: $\log(z)$ is continuous on $\mathbb{C} \setminus \{z \in \mathbb{C} : \text{Im}(z) = 0, \text{Re}(z) \leq 0\}$. This excluded line is the *Branch cut*. (Had we defined $\text{Arg}(z)$ differently, we would have a different branch cut).

Real and imaginary parts: $\log(z) = \log|z| + i\text{Arg}(z)$. If $z = x + iy$: If $x > 0$,

$$\log(z) = \log\sqrt{x^2 + y^2} + i \arctan\left(\frac{y}{x}\right)$$

where $u(x, y) = \log\sqrt{x^2 + y^2}$ and $v(x, y) = \arctan(y/x)$.

Easy to verify: $\log(z)$ is holomorphic on $\mathbb{C} \setminus \{z : \text{Im}(z) = 0, \text{Re}(z) \leq 0\}$ with $(\log z)' = \frac{1}{z}$.

Remark: For positive real numbers, $\log(z_1 \cdot z_2) = \log(z_1) + \log(z_2)$. For general complex numbers (due to discontinuity), this is **no longer true**. If $z_1 = z_2 = -1$, $\log(z_1) = \log(z_2) = i\pi$, but $\log(z_1 \cdot z_2) = \log(1) = 0$. But: It is true up to $2\pi i$.

2.5 Power function with complex exponent

Let $\gamma \in \mathbb{C}$. Define:

$$f(z) = z^\gamma = e^{\log(z^\gamma)} = e^{\gamma \cdot \log(z)}$$
$$= e^{\gamma \cdot \log|z| + i\gamma \cdot \text{Arg}(z)}.$$

It is holomorphic over $\mathbb{C} \setminus \{z : \text{Im}(z) = 0, \text{Re}(z) \leq 0\}$ (as the composition of $e^{\gamma z}$ with $\log z$). When $\gamma \in \mathbb{Z}$: it is continuous (in fact, holomorphic) for all $z \neq 0$.

3 Week 3 – 02.03.26 - Complex Integration

(Chapter 10 in the book)

Recall: Vector Analysis

If $\vec{F}$ is a continuous vector field on $\mathbb{R}^2$, so $\vec{F} : \mathbb{R}^2 \to \mathbb{R}^2$, it can be written as $\vec{F}(x_1, x_2) = (F_1(x_1, x_2), F_2(x_1, x_2))$.
Let $\gamma : [a, b] \longrightarrow \mathbb{R}^2$ be a C^1 curve with image Γ. The line integral is defined by:

$$\int_\Gamma \vec{F} \cdot d\vec{l} := \int_a^b \vec{F}(\gamma(t)) \cdot \gamma'(t)\, dt$$

Orientation will be important! The inner product is given by:

$$F_1(\gamma(t)) \cdot \gamma_1'(t) + F_2(\gamma(t)) \cdot \gamma_2'(t)$$

Key points:

- The integral is **independent of the parametrization**.
- If we **switch orientation**, the integral has the opposite sign.
- If the curve is **closed** (i.e., $\gamma(a) = \gamma(b)$) and the curl is zero (curl $\vec{F} = 0$) in the interior of Γ, then $\int_\Gamma \vec{F} \cdot d\vec{l} = 0$.
- A curve is **simple** if there are no self-intersections.
- A curve is **regular** if $\gamma'(t) \neq 0$.

We will define similarly the integral of complex functions $f : \mathbb{C} \simeq \mathbb{R}^2 \longrightarrow \mathbb{C} \simeq \mathbb{R}^2$.

Definition

Let $\gamma : [a, b] \to \mathbb{C}$ be a regular curve that parametrizes Γ.
Let $f : \mathbb{C} \to \mathbb{C}$ be a continuous function. Then the complex integral is defined by:

$$\int_\Gamma f(z)\, dz = \int_\gamma f(z)\, dz := \int_a^b f(\gamma(t)) \cdot \gamma'(t)\, dt$$

(Note: this involves complex multiplication!)

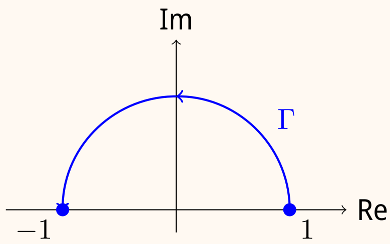
Usual notation: The integral over the curve with the opposite orientation (often denoted $-\Gamma$) has the opposite sign:

$$\int_{-\Gamma} f(z)\, dz = -\int_\Gamma f(z)\, dz$$

The value of the integral is independent of the parametrization, as long as the orientation is the same.

Example 1

Let Γ be the unit half-circle around $z = 0$, parametrized counter-clockwise.



Parametrization: $\gamma : [0, \pi] \to \mathbb{C}, \quad \gamma(t) = e^{it}$.
Derivative: $\gamma'(t) = ie^{it}$.
Integrate $f(z) = z^2$ over Γ:

$$\begin{aligned}\int_\Gamma z^2\,dz &= \int_0^\pi (e^{it})^2 \cdot ie^{it}\,dt \\ &= i\int_0^\pi e^{3it}\,dt \\ &= i\left[\frac{e^{3it}}{3i}\right]_{t=0}^{t=\pi} = \frac{e^{3i\pi} - e^0}{3} = \frac{-1-1}{3} = -\mathbf{\frac{2}{3}}\end{aligned}$$

Note: We used the fundamental theorem of calculus and the fact that $e^{\lambda t} = \frac{1}{\lambda}(e^{\lambda t})'$ for $\lambda \in \mathbb{C}$. For more general functions, we can always split into real and imaginary parts and compute the two (real) integrals separately.

Example 2

Let $\Gamma = \{z : |z| = 1\}$ counter-clockwise (full circle).
$\gamma(t) = e^{it}$ for $t \in [0, 2\pi]$, and $\gamma'(t) = ie^{it}$.

$$\begin{aligned}\int_\Gamma z^2\,dz &= \int_0^{2\pi} (e^{it})^2 ie^{it}\,dt \\ &= i\int_0^{2\pi} e^{3it}\,dt = i\left[\frac{e^{3it}}{3i}\right]_0^{2\pi} = \frac{1-1}{3} = \mathbf{0}\end{aligned}$$

Observation: Integrals of holomorphic functions are governed by similar rules as integrals of vector fields with curl = 0.

Examples 3 and 4

With the same Γ (counter-clockwise unit circle):

- For $f(z) = \frac{1}{z}$:
$$\int_\Gamma \frac{1}{z}\,dz = \int_0^{2\pi} \frac{1}{e^{it}} ie^{it}\,dt = i\int_0^{2\pi} 1\,dt = \mathbf{2\pi i \neq 0}$$
- For $f(z) = \frac{1}{z^2}$:
$$\int_\Gamma \frac{1}{z^2}\,dz = \int_0^{2\pi} \frac{1}{(e^{it})^2} ie^{it}\,dt = i\int_0^{2\pi} e^{-it}\,dt = i\left[\frac{e^{-it}}{-i}\right]_0^{2\pi} = -(1-1) = \mathbf{0}$$

3.1 Cauchy's Theorem

Theorem

Let U be an open and **simply connected** set, and let Γ be a closed and piecewise regular curve inside U.
If $f : U \to \mathbb{C}$ is **holomorphic**, then:
$$\int_\Gamma f(z)\,dz = 0$$

Remarks:

1. **Simply connected:** Means it has only one component and has no "holes".
   - $\mathbb{C}$: Simply connected.
   - $\{z : |z| < 1\}$ (Disk): Simply connected.
   - $\mathbb{C} \setminus \{0\}$: **NOT** simply connected.
2. **In practice:** We are given f and Γ, and we try to find a simply connected set U which contains Γ and lies inside the domain where f is holomorphic.

Proof (Outline)

Let $\gamma(t) = \alpha(t) + i\beta(t)$ be the parametrization, and $f(x+iy) = u(x,y) + iv(x,y)$.

$$\begin{aligned}\int_\Gamma f(z)\,dz &= \int_a^b f(\gamma(t))\cdot\gamma'(t)\,dt \\ &= \int_a^b \Big(u(\gamma(t)) + iv(\gamma(t))\Big)\Big(\alpha'(t) + i\beta'(t)\Big)\,dt \\ &= \underbrace{\int_a^b (u\alpha' - v\beta')\,dt}_{\text{Real part}} + i\underbrace{\int_a^b (v\alpha' + u\beta')\,dt}_{\text{Imaginary part}}\end{aligned}$$

We can rewrite this as the integral of two real vector fields. Let $\vec{F} = (u, -v)$ and $\vec{G} = (v, u)$. The equation becomes:

$$= \int_\Gamma \vec{F}\cdot d\vec{l} + i\int_\Gamma \vec{G}\cdot d\vec{l}$$

Since U is simply connected and $\Gamma \subseteq U$, the interior of Γ (denoted int(Γ)) is entirely contained in U. $\vec{F}$ and $\vec{G}$ are well defined on int(Γ). Applying **Green's theorem**:

$$= \iint_{\text{int}(\Gamma)} \text{curl}\,\vec{F}\,dxdy + i\iint_{\text{int}(\Gamma)} \text{curl}\,\vec{G}\,dxdy$$

Now, let's calculate the curls:

- curl $\vec{F} = \partial_x(-v) - \partial_y(u) = -v_x - u_y$
- curl $\vec{G} = \partial_x(u) - \partial_y(v) = u_x - v_y$

By the **Cauchy-Riemann equations** (since f is holomorphic, we have $u_x = v_y$ and $u_y = -v_x$). So curl $\vec{F} = 0$ and curl $\vec{G} = 0$. The integral is therefore $0 + i0 = 0$.

Examples

- For $\Gamma = \{z : |z| = 1\}$ and $f(z) = z^2$. Applying Cauchy's theorem for $U = \mathbb{C}$: $\int_\Gamma z^2\,dz = 0$.
- For the same Γ and $f(z) = \frac{1}{z}$. f is holomorphic on $\mathbb{C}\setminus\{0\}$. There is **no** simply connected set $U \subseteq \mathbb{C}\setminus\{0\}$ which contains Γ (because Γ surrounds the "hole" at 0). Cauchy's theorem does not apply (and we saw earlier that the integral is $2\pi i \neq 0$).

Example: Shifted circle

Let $f(z) = \frac{1}{z}$ and $\Gamma = \{z : |z-2| = 1\}$.
If we choose $U = \{z : \text{Re}(z) > 0\}$ (the right half-plane), it is simply connected, f is holomorphic there, and $\Gamma \subset U$. So by Cauchy's theorem: $\int_\Gamma \frac{1}{z}\,dz = \mathbf{0}$.
Explicit verification: $\gamma(t) = 2 + e^{it}, \quad \gamma'(t) = ie^{it}$.

$$\begin{aligned}\int_\Gamma f(z)\,dz &= \int_0^{2\pi} \frac{1}{2+e^{it}} \cdot ie^{it}\,dt \\ &= \int_0^{2\pi} \frac{d}{dt}\Big(\log(2+e^{it})\Big)\,dt \\ &= \Big[\log(2+e^{it})\Big]_0^{2\pi} = \mathbf{0}\end{aligned}$$

Explanation: We use the fact that the argument of log *does not go through the negative real line along the curve, so* $\log(z)$ *remains differentiable with* $(\log z)' = 1/z$.

3.2 An Extension of Cauchy's Theorem (Deformation)

Theorem: Contour Deformation

Let Γ_0, Γ_1 be closed, simple, regular curves with $\Gamma_1 \subset \text{int}(\Gamma_0)$ (one is inside the other). Let γ_0, γ_1 be parametrizations of Γ_0, Γ_1 with the **same orientation**.
If f is holomorphic on the region **between** the curves (i.e. on $\text{int}(\Gamma_0) \setminus \text{int}(\Gamma_1)$), then:

$$\int_{\Gamma_0} f(z)\,dz = \int_{\Gamma_1} f(z)\,dz$$

Proof (Idea)

Consider two new closed paths (say C_0 and C_1) by connecting Γ_0 and Γ_1 with line segments (a "cut"). Since f is holomorphic in the interior of both complete paths without holes, the integral over each is 0. By adding the two together, the integrals over the segments cover the same path with opposite orientations and cancel out. We are left with the integral over Γ_0 in the correct direction, and the integral over Γ_1 in the opposite direction. Hence the result.

Usefulness: Very useful result. It can be used to "shift" the curve of integration and obtain a simpler integral.

Example: The Square

Compute $\int_\Gamma \frac{1}{z}\,dz$ where Γ is the square with vertices $2+2i, -2+2i, -2-2i, 2-2i$, oriented counter-clockwise.
Since $f(z) = 1/z$ is holomorphic on $\mathbb{C} \setminus \{0\}$, we can deform this square into the unit circle $\Gamma_1 = \{z : |z| = 1\}$. The origin 0 is a problem, but it is located inside both curves, so the area "between the square and the circle" is perfectly safe and holomorphic.

$$\int_{\text{Square}} \frac{1}{z}\,dz = \int_{|z|=1} \frac{1}{z}\,dz = \mathbf{2\pi i}$$

(The same value we computed earlier!).

3.3 Cauchy Integral Formulas

Theorem: Cauchy Formula

Suppose that $D \subseteq \mathbb{C}$ is open and $f : D \to \mathbb{C}$ is holomorphic. If $z_0 \in D$, then:

$$f(z_0) = \frac{1}{2\pi i} \int_\gamma \frac{f(z)}{z - z_0}\,dz$$

for any closed, simple curve γ inside D which contains z_0 in its interior and is oriented counter-clockwise.

Proof (Idea)

The function defined by $z \mapsto \frac{f(z)}{z-z_0}$ is holomorphic in $D \setminus \{z_0\}$. By applying the extension of Cauchy's theorem, we can deform γ into a very small circle γ_1 around z_0 of radius $r > 0$. Parametrization: $\gamma_1(t) = z_0 + re^{it}$ for $t \in [0, 2\pi]$.

$$\begin{aligned}
\int_\gamma \frac{f(z)}{z - z_0}\,dz &= \int_{\gamma_1} \frac{f(z)}{z - z_0}\,dz \\
&= \int_0^{2\pi} \frac{f(z_0 + re^{it})}{re^{it}} \cdot \left(ire^{it}\right) dt \\
&= i \int_0^{2\pi} f(z_0 + re^{it})\,dt
\end{aligned}$$

Now, we let the radius approach zero ($r \to 0$). Since f is continuous at z_0, the integral becomes:

$$\lim_{r\to 0} i \int_0^{2\pi} f(z_0 + re^{it})\,dt = i \int_0^{2\pi} f(z_0)\,dt = i \cdot f(z_0) \cdot 2\pi = \mathbf{2\pi i f(z_0)}$$

Dividing by $2\pi i$, we obtain the formula.

4 Week 4 – 09.03.26 - Cauchy's Integral formula

4.1 Recall from last time

Theorem: Cauchy's Theorem

If $U \subseteq \mathbb{C}$ is an open and simply connected set, $f : U \to \mathbb{C}$ is a holomorphic function, and Γ is a piecewise regular, closed curve inside U, then:

$$\int_{\Gamma} f(z)\, dz = 0 \tag{1}$$

In practice: Given f and Γ, if f is holomorphic in a set slightly larger than the interior of Γ, we can apply the theorem.

Verification - Cauchy's Theorem

Status: Formally exact.
Proof/Context: This is the Cauchy-Goursat integral theorem, a fundamental result in complex analysis. Edouard Goursat proved in 1900 that the assumption of continuity for the derivative f' was not necessary; mere complex differentiability (holomorphy) is sufficient to guarantee that the integral over a closed loop in a simply connected domain is zero.

The theorem above allows us to shift and modify the path of integration when convenient. For instance:

- If Γ_0 and Γ_1 are two simple closed curves with the same orientation, where Γ_1 is in the interior of Γ_0, and f is holomorphic in the region contained between them (i.e., in int$(\Gamma_0) \setminus$ int(Γ_1)), then:

$$\int_{\Gamma_0} f(z)\, dz = \int_{\Gamma_1} f(z)\, dz \tag{2}$$

- If two paths Γ_1 and Γ_2 have the same endpoints, are oriented in the same direction, and f is holomorphic in the region bounded between Γ_1 and Γ_2, then:

$$\int_{\Gamma_1} f(z)\, dz = \int_{\Gamma_2} f(z)\, dz \tag{3}$$

  (This is proven by applying Cauchy's theorem to the closed loop formed by Γ_1 and $-\Gamma_2$).

4.2 Cauchy's Integral Formulas

Formula: Cauchy's Integral Formula (Classic)

Let $D \subseteq \mathbb{C}$ be an open set, $f : D \to \mathbb{C}$ be holomorphic, and $z_0 \in D$. Let γ be any simple closed curve such that int$(\gamma) \subseteq D$ and $z_0 \in$ int(γ) (oriented counter-clockwise).
Then:

$$f(z_0) = \frac{1}{2\pi i} \int_{\gamma} \frac{f(z)}{z - z_0}\, dz \tag{4}$$

Formula: Extended Cauchy's Integral Formula (Derivatives)

Under the same assumptions as above, the value of the n-th derivative of f at z_0 is given by:

$$f^{(n)}(z_0) = \frac{n!}{2\pi i} \int_\gamma \frac{f(z)}{(z - z_0)^{n+1}} \, dz \tag{5}$$

(Note: For $n = 0$, we recover the previous classic formula).

Idea of the proof: Start from the classic Cauchy's integral formula and differentiate with respect to z_0. On the right-hand side, the variable z_0 appears only in the term $\frac{1}{z-z_0}$. We therefore use the differentiation rule:

$$\frac{d}{dz_0}\left((z - z_0)^{-k}\right) = k \cdot (z - z_0)^{-k-1}$$

Fundamental Remark

The extended Cauchy formula is used to derive a spectacular statement made earlier in class: **Holomorphic functions are infinitely many times differentiable.** Indeed, the mere existence of the first derivative implies the existence of all higher-order derivatives in complex analysis—a behavior profoundly different from real analysis.

4.2.1 Application Examples

Example 1

Let $f(z) = \frac{\cos z}{z}$. Let γ be a simple closed curve, oriented counter-clockwise. What is $\int_\gamma f(z)\, dz$? We have to consider several cases, because f is holomorphic on $\mathbb{C} \setminus \{0\}$.

- If $0 \in \gamma$ (the point 0 lies on the curve), the integral is ill-defined.
- If 0 is **not** in the interior of γ, then by Cauchy's theorem: $\int_\gamma f(z)\, dz = 0$.
- If 0 **is** in the interior of γ: Note that $\int_\gamma f(z)\, dz = \int_\gamma \frac{g(z)}{z-0}\, dz$, where $g(z) = \cos z$, which is holomorphic on $\mathbb{C}$. By Cauchy's integral formula, $g(0) = \frac{1}{2\pi i} \int_\gamma \frac{g(z)}{z}\, dz$.
  Thus, $\int_\gamma f(z)\, dz = 2\pi i \cdot g(0) = 2\pi i \cdot \cos(0) = \mathbf{2\pi i}$.

Example 2

Let γ be the circle defined by $\{z : |z-2| = 1\}$ oriented counter-clockwise. Let us compute:

$$\int_\gamma \frac{e^z}{(z-2)^2}\,dz$$

Using the extended Cauchy's integral formula with $n = 1$, $z_0 = 2$, and $f(z) = e^z$ (which is holomorphic on $\mathbb{C}$):

$$\int_\gamma \frac{e^z}{(z-2)^{1+1}}\,dz = \frac{2\pi i}{1!} f^{(1)}(2)$$

Since the derivative of e^z is e^z, $f^{(1)}(2) = e^2$. The final result is therefore $\mathbf{2\pi i e^2}$.

4.3 Taylor and Laurent Series (Chapter 11)

4.3.1 Taylor Series Recall (Real Analysis)

Let $f : \mathbb{R} \to \mathbb{R}$ be a sufficiently regular function. For any $x_0 \in \mathbb{R}$, the Taylor polynomial of degree N is defined by:

$$P_N(x) = \sum_{n=0}^{N} \frac{f^{(n)}(x_0)}{n!}(x-x_0)^n$$

Theorem: Taylor's Theorem (Lagrange Remainder)

If $f \in C^{N+1}(\mathbb{R})$, then for all $x_0, x \in \mathbb{R}$, there exists $\theta \in (0,1)$ such that:

$$f(x) = P_N(x) + \frac{f^{(N+1)}(x_0 + \theta(x-x_0))}{(N+1)!}(x-x_0)^{N+1}$$

In practice, θ is unknown, but we can usually bound (estimate) this $(N+1)$-th derivative to prove that the remainder tends to 0. This allows us to approximate functions using Taylor polynomials.

If f has derivatives of all orders at x_0, we may consider the Taylor series:

$$\sum_{n=0}^{+\infty} \frac{f^{(n)}(x_0)}{n!}(x-x_0)^n$$

Question: Does the Taylor series converge? If yes, to what?

Example: Classic Example 1 (Real)

For $f(x) = e^x$ and $x_0 = 0$. The Taylor series is $\sum_{n=0}^{+\infty} \frac{x^n}{n!}$. It converges to e^x for all $x \in \mathbb{R}$.

Example: Classic Example 2 (Real)

For $f(x) = \frac{1}{1-x}$ and $x_0 = 0$. Since $f^{(n)}(x) = n!(1-x)^{-n-1}$, we have $f^{(n)}(0) = n!$. The Taylor series is:

$$\sum_{n=0}^{+\infty} \frac{f^{(n)}(0)}{n!} x^n = \sum_{n=0}^{+\infty} x^n$$

This is a geometric series that converges to $f(x)$ **only if** $|x| < 1$.
(Recall: $(1 + x + ... + x^N)(1-x) = 1 - x^{N+1} \implies \sum_{n=0}^{N} x^n = \frac{1-x^{N+1}}{1-x} \xrightarrow{N\to\infty} \frac{1}{1-x}$ if $|x| < 1$).

Analytic vs C^∞ Functions

Functions f for which the Taylor series converges to f in a neighborhood of x_0 are called **analytic**. It is crucial to note that *not all C^∞ functions are analytic*. **Classic counterexample by Augustin-Louis Cauchy (1822):**

$$f(x) = \begin{cases} e^{-\frac{1}{x^2}}, & \text{if } x \neq 0 \\ 0, & \text{if } x = 0 \end{cases}$$

All derivatives at $x = 0$ are exactly 0. The Taylor series is therefore formally $0 + 0x + 0x^2 + \cdots = 0$. However, $f(x) \neq 0$ for all $x \neq 0$. The series converges, but not to the function!

4.3.2 Taylor Series for Complex Functions

Theorem: Complex Taylor Series

Let $D \subseteq \mathbb{C}$ be an open set and $z_0 \in D$. Let $R > 0$ be such that the open disk $B_R(z_0) := \{z : |z - z_0| < R\}$ is contained in D.
If $f : D \to \mathbb{C}$ is holomorphic, then for all $z \in B_R(z_0)$:

$$f(z) = \sum_{n=0}^{+\infty} \frac{f^{(n)}(z_0)}{n!} (z - z_0)^n \tag{6}$$

The largest $R > 0$ satisfying this condition is called the **radius of convergence**.

Important Remarks:

1. Holomorphic functions are therefore **all analytic** (around every z_0 in their domain, there exists a disk on which the Taylor series converges to f).
2. **In practice:** The radius R corresponds exactly to the distance between z_0 and the "nearest" point where the function is not holomorphic (a singularity).
3. The coefficients of the Taylor series are directly linked to Cauchy's integral formula:

$$C_n = \frac{f^{(n)}(z_0)}{n!} = \frac{1}{2\pi i} \int_\gamma \frac{f(\zeta)}{(\zeta - z_0)^{n+1}} \, d\zeta$$

   for any simple closed curve γ around z_0, oriented counter-clockwise, within D.

Example 1 (Complex Taylor)

Entire function: $f(z) = e^z$.
f is entire (holomorphic on all of $\mathbb{C}$). The theorem applies for any $R > 0$ (the radius of convergence is infinite). The series is: $e^z = \sum_{n=0}^{+\infty} \frac{z^n}{n!}$.

Example 2 (Complex Taylor)

Simple singularity: $f(z) = \frac{1}{1-z}$ **centered at** $z_0 = 0$.
The domain is $D = \mathbb{C} \setminus \{1\}$. The largest possible radius to avoid the singularity at $z = 1$ is $R = 1$.
In this disk $B_1(0)$, we have: $\frac{1}{1-z} = \sum_{n=0}^{+\infty} z^n$.
(From now on, this series will be considered known and used without re-proving it for $|z| < 1$).

Example 3 (Complex Taylor)

Complex singularities: $f(z) = \frac{1}{1+z^2}$ **centered at** $z_0 = 0$.
The roots of the denominator are i and $-i$. The domain is therefore $D = \mathbb{C} \setminus \{i, -i\}$. The largest possible radius from 0 is the distance to $\pm i$, which is $R = 1$.
Instead of computing all derivatives, we use an algebraic substitution: let $w = -z^2$. We know that for $|w| < 1$, $\frac{1}{1-w} = \sum_{n=0}^{\infty} w^n$. We substitute to obtain:

$$f(z) = \frac{1}{1 - (-z^2)} = \sum_{n=0}^{+\infty} (-z^2)^n = \sum_{n=0}^{+\infty} (-1)^n z^{2n}$$

Verification: Uniqueness of Series Expansion

If you can express $f(z)$ in the form of a power series $\sum_{n=0}^{+\infty} C_n (z - z_0)^n$ on a disk of convergence, **this series must coincide with the Taylor series of f.** The representation is unique. This is why using algebraic tricks (like in Example 3) is a perfectly valid method and is often preferable to computing successive derivatives $\frac{f^{(n)}(z_0)}{n!}$.

4.3.3 Laurent Series

To go further than Taylor series (which only model holomorphic functions without singularities at z_0), we introduce negative powers to handle isolated singularities (such as a pole).

Theorem: Laurent Series

Let $D \subseteq \mathbb{C}$ be open and $z_0 \in D$. Suppose that $f : D \setminus \{z_0\} \to \mathbb{C}$ is holomorphic. Let $R > 0$ be such that $B_R(z_0) \subseteq D$.
Then for all $z \in B_R(z_0) \setminus \{z_0\}$ (the punctured disk):

$$f(z) = \sum_{n=-\infty}^{+\infty} C_n (z - z_0)^n \tag{7}$$

where the coefficients C_n are given by:

$$C_n = \frac{1}{2\pi i} \int_\gamma \frac{f(\zeta)}{(\zeta - z_0)^{n+1}} \, d\zeta \quad (\forall n \in \mathbb{Z}) \tag{8}$$

Anatomy of a Laurent Series:

$$f(z) = \underbrace{\cdots + \frac{C_{-2}}{(z - z_0)^2} + \frac{C_{-1}}{z - z_0}}_{\text{Singular Part (or Principal Part)}} + \underbrace{C_0 + C_1(z - z_0) + \ldots}_{\text{Regular Part (Taylor Series)}}$$

- **Singular part**: Contains the terms with negative exponents. This part characterizes the behavior of the function near the singularity.
- **Regular part**: Behaves like a Taylor series. If the singular part is zero, the function can be analytically extended at z_0, and the Laurent series is simply the Taylor series.
- As with Taylor, there is **uniqueness** of representation. Any algebraic method yielding a series of the above form allows us to formally identify the $C_n coefficients$.

Example 1: Regular Holomorphic Function

If f is holomorphic at z_0, the Laurent series coincides with the Taylor series. The singular part vanishes because for $n \leq -1$, the term $(n+1) \leq 0$, the integrand $\frac{f(z)}{(z-z_0)^{n+1}}$ is holomorphic, and Cauchy's integral evaluates to zero.
Example: $f(z) = e^z = \sum_{n=0}^{+\infty} \frac{z^n}{n!}$.

Example 2: Laurent Series Centered at $z_0 = 0$

For the following functions defined on $\mathbb{C} \setminus \{0\}$:

- $f(z) = \frac{1}{z}$: Is already in the form of a Laurent series. $C_{-1} = 1$, and all other $C_n = 0$.
- $f(z) = \frac{2}{z^2}$: Already in the form of a Laurent series. $C_{-2} = 2$, and all other $C_n = 0$.
- $f(z) = \frac{5z+1}{z^2} = \frac{1}{z^2} + \frac{5}{z}$: The Laurent series is explicit:

$$C_n = \begin{cases} 1, & \text{if } n = -2 \\ 5, & \text{if } n = -1 \\ 0, & \text{otherwise} \end{cases}$$

5 Week 5 – 16.03.26

5.1 Fundamental Integral Formulas and Series Expansions

5.1.1 Review and Cauchy's Extended Formula

Recall from the previous weeks:

As a consequence of Cauchy's theorem:
If γ_0, γ_1 are simple, closed curves oriented in the same direction, such that γ_1 is inside γ_0 and f is holomorphic between γ_0 and γ_1:

$$\int_{\gamma_0} f(z)\, dz = \int_{\gamma_1} f(z)\, dz$$

(*In practice: I can move the loop of integration, as long as I don't "cross" any singularity of f*).

Theorem: Cauchy's Integral Formula (Extended)

If Γ is a simple, closed, counter-clockwise oriented curve and f is holomorphic in the interior of Γ, then for any z_0 in the interior of Γ:

$$f^{(n)}(z_0) = \frac{n!}{2\pi i} \int_{\Gamma} \frac{f(z)}{(z - z_0)^{n+1}}\, dz$$

5.1.2 Taylor and Laurent Series Expansions

Theorem: Taylor Series

If $D \subseteq \mathbb{C}$ is open, $z_0 \in D$ and f is holomorphic in D:
$\forall R > 0$ such that the disk $B_R(z_0)$ lies entirely in D, we have:

$$f(z) = \sum_{n=0}^{+\infty} \frac{f^{(n)}(z_0)}{n!} (z - z_0)^n$$

for any z in the disk $B_R(z_0)$.

The largest such R is called the **"Radius of convergence"** or **"Radius of analyticity"**.

In practice: R is the distance from z_0 to the nearest "singular" point.

Unlike in the real case, for holomorphic functions we can also write down a series expansion even when f is not defined at z_0 (i.e., $z_0 \notin D$!).

Theorem: Laurent Series

With D as before, if $f : D \setminus \{z_0\} \longrightarrow \mathbb{C}$ is holomorphic, then for any $R > 0$ such that $B_R(z_0) \subseteq D$, the following equality holds for $z \in B_R(z_0) \setminus \{z_0\}$:

$$f(z) = \sum_{n=-\infty}^{+\infty} c_n \cdot (z - z_0)^n$$

where the constant c_n can be computed by the formula:

$$c_n = \frac{1}{2\pi i} \int_\gamma \frac{f(\zeta)}{(\zeta - z_0)^{n+1}} \, d\zeta$$

where γ is any simple, closed, counter-clockwise curve around z_0 such that int$(\gamma) \setminus \{z_0\} \subseteq D$.

Remark. When f (originally defined on $D \setminus \{z_0\}$) is actually extendible on z_0 as a holomorphic function there, then $c_n = 0$ for $n \leq -1$ and the Laurent series is simply the Taylor series.

In this case, the fact that the coefficient c_n given by the integral formula vanishes for $n \leq -1$ follows by Cauchy's theorem (since $(\zeta - z_0)^{-n-1}$ is holomorphic everywhere when $n \leq -1$).

Uniqueness and Calculations **Theorem (Uniqueness of Series).** Laurent and Taylor series are unique. So if I can express $f(z)$ as a power series $\sum_{n=-\infty}^{+\infty} c_n(z - z_0)^n$ using some other method (not necessarily using the integral formula), this is still the Laurent series.

Example. We can use the fact that $\frac{1}{1-z} = \sum_{n=0}^{+\infty} z^n$ for $|z| < 1$ to expand:

$$\frac{1}{1 + z^2} = \frac{1}{1 - (-z^2)} = \sum_{n=0}^{+\infty} (-z^2)^n = \sum_{n=0}^{+\infty} (-1)^n z^{2n} = \sum_{k=0}^{+\infty} c_k \cdot z^k$$

where

$$c_k = \begin{cases} 0, & k \text{ odd} \\ (-1)^{k/2}, & k \text{ even} \end{cases}$$

5.2 Classification of Singularities

Any Laurent series splits into its singular and regular parts:

$$f(z) = \sum_{n=-\infty}^{+\infty} c_n \cdot (z - z_0)^n = \underbrace{\cdots + \frac{c_{-2}}{(z - z_0)^2} + \frac{c_{-1}}{z - z_0}}_{\text{singular part}} + \underbrace{c_0 + c_1(z - z_0) + \cdots}_{\text{regular part}}$$

If f can be extended to z_0 so that it's holomorphic there: The singular part must vanish (and the regular part is just the Taylor series). (In fact: The opposite is also true).

5.2.1 Case Study of a Rational Function

Example. Let $f : \mathbb{C} \setminus \{0, -1\} \longrightarrow \mathbb{C}$, $f(z) = \frac{1}{z \cdot (z+1)}$. What is the Laurent series of f at z_0? What is the radius of convergence? We will consider cases:

Case $z_0 = 0$: Observe $f(z) = \frac{1}{z} - \frac{1}{z+1}$. Recall: $\frac{1}{1+z} = \frac{1}{1-(-z)} = \sum_{n=0}^{+\infty}(-z)^n = \sum_{n=0}^{+\infty}(-1)^n z^n$ for $|z| < 1$. So,

$$f(z) = \frac{1}{z} - \sum_{n=0}^{+\infty}(-1)^n z^n = \frac{1}{z} + \sum_{n=0}^{+\infty}(-1)^{n+1} z^n = \sum_{n=-1}^{+\infty}(-1)^{n+1} z^n$$

Radius of convergence: $R = 1$ (equal to the distance of $z_0 = 0$ from the nearest point of "singularity" of f, in this case $z = -1$).

Case $z_0 = -1$:

$$f(z) = \frac{1}{z \cdot (z+1)} = -\frac{1}{z+1} + \frac{1}{z} = -\frac{1}{z-(-1)} + \frac{1}{z}$$

Expand $\frac{1}{z}$ in a Taylor series around $z_0 = -1$:

$$\frac{1}{z} = \frac{1}{-1+(z+1)} = -\frac{1}{1-(z+1)} = -\sum_{n=0}^{+\infty}(z+1)^n$$

which converges for $|z-(-1)| < 1$. So,

$$f(z) = -\frac{1}{z-(-1)} - \sum_{n=0}^{+\infty}(z-(-1))^n = \sum_{n=-1}^{+\infty}(-1) \cdot (z-(-1))^n$$

Radius of convergence: $R = 1$.

Case $z_0 \neq 0, -1$: In this case: f is holomorphic on a set containing z_0, so it has a regular Taylor series. It converges in any disk centered at z_0 which does not contain 0 or -1. So, $R = \min\{|z_0|, |z_0 + 1|\}$.

5.2.2 Definitions and Characterizations

Definition (Regular Point). We say that f has a **regular point** at z_0 if the singular part of the Laurent series of f at z_0 vanishes.

Example. $f(z) = \sum_{n=0}^{+\infty} \frac{1}{n!} z^n \quad (= e^z)$

Remark. If f is holomorphic at z_0, then z_0 is also a regular point. The converse is also true: If z_0 is a regular point of f, then $\lim_{z \to z_0} f(z)$ exists and if we define f at z_0 so that $f(z_0) = \lim_{z \to z_0} f(z)$, then f is holomorphic at z_0.

Criterion that we will use: z_0 is a regular point of $f \iff \lim_{z \to z_0} f(z)$ exists.

Definition (Pole of Order m). We say that f has a **pole of order** m at z_0 if the coefficients of the Laurent series of f at z_0 satisfy $c_{-m} \neq 0$, and $c_n = 0$ for $n < -m$.

Example.

- $f(z) = \frac{1}{z}$ has a pole of order 1 at $z_0 = 0$.
- $f(z) = \frac{1}{(z+2)^2} + \frac{1}{z}$ has a pole of order 2 at $z_0 = -2$ and a pole of order 1 at $z_0 = 0$.

In the case of a pole of order m at z_0:

$$f(z) = \frac{c_{-m}}{(z - z_0)^m} + \frac{c_{-m+1}}{(z - z_0)^{m-1}} + \cdots + c_0 + c_1(z - z_0) + \cdots$$

Note: $g(z) = (z - z_0)^m \cdot f(z)$ has a regular point at z_0!

Definition (Essential Singularity). We say that f has an **essential singularity** at z_0 if the Laurent series has infinitely many coefficients $c_n \neq 0$ with $n \leq -1$.

Example. $f(z) = e^{1/z}$ at $z_0 = 0$.

$$f(z) = \sum_{n=0}^{+\infty} \frac{(1/z)^n}{n!} = \sum_{n=-\infty}^{0} \frac{1}{(-n)!} z^n$$

For $f(z) = z \cdot e^{1/z}$ at $z_0 = 0$, the Laurent series is:

$$z \cdot e^{1/z} = z \cdot \sum_{n=0}^{+\infty} \frac{(1/z)^n}{n!} = \underbrace{z + 1}_{\text{regular part}} + \underbrace{\frac{1}{2!\,z} + \frac{1}{3!\,z^2} + \cdots}_{\text{singular part}}$$

5.3 Study of Quotients

Suppose that $p, q : D \setminus \{z_0\} \longrightarrow \mathbb{C}$ are holomorphic with z_0 being regular for both, and their Laurent series are given by:

$$p(z) = a_0 + a_1(z - z_0) + a_2(z - z_0)^2 + \cdots$$
$$q(z) = b_0 + b_1(z - z_0) + b_2(z - z_0)^2 + \cdots$$

(so p, q extend holomorphically at $z = z_0$).

We want to study the quotient $\frac{p(z)}{q(z)}$ near z_0. We want to know what is the order of the poles, if any. We will first look at a few instructive examples.

5.3.1 Basic Singular Behaviors

Proposition (a) If $b_0 \neq 0$ (so $q(z_0) \neq 0$): $\frac{1}{q(z)}$ is holomorphic near z_0, and

$$\lim_{z \to z_0} f(z) = \lim_{z \to z_0} \frac{p(z)}{q(z)} = \frac{p(z_0)}{q(z_0)} = \frac{a_0}{b_0}$$

So z_0 is a regular point for f.

Proposition (b) If $a_0 = b_0 = 0$ **but** $b_1 \neq 0$**:**

$$f(z) = \frac{a_1(z - z_0) + a_2(z - z_0)^2 + \cdots}{b_1(z - z_0) + b_2(z - z_0)^2 + \cdots} = \frac{a_1 + a_2(z - z_0) + \cdots}{b_1 + b_2(z - z_0) + \cdots}$$

So

$$\lim_{z \to z_0} f(z) = \frac{a_1}{b_1} \quad \left(= \frac{p'(z_0)}{q'(z_0)} \right) \quad \text{(de L'Hôpital's rule)}$$

So z_0 is a regular point for f. Similar behavior holds if $a_0 = a_1 = 0$, $b_0 = b_1 = 0$ but $b_2 \neq 0$.

Example 1: Let $f(z) = \frac{z}{\sin(z)}$, $z_0 = 0$.

$$f(z) = \frac{z}{z - \frac{z^3}{3!} + \frac{z^5}{5!} - \cdots} = \frac{1}{1 - \frac{z^2}{3!} + \frac{z^4}{5!} - \cdots}$$

So $\lim_{z\to 0} f(z) = 1$. Regular point ✓

Example 2: Let $f(z) = \frac{e^z - 1 - z}{z^2}$, $z_0 = 0$. $e^z - 1 - z = \frac{z^2}{2!} + \frac{z^3}{3!} + \cdots$ So:

$$\lim_{z\to 0} f(z) = \lim_{z\to 0} \frac{z^2/2! + z^3/3! + \cdots}{z^2} = \frac{1}{2}$$

Regular point ✓

In Summary: Either expand in power series and take out as many common powers of $(z - z_0)$ from the numerator and the denominator as possible and take the limit, **or** use de L'Hôpital's rule.

5.3.2 General Pole Orders

Proposition (c) General Pole Orders. Suppose that

$$p(z) = a_k(z - z_0)^k + a_{k+1}(z - z_0)^{k+1} + \cdots \quad (a_k \neq 0)$$
$$q(z) = b_l(z - z_0)^l + b_{l+1}(z - z_0)^{l+1} + \cdots \quad (b_l \neq 0)$$

and $f(z) = \frac{p(z)}{q(z)}$.

- **If** $k \geq l$, then z_0 is a regular point:

$$\begin{aligned}
\lim_{z\to z_0} f(z) &= \lim_{z\to z_0} \frac{a_k(z - z_0)^k + a_{k+1}(z - z_0)^{k+1} + \cdots}{b_l(z - z_0)^l + b_{l+1}(z - z_0)^{l+1} + \cdots} \\
&= \lim_{z\to z_0} \frac{a_k(z - z_0)^{k-l} + a_{k+1}(z - z_0)^{k-l+1} + \cdots}{b_l + b_{l+1}(z - z_0) + \cdots} \\
&= \begin{cases} 0, & \text{if } k > l, \\ \frac{a_k}{b_l}, & \text{if } k = l. \end{cases}
\end{aligned}$$

  The limit exists, so it's a regular point.

- **If** $k < l$: Then z_0 is a pole of order $l - k$:

$$\begin{aligned}
f(z) &= \frac{a_k(z - z_0)^k + a_{k+1}(z - z_0)^{k+1} + \cdots}{b_l(z - z_0)^l + b_{l+1}(z - z_0)^{l+1} + \cdots} \\
&= \frac{a_k + a_{k+1}(z - z_0) + \cdots}{b_l(z - z_0)^{l-k} + b_{l+1}(z - z_0)^{l-k+1} + \cdots} \\
&= \frac{1}{(z - z_0)^{l-k}} \cdot \frac{a_k + a_{k+1}(z - z_0) + \cdots}{b_l + b_{l+1}(z - z_0) + \cdots}
\end{aligned}$$

  For this last factor, z_0 is a regular point, so its Laurent series has only a regular part: $= c_0 + c_1(z - z_0) + \cdots$ where $c_0 = \frac{a_k}{b_l}$.
  So,

$$f(z) = \frac{c_0}{(z - z_0)^{l-k}} + \frac{c_1}{(z - z_0)^{l-k-1}} + \cdots$$

  So the top order term is: $\frac{1}{(z - z_0)^{l-k}}$.

Example. Let $f(z) = \frac{\sin(z)}{2z^3+z^4}$. Show it has a pole of order 2 at $z_0 = 0$.

We know $\sin(z) = z - \frac{z^3}{3!} + \frac{z^5}{5!} - \cdots$ So,

$$\begin{aligned} f(z) &= \frac{z - \frac{z^3}{3!} + \frac{z^5}{5!} - \cdots}{2z^3 + z^4} \\ &= \frac{1 - \frac{z^2}{3!} + \frac{z^4}{5!} - \cdots}{2z^2 + z^3} \\ &= \frac{1}{z^2} \cdot \underbrace{\frac{1 - \frac{z^2}{3!} + \cdots}{2+z}}_{\text{Regular at } z_0=0} \end{aligned}$$

So near $z_0 = 0$:

$$f(z) = \frac{1}{2z^2} + \frac{c_{-1}}{z} + c_0 + \cdots$$

Hence, it has a pole of order 2.

6 Week 6 – 23.03.26 - Residues

Definition: Residue

The coefficient c_{-1} of the term $(z-z_0)^{-1}$ in the Laurent series for f at z_0 is called the **residue** of f at z_0.

$$\mathrm{Res}_{z_0}(f) := c_{-1}$$

In view of the definition of the coefficients of the Laurent series:

$$c_{-1} = \frac{1}{2\pi i}\int_\gamma f(z)dz$$

for any curve γ which is closed, simple, counter-clockwise oriented, contains z_0 in its interior, and f is holomorphic on int$(\gamma) \setminus \{z_0\}$.

Trivial & Basic Examples

Trivial example: If f is holomorphic on a set D containing z_0, then $\mathrm{Res}_{z_0}(f) = 0$ (Cauchy's theorem).
Basic example: $\mathrm{Res}_0\left(\frac{1}{z}\right) = 1$ (since $f(z) = \frac{1}{z}$ already is expressed as its Laurent series at $z_0 = 0$).

Note: Extracting the Residue for a Pole of Order m

Suppose that $f(z)$ has a pole of order m at z_0:

$$f(z) = c_{-m}(z-z_0)^{-m} + \cdots + c_{-1}(z-z_0)^{-1} + c_0 + \ldots$$

Then:

$$\begin{aligned} g(z) &= (z-z_0)^m f(z) \\ &= c_{-m} + c_{-m+1}(z-z_0) + \cdots + c_{-1}(z-z_0)^{m-1} + \ldots \end{aligned}$$

is the Taylor series of the regular function g at z_0.
Thus: We can extract the residue using the formula for the Taylor coefficients:

$$c_{-1} = \frac{1}{(m-1)!}\frac{d^{m-1}}{dz^{m-1}}(g(z))\bigg|_{z=z_0} = \lim_{z\to z_0}\frac{1}{(m-1)!}\frac{d^{m-1}}{dz^{m-1}}\left((z-z_0)^m f(z)\right)$$

The above formula is useful if we already know that f has a pole of order m at z_0.

Most useful: When $m = 1$. In that case:

$$\mathrm{Res}_{z_0}(f) = \lim_{z\to z_0}\left((z-z_0)\cdot f(z)\right)$$

Example

$f(z) = \frac{1}{\sin(z)}$, $z_0 = 0$.
Note:

$$f(z) = \frac{1}{z - \frac{z^3}{3!} + \dots} = \frac{1}{z} \cdot \frac{1}{1 - \frac{z^2}{3!} + \dots}$$

So f has a pole of order 1 at $z_0 = 0$. (It would also follow from our more general discussion last time for functions of the form $\frac{p(z)}{q(z)}$).
So:

$$\begin{aligned}
\text{Res}_0(f) &= \lim_{z \to 0} (z \cdot f(z)) \\
&= \lim_{z \to 0} \frac{z}{\sin(z)} \\
&= \lim_{z \to 0} \frac{z}{z - \frac{z^3}{3!} + \dots} \\
&= \lim_{z \to 0} \frac{1}{1 - \frac{z^2}{3!} + \dots} = 1
\end{aligned}$$

6.1 Residue Theorem and Applications

Recall: If f is holomorphic "around" z_0, but with a possible singularity at z_0:

$$\int_\gamma f(z)dz = 2\pi i \, \text{Res}_{z_0}(f)$$

if γ is simple, closed, counter-clockwise oriented, contains z_0 and no other singularity of f (so f is holomorphic on the set $\text{int}(\gamma) \setminus \{z_0\}$).

Theorem: Residue Theorem

Suppose that $D \subseteq \mathbb{C}$ is open and simply connected and let γ be a simple, closed curve in D which is counter-clockwise oriented. Let $z_1, z_2, \dots, z_m \in \text{int}(\gamma)$.
If $f : D \setminus \{z_1, z_2, \dots, z_m\} \longrightarrow \mathbb{C}$ is holomorphic, then

$$\int_\gamma f(z)dz = 2\pi i \left(\text{Res}_{z_1}(f) + \dots + \text{Res}_{z_m}(f)\right)$$

Remark: In order to compute $\int_\gamma f(z)dz$, it therefore suffices to compute the residues of f at the singular points in the interior of γ.

Sketch of the proof (by picture)

Assume (without loss of generality) that we only have two singular points: z_1 and z_2. By the extension of Cauchy's theorem, integrating over a deformed contour that isolates the singularities gives zero. When the "bridge" is too narrow: The integral over the straight path components cancels out, leaving us with the two integrals over the smaller loops γ_1 and γ_2 around z_1 and z_2, respectively:

$$\int_\gamma f(z)dz = \int_{\gamma_1} f(z)dz + \int_{\gamma_2} f(z)dz = 2\pi i \, \text{Res}_{z_1}(f) + 2\pi i \, \text{Res}_{z_2}(f)$$

6.1.1 Examples

Example 1

Consider $f(z) = \frac{1}{z}$, let γ be a simple, closed, counter-clockwise oriented curve in $\mathbb{C}$. Compute $\int_\gamma f(z)dz$.

- **Case 1:** $0 \in \text{int}(\gamma)$. Then $\int_\gamma f(z)dz = 2\pi i \, \text{Res}_0 \left(\frac{1}{z}\right) = 2\pi i$.
- **Case 2:** $0 \notin \overline{\text{int}(\gamma)} = \text{int}(\gamma) \cup \gamma$. Then $\int_\gamma f(z)dz = 0$.
- **Case 3:** $0 \in \gamma$. In this case, $\int_\gamma f(z)dz$ is not well-defined.

Note:

- $\text{int}(\gamma)$: The open set enclosed by γ (so it doesn't contain γ).
- $\overline{\text{int}(\gamma)} = \text{int}(\gamma) \cup \gamma$: The closed set enclosed by γ (so it contains the curve γ).

Example 2

Let $f : \mathbb{C} \setminus \{0, -1\} \longrightarrow \mathbb{C}$, $f(z) = \frac{1}{z \cdot (z+1)}$.
We have two poles of order 1 at $z_1 = 0$ and $z_2 = -1$.

$$\text{Res}_0(f) = \lim_{z \to 0}(z \cdot f(z)) = \lim_{z \to 0} \frac{1}{z+1} = 1$$
$$\text{Res}_{-1}(f) = \lim_{z \to -1}((z+1)f(z)) = \lim_{z \to -1} \frac{1}{z} = -1$$

Let γ be a simple, closed, counter-clockwise oriented curve. Compute $\int_\gamma f(z)dz =?$

- **Case A:** $0, -1 \notin \overline{\text{int}(\gamma)} \implies \int_\gamma f(z)dz = 0$.
- **Case B:** $0 \in \text{int}(\gamma), -1 \notin \overline{\text{int}(\gamma)} \implies \int_\gamma f(z)dz = 2\pi i \, \text{Res}_0(f) = 2\pi i$.
- **Case C:** $0 \notin \overline{\text{int}(\gamma)}, -1 \in \text{int}(\gamma) \implies \int_\gamma f(z)dz = 2\pi i \, \text{Res}_{-1}(f) = -2\pi i$.
- **Case D:** $0, -1 \in \text{int}(\gamma) \implies \int_\gamma f(z)dz = 2\pi i(\text{Res}_0(f) + \text{Res}_{-1}(f)) = 2\pi i(1 + (-1)) = 0$.
- **Case E:** $0 \in \gamma \vee -1 \in \gamma$. In this case, the integral is not well-defined.

Example 3

$f : \mathbb{C} \setminus \{0, 1\} \longrightarrow \mathbb{C}$, $f(z) = \frac{1}{z^2} + \frac{2}{z} + \frac{3}{z-1}$.
Pole of order 2 at 0, pole of order 1 at 1.

$$\text{Res}_1(f) = \lim_{z \to 1}((z-1)f(z)) = \lim_{z \to 1}\left(\frac{z-1}{z^2} + \frac{2(z-1)}{z} + 3\right) = 3$$
$$\text{Res}_0(f) = \lim_{z \to 0}\left(\frac{1}{1!}\frac{d}{dz}(z^2 f(z))\right) = \lim_{z \to 0}\left(\frac{d}{dz}\left(1 + 2z + \frac{3z^2}{z-1}\right)\right)$$
$$= \lim_{z \to 0}\left(2 + \frac{6z}{z-1} - \frac{3z^2}{(z-1)^2}\right) = 2$$

(*I could have also deduced that by noting that* $f(z) = \frac{1}{z^2} + \frac{2}{z} + \frac{3}{z-1}$*, so from the singular part at* $z = 0$ *we can trivially read off* $c_{-1} = 2$).
If $\gamma(t) = 10e^{it}, t \in [0, 2\pi]$: $\text{int}(\gamma)$ contains both poles:

$$\int_\gamma f(z)dz = 2\pi i(\text{Res}_0(f) + \text{Res}_1(f)) = 10\pi i$$

Examples 4 & 5

Example 4: $\exp\left(\frac{1}{z}\right) = 1 + z^{-1} + \frac{z^{-2}}{2!} + \ldots$
$\text{Res}_0\left(\exp\left(\frac{1}{z}\right)\right) = c_{-1} = 1$.

Example 5: $\exp\left(\frac{1}{z^2}\right) = 1 + z^{-2} + \frac{z^{-4}}{2!} + \ldots$
$\text{Res}_0\left(\exp\left(\frac{1}{z^2}\right)\right) = 0$.

6.2 Applications

The residue theorem allows us to easily compute integrals over simple closed curves. Sometimes, we can write integrals over the real line as limits of integrals over closed curves in $\mathbb{C}$.

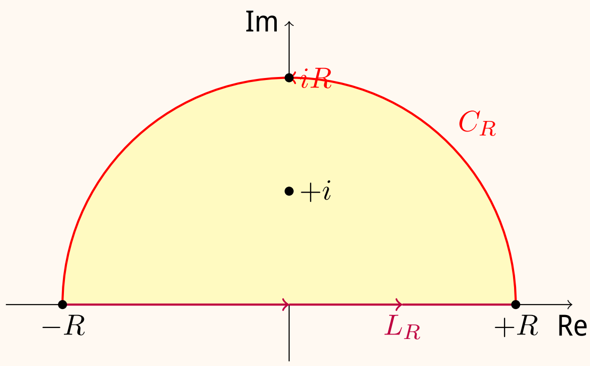
Application Example: Integral over the Real Line Part 1

Compute: $\int_{-\infty}^{+\infty} \frac{1}{1+x^2} dx$
Set $f(z) = \frac{1}{z^2+1}$. For any $R > 1$, consider the segment:

$$L_R : [-R, R] \longrightarrow \mathbb{C}, \quad t \longmapsto t \quad \text{(segment on the real axis)}$$

and the half circle:

$$C_R : [0, \pi] \longrightarrow \mathbb{C}, \quad t \longmapsto Re^{it}$$



Application Example: Integral over the Real Line Part 2

Consider the closed curve $\gamma = L_R \cup C_R$ (boundary of the half-disc of radius R). The function $f(z) = \frac{1}{z^2+1} = \frac{1}{(z+i)(z-i)}$ has poles of order 1 at $\pm i$.
From the residue theorem:

$$\int_\gamma f(z)dz = 2\pi i \operatorname{Res}_i(f) \quad (\text{since } \operatorname{int}(\gamma) \text{ contains only } i)$$

$$\operatorname{Res}_i(f) = \lim_{z\to i}((z-i)f(z)) = \lim_{z\to i}\frac{1}{z+i} = \frac{1}{2i}$$

So:

$$\int_\gamma f(z)dz = \pi \tag{9}$$

But, we also know:

$$\int_\gamma f(z)dz = \int_{L_R} f(z)dz + \int_{C_R} f(z)dz$$

- $\int_{L_R} f(z)dz = \int_{-R}^{R} \frac{1}{t^2+1}dt \xrightarrow{R\to+\infty} \int_{-\infty}^{+\infty} \frac{1}{x^2+1}dx$
- $\int_{C_R} f(z)dz = \int_0^\pi \frac{R\cdot ie^{it}}{1+R^2e^{2it}}dt$

We will show that $\int_{C_R} f(z)dz \xrightarrow{R\to+\infty} 0$.

$$\begin{aligned}\left|\int_{C_R} f(z)dz\right| &= \left|\int_0^\pi \frac{R\cdot ie^{it}}{1+R^2e^{2it}}dt\right| \\ &\le \int_0^\pi \frac{R\cdot|ie^{it}|}{|1+R^2e^{2it}|}dt\end{aligned}$$

Note: $|ie^{it}| = 1$ and, when $R\to+\infty$, $|1+R^2e^{2it}| \approx R^2$. Since, by the triangle inequality,

$$|R^2e^{2it}| - 1 \le |R^2e^{2it}+1| \le |R^2e^{2it}| + 1$$

We have $R^2 - 1 \le |1+R^2e^{2it}|$. Thus:

$$\left|\int_{C_R} f(z)dz\right| \le \int_0^\pi \frac{R}{R^2-1}dt \approx \int_0^\pi \frac{R}{R^2}dt = \frac{\pi}{R} \xrightarrow{R\to+\infty} 0$$

From (9), we deduce that:

$$\int_{-\infty}^{+\infty} \frac{1}{1+x^2}dx = \pi$$

7 Week 7 – 30.03.26 - Applications of the Residue Theorem

We can use the residue theorem to compute certain integrals over the real line. A classic example is $\int_{-\infty}^{+\infty} \frac{1}{1+x^2}\,dx$. This method works for functions $f(z)$ that, as $|z| \to +\infty$, decay faster than $\frac{1}{|z|}$.

7.1 Integrals Over the Real Line

Let $R(x) = \frac{P(x)}{Q(x)}$, where $P(x)$ and $Q(x)$ are real polynomials such that:

1. $\deg Q \geq \deg P + 2$
2. $R(x)$ is defined on all of $\mathbb{R}$ (i.e., $Q(x) \neq 0$ for all $x \in \mathbb{R}$).

Remarks:

- The condition $Q(x) \neq 0$ on $\mathbb{R}$ implies that $\deg Q$ is even, since an odd-degree real polynomial always has at least one real root.
- $Q(x) = a_k x^k + a_{k-1}x^{k-1} + \cdots + a_0$ has real coefficients. Thus, if $Q(z) = 0$ for some $z \in \mathbb{C}$, then taking the complex conjugate yields $Q(\bar{z}) = 0$. The roots of Q in $\mathbb{C}$ come in complex conjugate pairs.

For any $a \geq 0$, we are interested in computing integrals of the form:

$$\int_{-\infty}^{+\infty} R(x)\cos(ax)\,dx \quad \text{and} \quad \int_{-\infty}^{+\infty} R(x)\sin(ax)\,dx$$

Since $e^{iax} = \cos(ax) + i\sin(ax)$, we can compute them simultaneously by evaluating:

$$I = \int_{-\infty}^{+\infty} R(x)e^{iax}\,dx \tag{10}$$

Applications:

- **Fourier transform (Signal Analysis):** $\hat{f}(a) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} f(t)e^{-iat}\,dt$
- Study of resonances in harmonic systems with friction.

7.2 Evaluation Contour

Consider the function $f(z) = R(z)e^{iaz}$. If $z_1, \bar{z}_1, \ldots, z_k, \bar{z}_k$ are the zeroes of $Q(z)$, then $f : \mathbb{C} \setminus \{z_1, \ldots, z_k\} \to \mathbb{C}$ is holomorphic.

Let $r > 0$ be large enough so that all zeroes $z_1, \ldots, z_k$ in the upper half-plane are contained in the semi-disc $B_r(0) \cap \{\text{Im}(z) > 0\}$. We define the curves:

- $L_r = \{z : \text{Im}(z) = 0, -r \le \text{Re}(z) \le r\}$ parameterized by $\gamma_1(t) = t,\ t \in [-r, r]$.
- $C_r = \{z : z = re^{i\theta},\ \theta \in [0, \pi]\}$ (half-circle in the upper half-plane).

Let $\Gamma_r = L_r \cup C_r$ be the simple closed contour. Then:

$$\int_{\Gamma_r} f(z)\,dz = \int_{-r}^{r} R(x)e^{iax}\,dx + \int_{C_r} f(z)\,dz \tag{11}$$

As $r \to +\infty$:

1. $\int_{\Gamma_r} f(z)\,dz = 2\pi i \sum_{\text{Im}(z_j)>0} \text{Res}_{z_j}(f)$ (by the Residue Theorem).
2. $\int_{-r}^{r} R(x)e^{iax}\,dx \to I$.
3. $\int_{C_r} f(z)\,dz \to 0$.

Proof that $\int_{C_r} f(z)dz \to 0$

For $\text{Im}(z) \ge 0$ and $a \ge 0$, we have $|e^{iaz}| \le 1$. Splitting $z = x + iy$:

$$|e^{iaz}| = |e^{ia(x+iy)}| = |e^{iax}e^{-ay}| = e^{-ay} \le 1$$

This is the reason we close the loop in the upper half-plane. Furthermore, since deg $Q \ge$ deg $P + 2$, we have for large r:

$$\left|\frac{P(re^{i\theta})}{Q(re^{i\theta})}\right| \le \frac{C}{r^{\text{deg}\,Q - \text{deg}\,P}} \le \frac{C}{r^2}$$

Therefore:

$$\begin{aligned}\left|\int_{C_r} f(z)\,dz\right| &\le \int_0^{\pi} \left|f(re^{i\theta})\right| \cdot \left|ire^{i\theta}\right|\,d\theta \\ &\le \int_0^{\pi} \frac{C}{r^2} \cdot 1 \cdot r\,d\theta = \frac{C\pi}{r} \xrightarrow{r\to+\infty} 0\end{aligned}$$

Conclusion for Real Integrals:

$$\int_{-\infty}^{+\infty} R(x)e^{iax}\,dx = 2\pi i \sum_{\text{Im}(z_j)>0} \text{Res}_{z_j}\left(R(z)e^{iaz}\right) \quad \text{for } a \ge 0 \tag{12}$$

Note: If $a < 0$, set $\tilde{x} = -x$ and compute similarly.

7.3 Example 1

Compute $A = \int_{-\infty}^{+\infty} \frac{x^2}{16+x^4} \cos(x)\,dx$.

Here $R(x) = \frac{x^2}{16+x^4}$ and $a = 1$. $A = \text{Re}\left(\int_{-\infty}^{+\infty} \frac{x^2}{16+x^4} e^{ix}\,dx\right)$.
Roots of $Q(z) = 16 + z^4 = 0 \implies z^4 = 16e^{i\pi}$. The roots are $z_k = 2e^{i(\pi/4 + k\pi/2)}$ for $k = 0, 1, 2, 3$.

- $z_1 = \sqrt{2} + i\sqrt{2}$ (upper half-plane)
- $z_2 = -\sqrt{2} + i\sqrt{2}$ (upper half-plane)

Let $f(z) = \frac{z^2}{16+z^4} e^{iz}$. The residues at simple poles z_1, z_2 can be calculated via $P(z_0)/Q'(z_0)$ or factoring.

$$\text{Res}_{z_1}(f) = \lim_{z \to z_1} (z - z_1) f(z) = \frac{z_1^2 e^{iz_1}}{4z_1^3} = \frac{e^{iz_1}}{4z_1}$$
$$\text{Res}_{z_2}(f) = \lim_{z \to z_2} (z - z_2) f(z) = \frac{z_2^2 e^{iz_2}}{4z_2^3} = \frac{e^{iz_2}}{4z_2}$$

Applying the Residue Theorem and extracting the real part yields:

$$A = \text{Re}\left(2\pi i (\text{Res}_{z_1}(f) + \text{Res}_{z_2}(f))\right) = \frac{\pi\sqrt{2}}{4} e^{-\sqrt{2}} (\cos\sqrt{2} - \sin\sqrt{2}) \tag{13}$$

7.4 Integrals of Rational Trigonometric Functions

Suppose $P(x, y)$ and $Q(x, y)$ are polynomials, such that $Q(x, y) \neq 0$ on the unit circle (i.e., for $x^2 + y^2 = 1$). We want to compute:

$$\int_0^{2\pi} \frac{P(\cos t, \sin t)}{Q(\cos t, \sin t)} \, dt \tag{14}$$

We transform this into a complex contour integral over the unit circle $|z| = 1$, utilizing the parameterization $z = e^{it}$. Thus, $dz = ie^{it} \, dt \implies dt = \frac{dz}{iz}$.

$$\cos t = \frac{e^{it} + e^{-it}}{2} = \frac{z + z^{-1}}{2}, \quad \sin t = \frac{e^{it} - e^{-it}}{2i} = \frac{z - z^{-1}}{2i}$$

The integral becomes $\int_{|z|=1} f(z) \, dz$ where:

$$f(z) = \frac{1}{iz} \cdot \frac{P\left(\frac{z+z^{-1}}{2}, \frac{z-z^{-1}}{2i}\right)}{Q\left(\frac{z+z^{-1}}{2}, \frac{z-z^{-1}}{2i}\right)} \tag{15}$$

Since P and Q are polynomials, f is a **meromorphic function** (holomorphic except for isolated poles). By the Residue Theorem:

$$\int_0^{2\pi} \frac{P(\cos t, \sin t)}{Q(\cos t, \sin t)} \, dt = 2\pi i \sum_{|z_k|<1} \text{Res}_{z_k}(f) \tag{16}$$

7.4.1 Example 2

Compute $\int_0^{2\pi} \frac{1}{2+\cos\theta} \, d\theta$.
Here $P(x, y) = 1$ and $Q(x, y) = 2 + x$. Substituting $\cos\theta = \frac{z+z^{-1}}{2}$ and $d\theta = \frac{dz}{iz}$:

$$f(z) = \frac{1}{iz} \cdot \frac{1}{2 + \frac{z+1/z}{2}} = \frac{1}{iz} \cdot \frac{2z}{z^2 + 4z + 1} \cdot \frac{1}{i} = \frac{2}{i(z^2 + 4z + 1)}$$

Singular points (roots of $z^2 + 4z + 1 = 0$):

$$z_1 = -2 + \sqrt{3}, \quad z_2 = -2 - \sqrt{3}$$

Only $z_1 \approx -0.268$ lies inside the unit circle $|z| = 1$. $f(z)$ has a simple pole at z_1.

$$\text{Res}_{z_1}(f) = \lim_{z \to z_1} (z - z_1) f(z) = \frac{2}{i(z_1 - z_2)} = \frac{2}{i(-2+\sqrt{3} - (-2-\sqrt{3}))} = \frac{2}{i(2\sqrt{3})} = \frac{1}{i\sqrt{3}}$$

Evaluating the integral:

$$\int_0^{2\pi} \frac{1}{2 + \cos\theta}\, d\theta = 2\pi i \left(\text{Res}_{z_1}(f)\right) = 2\pi i \left(\frac{1}{i\sqrt{3}}\right) = \frac{2\pi}{\sqrt{3}} \tag{17}$$

(Sanity check: the original integrand is purely real and positive, so the final integral yields a positive real number).

Summary Table: Contour Integration Strategies

Integral Type	Domain	Substitution / Contour	Relevant Singularities
Rational Functions over $\mathbb{R}$	$\mathbb{R} \to \pm\infty$	Semicircle in upper half-plane	Poles where $\text{Im}(z) > 0$
Rational Trigonometric	$[0, 2\pi]$	$z = e^{i\theta}$, Unit circle $|z| = 1$	Poles where $|z| < 1$

8 Week 8 – 13.04.26 - Review of Fourier Transform

Let $f : \mathbb{R} \to \mathbb{C}$ be a (piecewise) continuous function.

Definition: Fourier Transform

The Fourier transform of f is defined as:

$$\hat{f}(a) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} e^{-iax} f(x) \, \mathrm{d}x \tag{18}$$

Sometimes we also use the notation $\mathcal{F}[f](a)$ for $\hat{f}(a)$.

In order for the integral to be well-defined, we need $\int_{-\infty}^{+\infty} |f(x)| \, \mathrm{d}x < \infty$ (i.e., $f \in L^1(\mathbb{R})$).

Definition: Inverse Fourier Transform

The Inverse Fourier transform is defined as:

$$\mathcal{F}^{-1}[\hat{f}](x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} e^{+iax} \hat{f}(a) \, \mathrm{d}a \tag{19}$$

It is the inverse of $\mathcal{F}$, meaning $\mathcal{F}^{-1}[\mathcal{F}[f]] = f$.

Note: The Fourier transform maps $\{f : \mathbb{R} \to \mathbb{C}\}$ to $\{\hat{f} : \mathbb{R} \to \mathbb{C}\}$. In general, even if f is real-valued, $\hat{f}$ might be $\mathbb{C}$-valued. (But, if f is real-valued and *even*, $\hat{f}$ is also real-valued and even).

The transform bridges two domains:

Physical Domain (time-domain or space-domain) $f \longleftrightarrow$ **Frequency Domain** $\hat{f}$

8.1 Properties of the Fourier Transform

Properties

1. **Linearity:** $\mathcal{F}[af+bg] = a\mathcal{F}[f] + b\mathcal{F}[g]$ for $a, b \in \mathbb{C}$.

2. **Time Shift:** Let $g(x) = f(x - x_0)$. Then:

$$\begin{aligned}\hat{g}(a) &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} e^{-iax} g(x)\,\mathrm{d}x = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} e^{-iax} f(x - x_0)\,\mathrm{d}x \\ &\overset{y=x-x_0}{=} \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} e^{-ia(y+x_0)} f(y)\,\mathrm{d}y \\ &= \frac{1}{\sqrt{2\pi}} e^{-iax_0} \int_{-\infty}^{+\infty} e^{-iay} f(y)\,\mathrm{d}y = e^{-iax_0}\hat{f}(a)\end{aligned}$$

3. **Frequency Shift:** Let $g(x) = e^{ia_0x} f(x)$. Then:

$$\begin{aligned}\hat{g}(a) &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} e^{-iax} g(x)\,\mathrm{d}x = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} e^{-iax} \cdot e^{ia_0x} f(x)\,\mathrm{d}x \\ &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} e^{-i(a-a_0)x} f(x)\,\mathrm{d}x = \hat{f}(a - a_0)\end{aligned}$$

Note: Property 3 is essentially like Property 2 but for the inverse Fourier transform. This is to be expected, since the inverse Fourier transform is defined with a similar expression as the Fourier transform (with e^{+iax} in place of e^{-iax}).

4. **Re-scaling (or Dilation):** Let $g(x) = f(\lambda x)$ for some $\lambda \neq 0$. We want to substitute $y = \lambda x$.

   - If $\lambda > 0$, then $\mathrm{d}y = \lambda\,\mathrm{d}x \implies \mathrm{d}x = \mathrm{d}y/\lambda$.
   - If $\lambda < 0$, we get $\int_{+\infty}^{-\infty} \dots \frac{\mathrm{d}y}{\lambda} = \int_{-\infty}^{+\infty} \dots \frac{\mathrm{d}y}{|\lambda|}$.

In both cases:

$$\begin{aligned}\hat{g}(a) &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} e^{-ia\frac{y}{\lambda}} \cdot f(y) \frac{\mathrm{d}y}{|\lambda|} \\ &= \frac{1}{|\lambda|} \cdot \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} e^{-i\frac{a}{\lambda}y} f(y)\,\mathrm{d}y = \frac{1}{|\lambda|}\hat{f}\left(\frac{a}{\lambda}\right)\end{aligned}$$

Remark: When f is very concentrated around a point, $\mathcal{F}[f]$ is very spread out. This is a manifestation of the **uncertainty principle**.

5. **Derivatives and Fourier Transform:** Let $g(x) = f'(x)$. Then:

$$\hat{g}(a) = ia\hat{f}(a)$$

More generally:

$$\mathcal{F}[f^{(n)}](a) = (ia)^n \mathcal{F}[f](a)$$

So the Fourier transform maps differential equations with constant coefficients to algebraic equations on the frequency domain.

8.2 Convolution

Definition: Convolution

Given $f, g : \mathbb{R} \to \mathbb{C}$, the convolution $f * g$ is defined by:

$$(f * g)(x) = \int_{-\infty}^{+\infty} f(y) g(x - y) \, \mathrm{d}y \tag{20}$$

8.3 Important Properties of Convolution

i) **Commutativity:** Using the change of variables $\bar{y} = x - y$, one can see that:

$$\int_{-\infty}^{+\infty} f(y) g(x - y) \, \mathrm{d}y = \int_{-\infty}^{+\infty} f(x - \bar{y}) g(\bar{y}) \, \mathrm{d}\bar{y}$$

So $f * g = g * f$.

ii) **Associativity:** $f * (g * h) = (f * g) * h$.

iii) **Distributivity:** $f * (g + h) = f * g + f * h$.

iv) **Derivative:** $(f * g)' = f' * g = f * g'$. So $f * g$ is at least as smooth as the smoother among f and g.

One way to interpret $f * g$ is as the **"local average"** of f with weights g.

Important Example:

If $g(x) = \begin{cases} \frac{1}{2\varepsilon}, & x \in [-\varepsilon, \varepsilon] \\ 0, & \text{else} \end{cases}$

Then:

$$(f * g)(x) = \int_{-\infty}^{+\infty} f(y) g(x - y) \, \mathrm{d}y = \frac{1}{2\varepsilon} \int_{x-\varepsilon}^{x+\varepsilon} f(y) \, \mathrm{d}y$$

So it is really the local average of f. Note that, in this case, as $\varepsilon \to 0$, $(f * g)(x) \to f(x)$.

The last point is true more generally: If $g_\lambda(x)$ is a family of functions with $\int_{-\infty}^{+\infty} g_\lambda(x) \, \mathrm{d}x = 1$ and $g_\lambda(x) = 0$ outside $[-\lambda, \lambda]$, then $f * g_\lambda(x) \to f(x)$ as $\lambda \to 0$. (In this case, as $\lambda \to 0$, $g_\lambda(x)$ looks like a Dirac δ-function).

8.4 Fourier Transform and Convolutions

Theorem: Convolution Theorem

Let $f, g : \mathbb{R} \to \mathbb{C}$. Then:

$$\mathcal{F}[f * g](a) = \sqrt{2\pi} \cdot \mathcal{F}[f](a) \cdot \mathcal{F}[g](a)$$

Conclusion: $\mathcal{F}$ transforms convolutions into products (and the opposite as well, since $\mathcal{F}^{-1}$ has similar properties). In applications, we can often express $\mathcal{F}[f]$ as the product of known functions, in which case we recover f as a convolution.

8.5 The Gaussian Function and Plancherel's Theorem

Theorem: Gaussian Transform

For the Gaussian function $f(x) = e^{-x^2/2}$, we have:

$$\hat{f}(a) = e^{-a^2/2}$$

(We will prove it in the exercises). So from Property 4 (Re-scaling), for $\lambda \neq 0$: If $f_\lambda(x) = e^{-\lambda^2 \frac{x^2}{2}} \implies \hat{f}_\lambda(a) = \frac{1}{|\lambda|} e^{-\frac{a^2}{2\lambda^2}}$.

Theorem: Plancherel's Theorem

$$\int_{-\infty}^{+\infty} |f(x)|^2 \, \mathrm{d}x = \int_{-\infty}^{+\infty} |\hat{f}(a)|^2 \, \mathrm{d}a$$

(The signal and its Fourier transform have the same energy).

8.6 Complex Analysis and the Fourier Transform

How does complex analysis enter the study of the Fourier transform? Let's see an example.

Let $f(x) = \frac{1}{1+x^2}$. Then:

$$\hat{f}(a) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} e^{-iax} \frac{1}{1+x^2} \, \mathrm{d}x$$

We can evaluate this integral as an application of the residue theorem. Let $a \leq 0$ (similar analysis if $a > 0$). Let:

$$h(z) = \frac{e^{-iaz}}{1+z^2} = \frac{e^{-iaz}}{(z-i)(z+i)}$$

We integrate over a semi-circle contour in the upper half-plane bounded by $[-R, R]$. Since $a \leq 0$, on $\{z : |z| = R \cap \mathrm{Im}(z) > 0\}$, $h(z)$ decays to 0 as $R \to \infty$.

From the examples we have already calculated in class (for $P(x)/Q(x) \cdot e^{iax}$):

$$\begin{aligned}\int_{-\infty}^{+\infty} h(x) \, \mathrm{d}x &= 2\pi i \cdot \mathrm{Res}(h, i) \\ &= 2\pi i \cdot \lim_{z \to i} ((z-i)h(z)) \\ &= 2\pi i \frac{e^{-ia(i)}}{2i} = \pi e^{a}\end{aligned}$$

So, for $a \leq 0$, $\pi e^{a} = \pi e^{-|a|}$. Thus:

$$\mathcal{F}\left[\frac{1}{1+x^2}\right](a) = \frac{1}{\sqrt{2\pi}} \cdot \pi e^{-|a|} = \sqrt{\frac{\pi}{2}} e^{-|a|}$$

For $a \geq 0$: The same computation (but with the semi-circle now in the lower half-plane) gives the same formula. (Or use that, since f is real and even, the same must be true for $\hat{f}$).

8.7 The Laplace Transform

The Fourier transform is used for signals that are defined on the whole real line ($\mathbb{R}$). The Laplace transform is for signals defined on $t \geq 0$. Both transform differential equations into algebraic equations.

Let $f : [0, +\infty) \to \mathbb{C}$ be a piecewise continuous function.

Definition: Laplace Transform

The Laplace transform of f is:

$$\mathcal{L}[f](z) = \int_0^{+\infty} f(t) e^{-zt} \, \mathrm{d}t \quad (z : \text{complex}) \tag{21}$$

Sometimes, we use capital letters to denote the Laplace transform, for instance: $F(z) = \mathcal{L}[f](z)$, $G(z) = \mathcal{L}[g](z)$.

When is the integral well-defined (i.e., "converges")?

We need $\int_0^{+\infty} |f(t)| \cdot |e^{-zt}| \, \mathrm{d}t < +\infty$.

Suppose that, for some $\gamma_0 \in \mathbb{R}$, we have:

$$\int_0^{+\infty} |f(t)| e^{-\gamma_0 t} \, \mathrm{d}t < +\infty$$

(Naively, this condition says that, as $t \to \infty$, $|f(t)| \lesssim e^{\gamma_0 t}$). Then, for $\mathrm{Re}(z) > \gamma_0$:

$$\begin{aligned} |\mathcal{L}[f](z)| &\leq \int_0^{+\infty} |f(t)||e^{-zt}| \, \mathrm{d}t \\ &= \int_0^{+\infty} |f(t)| \cdot \left| e^{-\mathrm{Re}(z)t} \cdot e^{-i\,\mathrm{Im}(z)t} \right| \, \mathrm{d}t \\ &= \int_0^{+\infty} |f(t)| \cdot e^{-\mathrm{Re}(z)t} \, \mathrm{d}t < +\infty \end{aligned}$$

since $e^{-\mathrm{Re}(z)t} < e^{-\gamma_0 t}$ (in view of our assumption for γ_0). Therefore, if such a γ_0 exists, $\mathcal{L}[f](z)$ is defined at least over $U = \{z \in \mathbb{C} : \mathrm{Re}(z) > \gamma_0\}$.

Definition: Abscissa of Convergence

A γ_0 such that $\int_0^{+\infty} |f(t)| e^{-\gamma t} \, \mathrm{d}t < +\infty$ for all $\gamma > \gamma_0$ is called an **abscissa of convergence** for f. In such a case, $\mathcal{L}[f](z)$ is well-defined for $\mathrm{Re}(z) > \gamma_0$.

Example: If $f(t) = 0$ for $t > M$, then:

$$\int_0^{+\infty} |f(t)|e^{-\gamma_0 t}\,\mathrm{d}t = \int_0^M |f(t)|e^{-\gamma_0 t}\,\mathrm{d}t < \infty \quad \text{for all } \gamma_0 \in \mathbb{R}$$

So in that case, $\mathcal{L}[f](z)$ is defined for all $z \in \mathbb{C}$.

Remark: The "faster" the rate of growth for $f(t)$ (or the "slower" the rate of decay), the smaller the domain where $\mathcal{L}[f](z)$ is well-defined.

If γ_0 is an abscissa of convergence and $U = \{z : \mathrm{Re}(z) > \gamma_0\}$, then $\mathcal{L}[f]$ is holomorphic on U. Let $F(z) = \mathcal{L}[f](z)$. Then:

$$\begin{aligned}\frac{\mathrm{d}}{\mathrm{d}z}F(z) &= \frac{\mathrm{d}}{\mathrm{d}z}\int_0^{+\infty} f(t)e^{-zt}\,\mathrm{d}t \\ &= \int_0^{+\infty} f(t)(-t)e^{-zt}\,\mathrm{d}t \\ &= -\mathcal{L}[t \cdot f(t)](z)\end{aligned}$$

Theorem: Derivative of Laplace Transform

If $F(z) = \mathcal{L}[f](z)$, then:

$$F'(z) = -\mathcal{L}[t \cdot f(t)](z) \tag{22}$$

We will also use the following lemma (proven in the exercises):

Theorem: Lemma

If γ_0 is an abscissa of convergence for $\mathcal{L}[f(t)]$, it is also an abscissa of convergence for $\mathcal{L}[t \cdot f(t)]$.

8.8 Examples of Laplace Transforms

Example 1:

Let $f : [0, +\infty) \to \mathbb{R}$ be $f(t) = 1$ (constant). Then, for any $\gamma > 0$:

$$\int_0^{+\infty} |f(t)|e^{-\gamma t}\,\mathrm{d}t = \int_0^{+\infty} e^{-\gamma t}\,\mathrm{d}t = \left[\frac{e^{-\gamma t}}{-\gamma}\right]_{t=0}^{t=+\infty} = \frac{1}{\gamma} < \infty$$

So any $\gamma_0 \geq 0$ is an abscissa of convergence. $\mathcal{L}[f](z)$ is defined (and holomorphic) for $\mathrm{Re}(z) > 0$. And:

$$\begin{aligned}\mathcal{L}[1](z) &= \int_0^{+\infty} 1 \cdot e^{-zt}\,\mathrm{d}t = \left[\frac{e^{-zt}}{-z}\right]_{t=0}^{t=+\infty} \\ &= -\frac{1}{z}\left(\lim_{t\to+\infty}(e^{-zt}) - e^0\right)\end{aligned}$$

But $\lim_{t\to+\infty}(e^{-zt}) = 0$ for $\mathrm{Re}(z) > 0$ since $|e^{-zt}| = e^{-\mathrm{Re}(z)t} \cdot |e^{-i\,\mathrm{Im}(z)t}| = e^{-\mathrm{Re}(z)t} \to 0$ as $t \to +\infty$. So:

$$\mathcal{L}[1](z) = \frac{1}{z}$$

Example 2:

Let $f : [0, +\infty) \to \mathbb{R}$ be $f(t) = t$. Then $\frac{\mathrm{d}}{\mathrm{d}z}\mathcal{L}[1](z) = -\mathcal{L}[t](z)$, so:

$$\mathcal{L}[t](z) = -\frac{\mathrm{d}}{\mathrm{d}z}\left(\frac{1}{z}\right) = \frac{1}{z^2}$$

(Also holomorphic for $\mathrm{Re}(z) > 0$, since as we saw in the lemma, if γ_0 is an abscissa of convergence for $\mathcal{L}[f]$, the same is true for $\mathcal{L}[t \cdot f(t)]$, and any $\gamma_0 > 0$ is an abscissa of convergence for $\mathcal{L}[1]$).

Repeating this construction yields the following general theorem:

Theorem: Laplace Transform of Monomials

For any $n \in \mathbb{N}$,

$$\mathcal{L}[t^n](z) = \frac{n!}{z^{n+1}}$$

defined for $\mathrm{Re}(z) > 0$.

9 Week 9 – 20.04.26 - Laplace Transform and Convolution

Recall from last time:

Laplace transform: If $f : [0, +\infty) \longrightarrow \mathbb{C}$ is piecewise continuous, then

$$\mathcal{L}[f](z) = \int_0^{+\infty} f(t) \cdot e^{-tz} dt$$

whenever the integral converges.

Note: Morally speaking

The integral converges if as $t \to +\infty$, $|f(t)| \leq Ce^{\text{Re}(z)t}$ at a rate which is "integrable".

9.1 Abscissa of Convergence

Definition: Abscissa of convergence

$\gamma_0 \in \mathbb{R}$ is called an **abscissa of convergence** for $\mathcal{L}[f]$ if, for any $\gamma > \gamma_0$, we have

$$\int_0^{+\infty} |f(t)| \cdot e^{-\gamma t} dt < +\infty.$$

So if γ_0 is an abscissa of convergence for $\mathcal{L}[f]$, then $\mathcal{L}[f](z)$ is holomorphic in the half-plane

$$U = \{z \in \mathbb{C} : \text{Re}(z) > \gamma_0\}.$$

- **Sufficient condition:** If $\int_0^{+\infty} |f(t)|e^{-\gamma_0 t} dt < +\infty$, then γ_0 is an abscissa of convergence (since the same integral for $\gamma > \gamma_0$ then necessarily is bounded).
- **Lemma from the exercises last week:** If γ_0 is an abscissa of convergence for $\mathcal{L}[f]$, then it is also an abscissa of convergence for $\mathcal{L}[t^n f(t)]$, $n \in \mathbb{N}$.
- **We also proved last week:**
$$\frac{d}{dz}\mathcal{L}[f](z) = -\mathcal{L}[t \cdot f(t)](z) \tag{*}$$
- We also showed that, if $f(t) = 1$ (constant), then 0 is an abscissa of convergence for $\mathcal{L}[1]$ (since, for any $\gamma > 0$, $\int_0^{+\infty} 1 \cdot e^{-\gamma t} dt < +\infty$) and
$$\mathcal{L}[1](z) = \frac{1}{z}$$
(Since 0 is an abscissa of convergence, the above relation holds for $\text{Re}(z) > 0$).
- Using (*) and differentiating the above relation, we have for any z with $\text{Re}(z) > 0$ and any $n \in \mathbb{N}$:

$$\mathcal{L}[t^n](z) = \left(-\frac{d}{dz}\right)^n \left(\frac{1}{z}\right) = \frac{n!}{z^{n+1}}$$

9.2 More properties of the Laplace transform:

Theorem: Linearity

Let $f, g : [0, +\infty) \longrightarrow \mathbb{C}$ be such that $\mathcal{L}[f]$ and $\mathcal{L}[g]$ are defined for Re$(z) > \gamma_0$. Then, for any constants $a, b \in \mathbb{C}$,

$$\mathcal{L}[af + bg](z) = a\mathcal{L}[f](z) + b\mathcal{L}[g](z)$$

is defined for Re$(z) > \gamma_0$.

Theorem: Derivative Property

Let $f : [0, +\infty) \longrightarrow \mathbb{C}$ be of class C^1 (that is to say, f' is continuous). Assume that γ_0 is an abscissa of convergence for $\mathcal{L}[f]$ and $\mathcal{L}[f']$. Then, for Re$(z) > \gamma_0$:

$$\mathcal{L}[f'](z) = z \cdot \mathcal{L}[f](z) - f(0)$$

Remark

Similarly as in the case of the Fourier transform, the Laplace transform transforms derivatives into multiplication with z (so differential equations are transformed to algebraic equations). A crucial difference is that the Laplace transform of f' also picks up the "initial value" $f(0)$.

Proof of the derivative theorem:

$$\mathcal{L}[f'](z) = \int_0^{+\infty} f'(t) \cdot e^{-tz} dt$$

We would like to integrate by parts. We will do this carefully, in order to deal with the boundary term at $t = +\infty$ appropriately.
Since Re$(z) > \gamma_0$ and γ_0 is an abscissa of convergence for $\mathcal{L}[f']$, $\int_0^{+\infty} f'(t)e^{-tz}dt$ converges absolutely. So, for any sequence $T_n \xrightarrow{n\to+\infty} +\infty$ (which we will define precisely later), we have:

$$\begin{aligned}\mathcal{L}[f'](z) &= \int_0^{+\infty} f'(t)e^{-tz}dt \\ &= \lim_{n\to+\infty} \int_0^{T_n} f'(t)e^{-tz}dt\end{aligned}$$

Integration by parts:

$$\begin{aligned}&= \lim_{n\to+\infty} \left(\left[f(t) \cdot e^{-tz}\right]_{t=0}^{t=T_n} - \int_0^{T_n} f(t) \cdot (-z)e^{-tz}dt \right) \\ &= \lim_{n\to+\infty} \left(f(T_n)e^{-T_n z} - f(0) \cdot e^0\right) + z \cdot \int_0^{+\infty} f(t)e^{-tz}dt \\ &= \lim_{n\to+\infty} f(T_n) \cdot e^{-T_n z} - f(0) + z \cdot \mathcal{L}[f](z)\end{aligned}$$

We will now show that we can choose the sequence T_n so that $\lim_{n\to+\infty} f(T_n) \cdot e^{-T_n z} = 0$ (this will give the formula that we want).
Recall that γ_0 is an abscissa of convergence for $\mathcal{L}[f]$ and Re$(z) > \gamma_0$, so that

$$\int_0^{+\infty} |f(t)| \cdot |e^{-z\cdot t}|dt = \int_0^{+\infty} |f(t)| \cdot e^{-\text{Re}(z)t}dt < +\infty.$$

Therefore, there must exist a sequence $T_n \to +\infty$ such that

$$|f(T_n)| \cdot |e^{-z\cdot T_n}| \xrightarrow{n\to+\infty} 0$$

(otherwise, the function $|f(t)| \cdot |e^{-zt}|$ must stay bounded away from 0 as $t \to +\infty$, which would contradict the bound $\int_0^{+\infty} |f(t)| \cdot |e^{-zt}|dt < +\infty$).
For this sequence T_n: $\lim_{n\to+\infty} f(T_n) \cdot e^{-zT_n} = 0$.

Verification: Limit Inferior and the Existence of Sequence T_n

The text states: *"otherwise, the function must stay bounded away from 0 as $t \to +\infty$, which would contradict the bound."*
From a strict analytical standpoint, a function $h(t) \geq 0$ being integrable on $[0, \infty)$ **does not** universally imply $\lim_{t\to\infty} h(t) = 0$. It is entirely possible to construct an integrable function with arbitrarily high "spikes" that do not converge to zero.
However, the integrability $\int_0^\infty h(t)dt < \infty$ **does** strictly guarantee that the *limit inferior* goes to zero:

$$\liminf_{t\to\infty} h(t) = 0$$

If $\liminf_{t\to\infty} h(t) = c > 0$, then there exists some T such that for all $t > T$, $h(t) > c/2$. This would imply $\int_T^\infty h(t)dt > \int_T^\infty (c/2)dt = \infty$, a contradiction. Because the limit inferior is 0, by definition, there must exist *some* monotonically increasing sequence $(T_n)_{n\in\mathbb{N}}$ such that $T_n \to \infty$ and $h(T_n) \to 0$. This rigorously validates the integration by parts boundary step without falsely assuming the global limit converges to zero.

More generally, iterating the above formula: if $f \in C^n([0, +\infty), \mathbb{C})$ (this notation means that $f : [0, +\infty) \to \mathbb{C}$ is n-times differentiable and $f, f', \ldots, f^{(n)}$ are continuous), we have:

$$\begin{aligned}
\mathcal{L}[f'](z) &= z \cdot \mathcal{L}[f](z) - f(0) \\
\mathcal{L}[f''](z) &= z \cdot \mathcal{L}[f'](z) - f'(0) \\
&= z^2 \cdot \mathcal{L}[f](z) - z \cdot f(0) - f'(0) \\
&\vdots \\
\mathcal{L}[f^{(n)}](z) &= z^n \cdot \mathcal{L}[f](z) - z^{n-1} \cdot f(0) - \cdots - z \cdot f^{(n-2)}(0) - f^{(n-1)}(0) \qquad \text{(**)} \\
&= z^n \cdot \mathcal{L}[f](z) - \sum_{j=0}^{n-1} z^{n-1-j} f^{(j)}(0).
\end{aligned}$$

9.3 Examples

Example

Consider $f(t) = t^2$, $f'(t) = 2t$, $f''(t) = 2$. Their Laplace transforms are defined for Re$(z) > 0$ (we already calculated them, but we can rederive the formulas using (**)):

$$\begin{aligned} f(t) = t^2 &\implies \mathcal{L}[f](z) = \frac{2}{z^3} \\ f'(t) = 2t &\implies \mathcal{L}[f'](z) = \frac{2}{z^2} \\ f''(t) = 2 &\implies \mathcal{L}[f''](z) = \frac{2}{z} \end{aligned}$$

Verifying with the formula:

$$\mathcal{L}[f''](z) = z^2 \cdot \mathcal{L}[f](z) - z \cdot f(0) - f'(0) = z^2 \left(\frac{2}{z^3}\right) - z(0) - 0 = \frac{2}{z}$$

9.4 Laplace transform Properties

Theorem: Integral Property

Let $f : [0, +\infty) \to \mathbb{C}$ be piecewise continuous and $\gamma_0 \geq 0$ an abscissa of convergence for $\mathcal{L}[f]$.
If $g(t) = \int_0^t f(s)ds \implies \mathcal{L}[g](z) = \frac{1}{z} \cdot \mathcal{L}[f](z)$ for Re$(z) > \gamma_0$.

Remark

In general, the abscissa of convergence for $\mathcal{L}[g]$ cannot be negative (even if it is for f).

Sketch of proof:

If $g(t) = \int_0^t f(s)ds$, then $f(t) = g'(t)$ and $g(0) = 0$.
So:

$$\begin{aligned} \mathcal{L}[g'](z) &= z \cdot \mathcal{L}[g](z) - g(0) \\ &= z \cdot \mathcal{L}[g](z) \end{aligned}$$

Thus, $\mathcal{L}[g](z) = \frac{1}{z} \cdot \mathcal{L}[g'](z) = \frac{1}{z} \cdot \mathcal{L}[f](z)$.
(We will not show why $\mathcal{L}[g](z)$ is indeed defined for Re$(z) > \gamma_0$ when $\gamma_0 \geq 0$).

9.4.1 Time Scaling and Shifting

Theorem: Time Scaling and Shifting

Let $a > 0$ and $b \in \mathbb{R}$. Let $f : [0, +\infty) \to \mathbb{C}$ be piecewise continuous with abscissa of convergence $\gamma_0 \in \mathbb{R}$. Let

$$g(t) = e^{-bt} \cdot f(at)$$

Then

$$\mathcal{L}[g](z) = \frac{1}{a}\mathcal{L}[f]\left(\frac{z+b}{a}\right)$$

for any z with Re $\left(\frac{z+b}{a}\right) > \gamma_0 \iff$ Re$(z) > a\gamma_0 - b$ (and, in particular, $a\gamma_0 - b$ is an abscissa of convergence for $\mathcal{L}[g]$).

Sketch of the proof:

We will not show that $a\gamma_0 - b$ is an abscissa of convergence (Exercise!), but we will compute the formula for $\mathcal{L}[g]$:

$$\begin{aligned}\mathcal{L}[g](z) &= \int_0^{+\infty} e^{-bt} f(at) e^{-tz} dt \\ &= \int_0^{+\infty} f(at) \cdot e^{-(z+b)t} dt\end{aligned}$$

Substitute $s = at \implies dt = \frac{ds}{a}$:

$$\begin{aligned} &= \int_0^{+\infty} f(s) \cdot e^{-(z+b)\cdot\frac{s}{a}} \cdot \frac{ds}{a} \\ &= \frac{1}{a} \cdot \int_0^{+\infty} f(s) \cdot e^{-\left(\frac{z+b}{a}\right)s} ds \\ &= \frac{1}{a}\mathcal{L}[f]\left(\frac{z+b}{a}\right).\end{aligned}$$

Remark

Compare the above formula with the formula for the Fourier transform of the rescaling $f(t) \to f(\lambda \cdot t)$.

Example

$g(t) = e^{ct} = e^{ct} \cdot 1$. Here $a = 1$, $b = -c$. According to the theorem:

$$\mathcal{L}[g](z) = \mathcal{L}[1](z - c) = \frac{1}{z - c}$$

for Re$(z) > c$.

Remark

In this case, we can compute that $\gamma_0 = c$ is an abscissa of convergence for $\mathcal{L}[e^{ct}]$ without using the theorem:
For any $\gamma > c$, we have:

$$\begin{aligned}
\int_0^{+\infty} |e^{ct}| \cdot |e^{-\gamma t}| dt &= \int_0^{+\infty} e^{ct} \cdot e^{-\gamma t} dt \\
&= \int_0^{+\infty} e^{-(-c+\gamma)t} dt \\
&= \left[\frac{e^{-(\gamma - c)t}}{-(\gamma - c)} \right]_{t=0}^{t=+\infty} = \frac{1}{\gamma - c} < +\infty
\end{aligned}$$

Since $\gamma > c \iff \gamma - c > 0$.

Example

Find $f : [0, +\infty) \to \mathbb{C}$ such that

$$\mathcal{L}[f](z) = \frac{1}{(z-a)(z-b)}, \quad a, b \in \mathbb{R}, \quad (a \neq b).$$

Notice by partial fraction decomposition:

$$\begin{aligned}
\mathcal{L}[f](z) &= \frac{1}{(z-a)(z-b)} \\
&= \frac{1}{b-a} \left(\frac{1}{z-b} - \frac{1}{z-a} \right) \\
&= \frac{1}{b-a} \cdot \left(\mathcal{L}[e^{bt}](z) - \mathcal{L}[e^{at}](z) \right) \\
&= \frac{1}{b-a} \cdot \mathcal{L}[e^{bt} - e^{at}](z) \\
&= \mathcal{L}\left[\frac{e^{bt} - e^{at}}{b-a} \right](z)
\end{aligned}$$

Defined for Re$(z) >$ max$\{a, b\}$.

9.5 Convolutions

Let $f, g : [0, +\infty) \to \mathbb{C}$ be piecewise continuous functions. We extend f, g on all of $\mathbb{R}$ by assuming they are $= 0$ on $(-\infty, 0)$.

$$(f * g)(t) = \int_{-\infty}^{+\infty} f(t-s) g(s) ds$$

Since $g(s) = 0$ for $s < 0$, and $f(t-s) = 0$ for $t - s < 0 \implies s > t$, we have:

$$(f * g)(t) = \begin{cases} \int_0^t f(t-s) g(s) ds & \text{if } t > 0 \\ 0 & \text{if } t \leq 0 \end{cases}$$

So, $f * g$ can be restricted back on $[0, +\infty)$ (its values on $(-\infty, 0)$ are trivially zero). In this way, $f * g$ is well-defined as an operation for functions on $[0, +\infty)$.

Theorem: Convolution Theorem for Laplace

Let $f, g : [0, +\infty) \to \mathbb{C}$ be piecewise continuous functions with $\gamma_0 \in \mathbb{R}$ an abscissa of convergence for $\mathcal{L}[f]$ and $\mathcal{L}[g]$.
Then

$$\mathcal{L}[f * g] = \mathcal{L}[f] \cdot \mathcal{L}[g]$$

has abscissa of convergence γ_0.

Nuance regarding the Convolution Abscissa

While the theorem states $\mathcal{L}[f * g]$ has an abscissa of convergence γ_0, strict analytical treatment dictates that the abscissa of convergence for the convolution is bounded above by the maximum of the individual abscissas. That is, if γ_f and γ_g are the abscissas of absolute convergence for $\mathcal{L}[f]$ and $\mathcal{L}[g]$, then $\mathcal{L}[f * g]$ converges absolutely for Re$(z) >$ max(γ_f, γ_g).
In our case, since $\gamma_f \leq \gamma_0$ and $\gamma_g \leq \gamma_0$, the convolution converges for Re$(z) > \gamma_0$. However, due to possible cancellations in the convolution integral, the true strict abscissa of convergence for $f * g$ could actually be *strictly less* than max(γ_f, γ_g). The formulation in the notes operates safely on the assertion that γ_0 remains a *valid* upper bound for the abscissa for the operation.

10 Week 10 - 27.04.2026 - The Laplace Transform

Recall

Let us recall from before the break:
Laplace transform: If $f : [0, +\infty) \to \mathbb{C}$ is piecewise continuous,

$$\mathcal{L}[f](z) = \int_0^{+\infty} f(t)e^{-zt}\mathrm{d}t$$

Domain of definition: If $\int_0^{+\infty} |f(t)|e^{-\gamma t}\mathrm{d}t < +\infty$ for all $\gamma > \gamma_0$, then γ_0 is called an **abscissa of convergence**, and $\mathcal{L}[f](z)$ is holomorphic on the half-plane:

$$\{z \in \mathbb{C} : \mathrm{Re}(z) > \gamma_0\}$$

Rigorous Definition of the Abscissa of Convergence

The formulation above clarifies a common typo in handwritten notes. The integral of the module $\int_0^{\infty} |f(t)|e^{-\gamma t}dt$ dictates *absolute* convergence. The infimum of such $\gamma \in \mathbb{R}$ where this integral is finite is the *abscissa of absolute convergence*, usually denoted σ_a. The standard abscissa of convergence σ_c (where the integral converges conditionally) is governed by γ_0, with the relation $\sigma_c \leq \sigma_a$. Inside the region $\mathrm{Re}(z) > \sigma_c$, the transform represents an analytic (holomorphic) function.

10.1 Explicit Examples and Useful Identities

Example

- $\mathcal{L}[e^{-at}](z) = \int_0^{+\infty} e^{-at}e^{-zt}\mathrm{d}t = \dfrac{1}{z+a}$ (Valid for $a \in \mathbb{C}$, provided $\mathrm{Re}(z) > -\mathrm{Re}(a)$).
- When $a = 0$: $\mathcal{L}[1](z) = \dfrac{1}{z}$.

Useful Identities

These allow us to compute the Laplace transform of several explicit examples:

- **Linearity:** $\mathcal{L}[af + bg] = a\mathcal{L}[f] + b\mathcal{L}[g]$ for $a, b \in \mathbb{C}$.
- **Derivative with respect to z:** $\frac{\mathrm{d}}{\mathrm{d}z}\mathcal{L}[f](z) = -\mathcal{L}[t \cdot f(t)](z)$
- **Transform of a Derivative:**

$$\mathcal{L}[f'](z) = z \cdot \mathcal{L}[f](z) - f(0)$$
$$\mathcal{L}[f^{(n)}](z) = z^n\mathcal{L}[f](z) - z^{n-1}f(0) - z^{n-2}f'(0) - \cdots - f^{(n-1)}(0)$$

From the above properties, we can quickly derive:

$$\mathcal{L}[t^n](z) = \frac{n!}{z^{n+1}}$$

Alternatively, applied to exponential functions:

$$\mathcal{L}[t \cdot e^{-at}](z) = -\frac{\mathrm{d}}{\mathrm{d}z}\mathcal{L}[e^{-at}](z) = -\frac{\mathrm{d}}{\mathrm{d}z}\left(\frac{1}{z+a}\right) = \frac{1}{(z+a)^2}$$

Transform of an Integral

$$\mathcal{L}\left[\int_0^t f(s)\mathrm{d}s\right](z) = \frac{1}{z}\mathcal{L}[f](z)$$

Proof outline: Let $g(t) = \int_0^t f(s)\mathrm{d}s$. Then $g'(t) = f(t)$ and $g(0) = 0$. By the derivative formula, $\mathcal{L}[f](z) = \mathcal{L}[g'](z) = z\mathcal{L}[g](z) - g(0) = z\mathcal{L}[g](z)$.

Asymptotic Behavior of Integrals

Even if $f(s) \to 0$ as $s \to +\infty$, in general, the integral $\int_0^t f(s)\mathrm{d}s \not\to 0$ as $t \to +\infty$. So the abscissa of convergence for $\mathcal{L}\left[\int_0^t f\right]$ is usually ≥ 0.

Frequency Shifts, Rescalings, and Convolutions

- **Shifts:** $\mathcal{L}[e^{-bt}f(t)](z) = \mathcal{L}[f](z+b)$
- **Rescaling:** $\mathcal{L}[f(at)](z) = \frac{1}{a}\mathcal{L}[f]\left(\frac{z}{a}\right)$
- **Convolutions:** If $f, g : [0, +\infty) \to \mathbb{C}$ are piecewise continuous, recall that:

$$(f * g)(t) = \int_{-\infty}^{+\infty} f(t-s)g(s)\mathrm{d}s = \int_0^t f(t-s)g(s)\mathrm{d}s$$

(since $f, g = 0$ for $t < 0$). Then:

$$\mathcal{L}[f * g] = \mathcal{L}[f] \cdot \mathcal{L}[g]$$

Convolution Theorem Rigor

The statement $\mathcal{L}[f * g] = \mathcal{L}[f]\mathcal{L}[g]$ relies heavily on Fubini's Theorem to exchange the order of integration in the double integral defined by the Laplace transform of the convolution. Absolute convergence of $\mathcal{L}[f]$ and $\mathcal{L}[g]$ is strictly required to satisfy Fubini's conditions.

Using the above formulas, we can invert the Laplace transform of many explicit functions. (We will later see a formula for the inverse of the Laplace transform in general, but it is more involved than the corresponding formula for the Fourier transform).

10.2 Applications to 2nd Order Differential Equations

Consider the following initial value problem: For $\omega \in \mathbb{R}^*$, $a > 0$:

$$\begin{cases} y''(t) + 2ay'(t) + \omega^2 y(t) = f(t) & \text{for } t > 0 \\ y(0) = y_0 \\ y'(0) = v_0 \end{cases} \tag{23}$$

Since this is a 2nd order equation, we need to specify 2 initial (or boundary) conditions to get a unique solution.

Physical Interpretation

The above equation governs a **damped harmonic oscillator**:

- $f(t)$: Forcing term
- $a > 0$: Damping parameter (friction / resistance)
- ω: Natural angular frequency of the undamped system

10.3 Uniqueness of Solutions

Theorem

Let $y_1, y_2 : [0, +\infty) \to \mathbb{R}$ be two solutions of (23). Let $w(t) = y_1(t) - y_2(t)$. Then $w(t) = 0$ for all $t \geq 0$.

Proof

Subtracting (23) for y_2 from the equation for y_1, we get the homogeneous problem for w:

$$\begin{cases} w''(t) + 2aw'(t) + \omega^2 w(t) = 0 \\ w(0) = 0 \\ w'(0) = 0 \end{cases}$$

Multiply the differential equation by w':

$$w'' \cdot w' + 2a(w')^2 + \omega^2 \cdot w \cdot w' = 0$$

Note the chain rules: $w'' \cdot w' = \frac{1}{2}\frac{\mathrm{d}}{\mathrm{d}t}((w')^2)$ and $w \cdot w' = \frac{1}{2}\frac{\mathrm{d}}{\mathrm{d}t}(w^2)$. Substituting these gives:

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(\underbrace{\frac{1}{2}(w')^2}_{\text{Kinetic energy}} + \underbrace{\frac{1}{2}\omega^2 w^2}_{\text{Potential energy}} \right) + 2a(w')^2 = 0$$

Let the total energy be $E(t) = \frac{1}{2}(w')^2 + \frac{1}{2}\omega^2 w^2$. Then $\frac{\mathrm{d}}{\mathrm{d}t}E(t) = -2a(w')^2 \leq 0$. Thus, $E(t)$ is non-increasing (the effect of damping).
Since $w(0) = 0$ and $w'(0) = 0$, we have $E(0) = 0$. Since $E(t)$ is a sum of squares, $E(t) \geq 0$ for all t. Because it is non-increasing and starts at 0, it must remain 0 at all times:

$$E(t) = 0 \quad \text{for all } t \geq 0$$

Since $E(t) \geq \frac{1}{2}\omega^2 w^2 \geq 0$, $E(t) = 0$ forces $w(t) = 0$ for all $t \geq 0$. This guarantees the unique solution.

10.4 Solving via the Laplace Transform

Apply the Laplace transform to $y''(t) + 2ay'(t) + \omega^2 y(t) = f(t)$:

$$\left[z^2 Y(z) - zy(0) - y'(0)\right] + 2a\left[zY(z) - y(0)\right] + \omega^2 Y(z) = F(z)$$

Substitute initial conditions $y(0) = y_0$, $y'(0) = v_0$ and rearrange by factoring out $Y(z)$:

$$(z^2 + 2az + \omega^2)Y(z) - zy_0 - v_0 - 2ay_0 = F(z)$$

$$(z^2 + 2az + \omega^2)Y(z) = F(z) + (z + a)y_0 + ay_0 + v_0$$

By "completing the square", the characteristic polynomial is $z^2 + 2az + \omega^2 = (z + a)^2 + \omega^2 - a^2$. Solving for $Y(z)$:

$$Y(z) = \frac{1}{(z+a)^2 + \omega^2 - a^2} \cdot F(z) + \frac{z+a}{(z+a)^2 + \omega^2 - a^2} \cdot y_0 + \frac{1}{(z+a)^2 + \omega^2 - a^2} \cdot (ay_0 + v_0)$$

We will consider three cases depending on the sign of $\omega^2 - a^2$:

10.4.1 Case I: $a^2 = \omega^2$ (Critically Damped)

In this case, $\omega^2 - a^2 = 0$.

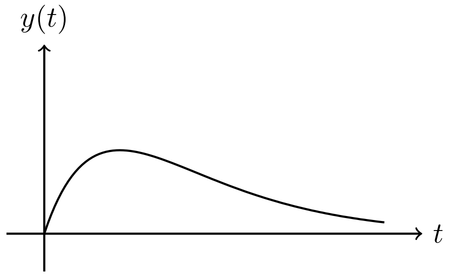
$$\frac{1}{(z+a)^2+\omega^2-a^2} = \frac{1}{(z+a)^2} \quad \text{and} \quad \frac{z+a}{(z+a)^2+\omega^2-a^2} = \frac{1}{z+a}$$

Recall: $\mathcal{L}[e^{-at}](z) = \frac{1}{z+a}$ and $\mathcal{L}[t \cdot e^{-at}](z) = \frac{1}{(z+a)^2}$. So:

$$Y(z) = \mathcal{L}[t \cdot e^{-at}](z) \cdot \mathcal{L}[f](z) + \mathcal{L}[e^{-at}](z) \cdot y_0 + \mathcal{L}[t \cdot e^{-at}](z) \cdot (ay_0 + v_0)$$

By taking the inverse Laplace transform (using the convolution theorem):

$$y(t) = \int_0^t (t-s) \cdot e^{-a(t-s)} f(s) \mathrm{d}s + e^{-at} y_0 + t \cdot e^{-at}(ay_0 + v_0)$$



Typical graph for $f = 0$ (Critically damped)

10.4.2 Case II: $\omega^2 > a^2$ (Under-Damped)

We start from the fundamental identities (derived via Euler's formula):

$$\mathcal{L}[\cos(\beta t)](z) = \frac{z}{z^2+\beta^2}, \qquad \mathcal{L}[\sin(\beta t)](z) = \frac{\beta}{z^2+\beta^2}$$

Let $\beta = \sqrt{\omega^2 - a^2} > 0$. Using the frequency shift property:

$$\frac{1}{(z+a)^2+\beta^2} = \frac{1}{\beta} \cdot \frac{\beta}{(z+a)^2+\beta^2} = \frac{1}{\beta} \cdot \mathcal{L}[e^{-at}\sin(\beta t)](z)$$

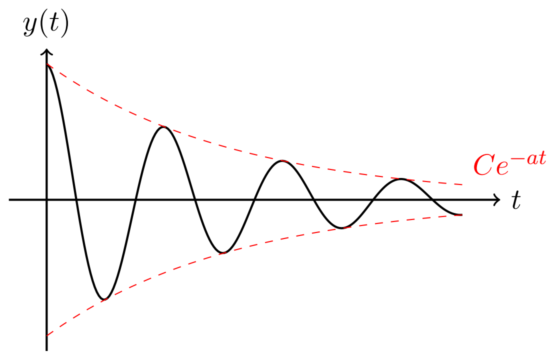
$$\frac{z+a}{(z+a)^2+\beta^2} = \mathcal{L}[e^{-at}\cos(\beta t)](z)$$

Therefore, $Y(z)$ becomes:

$$\begin{aligned} Y(z) = \mathcal{L}&\left[\frac{1}{\sqrt{\omega^2-a^2}} e^{-at}\sin(\sqrt{\omega^2-a^2}t)\right](z) \cdot F(z) \\ &+ \mathcal{L}\left[e^{-at}\cos(\sqrt{\omega^2-a^2}t)\right](z) \cdot y_0 \\ &+ \mathcal{L}\left[\frac{1}{\sqrt{\omega^2-a^2}} e^{-at}\sin(\sqrt{\omega^2-a^2}t)\right](z) \cdot (ay_0+v_0) \end{aligned}$$

Taking the inverse:

$$\begin{aligned} y(t) = &\int_0^t \frac{1}{\sqrt{\omega^2-a^2}} e^{-a(t-s)} \sin\left(\sqrt{\omega^2-a^2}(t-s)\right) f(s)\mathrm{d}s \\ &+ e^{-at}\cos\left(\sqrt{\omega^2-a^2}t\right) y_0 \\ &+ \frac{1}{\sqrt{\omega^2-a^2}} e^{-at}\sin\left(\sqrt{\omega^2-a^2}t\right)(ay_0+v_0) \end{aligned}$$



Typical graph for $f = 0$ (Under-damped)

10.4.3 Case III: $\omega^2 < a^2$ (Over-Damped)

We use the hyperbolic functions:

$$\mathcal{L}[\cosh(\beta t)](z) = \frac{z}{z^2 - \beta^2}, \qquad \mathcal{L}[\sinh(\beta t)](z) = \frac{\beta}{z^2 - \beta^2}$$

(These can be derived using $cosh(x) = \cos(ix)$ and $\sinh(x) = -i\sin(ix)$). Letting $\beta = \sqrt{a^2 - \omega^2} > 0$:

$$\frac{1}{(z+a)^2 - \beta^2} = \frac{1}{\beta}\mathcal{L}[\sinh(\beta t)e^{-at}](z)$$

$$\frac{z+a}{(z+a)^2 - \beta^2} = \mathcal{L}[\cosh(\beta t)e^{-at}](z)$$

Following the exact same structure as Case II:

$$\begin{aligned} y(t) = &\int_0^t \frac{1}{\sqrt{a^2-\omega^2}} e^{-a(t-s)} \sinh\left(\sqrt{a^2-\omega^2}(t-s)\right) f(s)\mathrm{d}s \\ &+ e^{-at}\cosh\left(\sqrt{a^2-\omega^2}t\right) y_0 \\ &+ \frac{1}{\sqrt{a^2-\omega^2}} e^{-at}\sinh\left(\sqrt{a^2-\omega^2}t\right)(ay_0 + v_0) \end{aligned}$$

The graph behavior for $f = 0$ is similar to the critically damped case (a potential single peak, followed by pure exponential decay without oscillation).

10.5 Physical Motivation and Resonance

From Hooke's law:

$$m \cdot y'' = f_{\text{ext}} - k \cdot y - d \cdot y' \implies y'' + \frac{d}{m}y' + \frac{k}{m}y = \frac{f_{\text{ext}}}{m}$$

Comparing coefficients gives $\omega^2 = \frac{k}{m}$ and $2a = \frac{d}{m}$.

Resonance: In the case when $a = 0$ (no damping), and with $y_0 = 0, v_0 = 0$:

$$y(t) = \int_0^t \frac{1}{|\omega|} \sin(|\omega|(t-s)) \cdot f(s)\mathrm{d}s$$

If the external forcing $f(s)$ is of the form $\sin(\gamma s)$:

- If $\gamma \neq \pm\omega$: $y(t)$ remains bounded as $t \to +\infty$ (”cancellations” occur inside the integral).
- If $\gamma = \pm\omega$: $|y(t)| \to +\infty$ as $t \to +\infty$. The forcing frequency exactly matches the eigen-frequency of the system, causing unbounded amplitude growth.

10.6 The Inverse Laplace Transform

Recall: If γ_0 is an abscissa of convergence for f (meaning, in practice, that asymptotically as $t \to \infty$, f behaves like $e^{\gamma_0 t}$ up to polynomial corrections), then $\mathcal{L}[f](z)$ is holomorphic (and thus has no singularities) on $\{z : \text{Re}(z) > \gamma_0\}$.

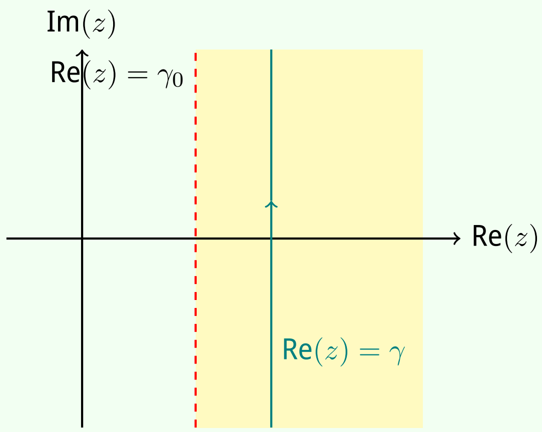
In some sense, the "opposite" is also true: If $F(z)$ is holomorphic in $\{z : \text{Re}(z) > \gamma_0\}$ and doesn't "grow" as $|z| \to \infty$, then γ_0 is an abscissa of convergence for some function f.

Mellin Inversion Formula

Suppose that $f : [0, +\infty) \to \mathbb{C}$ is a piecewise continuous function and, for some $\gamma \in \mathbb{R}$, $F(z) = \mathcal{L}[f](z)$ is holomorphic on the region $\{z : \text{Re}(z) \geq \gamma\}$, and

$$\lim_{T \to +\infty} \int_{-T}^{T} F(\gamma + i\zeta) e^{(\gamma + i\zeta)t} \mathrm{d}\zeta \quad \text{exists for all } t \geq 0.$$

(Both conditions are automatically true if some $\gamma_0 < \gamma$ is an abscissa of convergence for f).



Then, $f(t)$ is uniquely determined by the Bromwich integral:

$$f(t) = \frac{1}{2\pi} \int_{-\infty}^{+\infty} F(\gamma + i\zeta) e^{(\gamma + i\zeta)t} \mathrm{d}\zeta = \frac{1}{2\pi i} \int_{\gamma - i\infty}^{\gamma + i\infty} F(z) e^{zt} \mathrm{d}z$$

Absolute vs. Principal Value Convergence

Note: The above integral might not converge absolutely (e.g., if $f(t) = 1$ for $t \geq 0$, then $F(z) = \frac{1}{z}$ and $\int \frac{1}{|\gamma + i\zeta|} d\zeta = \infty$). So the integral is evaluated as a Cauchy Principal Value:

$$\frac{1}{2\pi} \lim_{T \to +\infty} \int_{-T}^{+T} F(\gamma + i\zeta) e^{(\gamma + i\zeta)t} \mathrm{d}\zeta$$

Proof

Extend $f : [0, +\infty) \to \mathbb{C}$ to a function on $(-\infty, +\infty)$ by setting $f(t) = 0$ for $t < 0$.
Let $g(t) = f(t)e^{-\gamma t}$. Take the Fourier transform of g:

$$\begin{aligned}
\hat{g}(\alpha) &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} g(t) e^{-i\alpha t} \mathrm{d}t \\
&= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} f(t) e^{-(\gamma + i\alpha)t} \mathrm{d}t \\
\text{(since } f(t) = 0 \text{ for } t < 0\text{)} \quad &= \frac{1}{\sqrt{2\pi}} \int_{0}^{+\infty} f(t) e^{-(\gamma + i\alpha)t} \mathrm{d}t \\
&= \frac{1}{\sqrt{2\pi}} F(\gamma + i\alpha) \quad \text{(Laplace transform evaluated at } z = \gamma + i\alpha\text{)}
\end{aligned}$$

By the inverse Fourier transform formula:

$$g(t) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} \hat{g}(\alpha) e^{i\alpha t} \mathrm{d}\alpha = \frac{1}{2\pi} \int_{-\infty}^{+\infty} F(\gamma + i\alpha) e^{i\alpha t} \mathrm{d}\alpha$$

Since $f(t) = g(t)e^{\gamma t}$, multiplying both sides by $e^{\gamma t}$ yields:

$$f(t) = \frac{1}{2\pi} \int_{-\infty}^{+\infty} F(\gamma + i\alpha) e^{(\gamma + i\alpha)t} \mathrm{d}\alpha$$

Lerch's Theorem

1. $f(t)$ is uniquely determined by $\mathcal{L}[f](z)$ through this formula (Lerch's theorem). Two functions having the same Laplace transform are identical almost everywhere.

2. The condition on the existence of $\lim_{T \to +\infty} \int_{-T}^{T} F(\gamma + i\zeta) e^{(\gamma + i\zeta)t} \mathrm{d}\zeta$ is automatically fulfilled if the absolute integrability condition $\int_{-\infty}^{+\infty} |F(\gamma + i\zeta)| \mathrm{d}\zeta < +\infty$ holds.

11 Week 11 - 04.05.2026 - The Inverse Laplace Transform and Application to Differential Equations

Recall from last time:

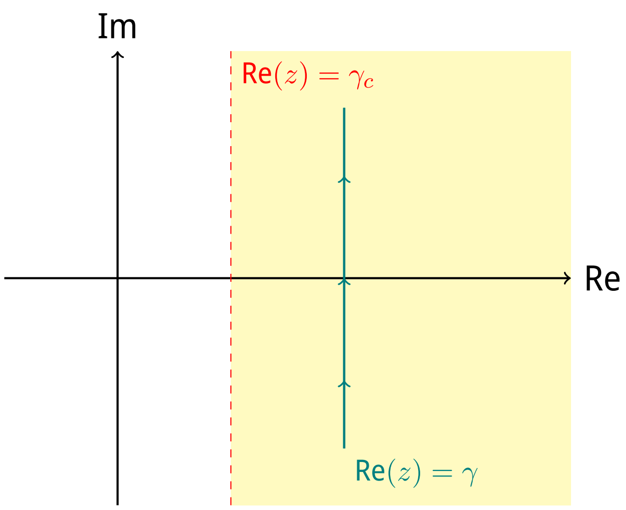
Lerch's Theorem for Inverse Laplace Transform

Suppose that $f : [0, +\infty) \to \mathbb{C}$ is a piecewise continuous function and for some $\gamma \in \mathbb{R}$, $F(z) = \mathcal{L}[f](z)$ is holomorphic on the region $\{z : \mathrm{Re}(z) \geq \gamma\}$ and the limit

$$\lim_{T\to+\infty} \int_{-T}^{T} F(\gamma + is) \cdot e^{(\gamma+is)t} \mathrm{d}s$$

exists for all $t \geq 0$. (Both conditions are automatically true if some $\gamma_c < \gamma$ is an abscissa of convergence for f). Then:

$$\begin{aligned} f(t) &= \frac{1}{2\pi} \int_{-\infty}^{+\infty} F(\gamma + is) \cdot e^{(\gamma+is)t} \mathrm{d}s \\ &= \frac{1}{2\pi i} \int_{\gamma-i\infty}^{\gamma+i\infty} F(z) \cdot e^{zt} \mathrm{d}z \end{aligned}$$



Note: The above integral might not converge absolutely (e.g., if $f(t) = 1$ for $t \geq 0$, $F(z) = \frac{1}{z}$). So we just mean the principal value:

$$\frac{1}{2\pi} \cdot \lim_{T\to+\infty} \int_{-T}^{+T} F(\gamma + is) e^{(\gamma+is)t} \mathrm{d}s.$$

Remarks

1. f is uniquely determined by $\mathcal{L}[f]$ by this formula (Lerch's theorem).
2. The condition on the existence of the limit $\lim_{T\to+\infty}\int_{-T}^{T} F(\gamma+is)\cdot e^{(\gamma+is)t}\mathrm{d}s$ is automatically true if:
$$\int_{-\infty}^{+\infty} |F(\gamma+is)|\mathrm{d}s < +\infty.$$

11.1 Example of Inverse Laplace Transform

Example

Let $F(z) = \frac{1}{(z+1)(z+2)^2}$. Find $f : [0,+\infty) \to \mathbb{R}$ such that $\mathcal{L}[f] = F$.
There are two ways to do it:
Method 1: Partial Fractions Note that

$$\frac{1}{(z+1)(z+2)^2} = \frac{1}{z+1} - \frac{1}{z+2} - \frac{1}{(z+2)^2}$$

Since we know that $\mathcal{L}[e^{-t}] = \frac{1}{z+1}$, $\mathcal{L}[e^{-2t}] = \frac{1}{z+2}$, and $\mathcal{L}[te^{-2t}] = \frac{1}{(z+2)^2}$, we find:

$$f(t) = e^{-t} - e^{-2t} - te^{-2t}$$

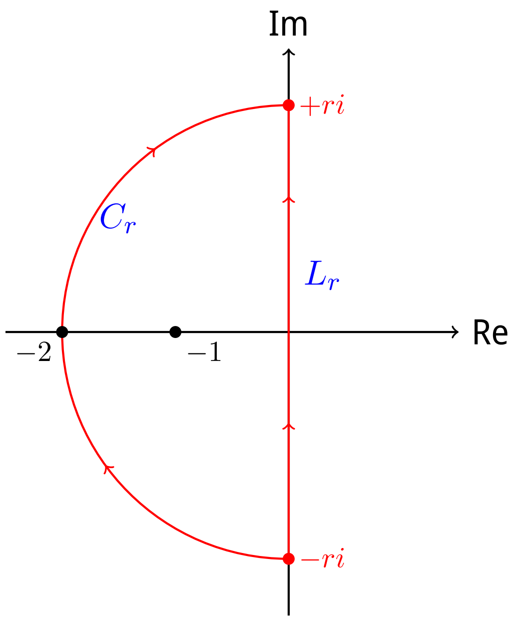
(By uniqueness of Laplace transforms, that's enough. Usually, this is the fastest way.)
Method 2: Contour Integration (Inverse Laplace transform combined with the Residue Theorem) Using the formula for the inverse Laplace transform:

$$f(t) = \frac{1}{2\pi}\int_{-\infty}^{+\infty} F(\gamma+is)e^{(\gamma+is)t}\mathrm{d}s$$

Let $h(z) = F(z)\cdot e^{zt} = \frac{e^{zt}}{(z+1)(z+2)^2}$. Poles for h: $z = -1, -2$. So we can choose $\gamma = 0$: F is holomorphic on $\{z : \mathrm{Re}(z) > -1\}$ which contains $\{\mathrm{Re}(z) \geq 0\}$. Also, $\int_{-\infty}^{+\infty} |F(\gamma+is)|\mathrm{d}s < +\infty$.

Inverse Laplace transform:

$$f(t) = \frac{1}{2\pi}\int_{-\infty}^{+\infty} h(\gamma+is)\mathrm{d}s = \frac{1}{2\pi i}\int_{-i\infty}^{+i\infty} h(z)\mathrm{d}z$$



Define the curves: $L_r : [-r, r] \to \mathbb{C}, s \mapsto is$ $C_r : [\frac{\pi}{2}, \frac{3\pi}{2}] \to \mathbb{C}, \theta \mapsto r \cdot e^{i\theta}$ $\Gamma = L_r \cup C_r$ (closed loop, counter-clockwise oriented).

When $r > 2$, Γ encloses all the poles of h (namely $z = -1, -2$). Applying the residue theorem:

$$\int_\Gamma h(z)\mathrm{d}z = 2\pi i \left(\mathrm{Res}_{z=-1}(h) + \mathrm{Res}_{z=-2}(h)\right).$$

Since $h(z) = \frac{e^{zt}}{(z+1)(z+2)^2}$:

- **At $z = -1$ (Simple pole):**

$$\mathrm{Res}_{z=-1}(h) = \lim_{z\to-1}((z+1)h(z)) = \lim_{z\to-1}\left(\frac{e^{zt}}{(z+2)^2}\right) = e^{-t}$$

- **At $z = -2$ (Pole of order 2):**

$$\begin{aligned}\mathrm{Res}_{z=-2}(h) &= \lim_{z\to-2}\left(\frac{\mathrm{d}}{\mathrm{d}z}\left((z+2)^2h(z)\right)\right)\\ &= \lim_{z\to-2}\left(\frac{\mathrm{d}}{\mathrm{d}z}\left(\frac{e^{zt}}{z+1}\right)\right)\end{aligned}$$

Correction of the derivative limit at $z = -2$

Let's rigorously evaluate the derivative using the quotient rule:

$$\frac{\mathrm{d}}{\mathrm{d}z}\left(\frac{e^{zt}}{z+1}\right) = \frac{te^{zt}(z+1) - e^{zt}\cdot 1}{(z+1)^2}$$

Taking the limit as $z \to -2$:

$$\lim_{z\to -2} \frac{te^{zt}(z+1) - e^{zt}}{(z+1)^2} = \frac{te^{-2t}(-1) - e^{-2t}}{(-1)^2} = -te^{-2t} - e^{-2t} = -e^{-2t}(t+1)$$

The original notes contained a sign error yielding $+e^{-2t}(t+1)$. The correct residue must be negative.

So:

$$\int_\Gamma h(z)\mathrm{d}z = 2\pi i(e^{-t} - e^{-2t} - te^{-2t})$$

Now:

$$\frac{1}{2\pi i}\int_\Gamma h(z)\mathrm{d}z = \underbrace{\frac{1}{2\pi i}\int_{L_r} h(z)\mathrm{d}z}_{\to f(t)} + \frac{1}{2\pi i}\int_{C_r} h(z)\mathrm{d}z$$

Indeed: $f(t) = e^{-t} - e^{-2t} - te^{-2t}$. It suffices to show that $\left|\int_{C_r} h(z)\mathrm{d}z\right| \xrightarrow{r\to+\infty} 0$.

$$\left|\int_{C_r} h(z)\mathrm{d}z\right| = \left|\int_{\frac{\pi}{2}}^{\frac{3\pi}{2}} h(re^{i\theta})\cdot ire^{i\theta}\mathrm{d}\theta\right| \le \int_{\frac{\pi}{2}}^{\frac{3\pi}{2}} |h(re^{i\theta})|\cdot r\mathrm{d}\theta$$

$$= \int_{\frac{\pi}{2}}^{\frac{3\pi}{2}} \frac{|e^{re^{i\theta}t}|}{|re^{i\theta}+1|\cdot|re^{i\theta}+2|^2}\cdot r\mathrm{d}\theta$$

Bounding the terms:

- $|e^{re^{i\theta}t}| = |e^{r(\cos\theta + i\sin\theta)t}| = |e^{r\cos\theta\cdot t}\cdot e^{ir\sin\theta\cdot t}| = e^{r\cos\theta\cdot t} \le 1$. (Since $r > 0$, $\cos\theta \le 0$ for $\theta \in [\frac{\pi}{2}, \frac{3\pi}{2}]$, and $t \ge 0$).
- $|re^{i\theta}| - 1 \le |re^{i\theta}+1| \le |re^{i\theta}| + 1 \implies |re^{i\theta}+1| \ge r-1$.
- Similarly, $|re^{i\theta}+2| \ge r-2$.

So:

$$\cdots \le \int_{\frac{\pi}{2}}^{\frac{3\pi}{2}} \frac{1}{(r-1)(r-2)^2}\cdot r\mathrm{d}\theta = \pi\frac{r}{(r-1)(r-2)^2} \xrightarrow{r\to+\infty} 0$$

Remark on the choice of the closed loop Γ: We chose the half-circle to be on the left (for Re$(z) < \gamma$) because in that region, $|e^{zt}| \le e^{\gamma t}$, ensuring the exponential term remains bounded and decays as required.

11.2 Applications to Partial Differential Equations

We will apply the theory of Fourier / Laplace transforms to solve initial-boundary value problems for linear partial differential equations with constant coefficients.

Example: Heat Equation

$$\begin{cases} \frac{\partial u}{\partial t}(x,t) - \frac{\partial^2 u}{\partial x^2}(x,t) = f(x,t) & \text{for } t > 0 \text{ and } 0 < x < L \\ u(x,0) = u_0(x) & \text{(initial condition)} \\ u(0,t) = 0, \quad u(L,t) = 0 & \text{(boundary conditions)} \end{cases} \tag{24}$$

Describes the evolution of the temperature of a 1-dimensional rod (interval $0 < x < L$) with heat source f and fixed temperature at the endpoints $x = 0, L$.

General procedure that we will follow for such problems:

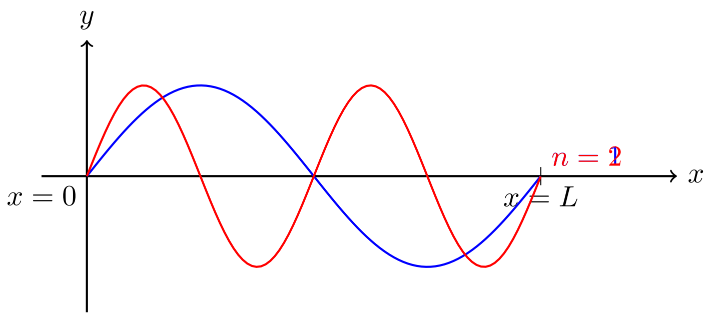
- **Fourier series decomposition** in directions corresponding to bounded intervals ($x \in [0, L]$).
- **Fourier transform** in doubly infinite directions ($x \in (-\infty, +\infty)$).
- **Laplace transform** for semi-infinite directions ($t \in [0, +\infty)$).

11.2.1 Recap on Fourier Series

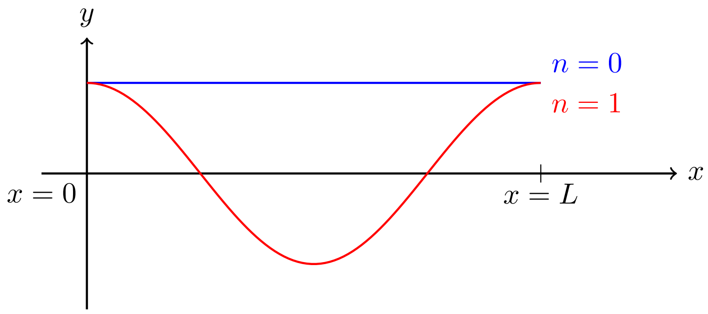
We want to decompose functions $f : [0, L] \to \mathbb{R}$ into an infinite sum of simple oscillating functions.

Basis functions for arbitrary periodic functions:

- $\sin\left(\frac{2\pi n}{L} x\right), \quad n \geq 1$
- $\cos\left(\frac{2\pi n}{L} x\right), \quad n \geq 0$



Sine modes



Cosine modes

Orthogonality relations: If we denote $\langle f, g\rangle := \int_0^L f(x)g(x)\mathrm{d}x$ (this is indeed an "inner-product" on the space of functions on $[0, L]$):

$$\left\langle \sin\left(\frac{2\pi n}{L}x\right), \sin\left(\frac{2\pi m}{L}x\right)\right\rangle = \begin{cases} \frac{L}{2}, & n = m \neq 0 \\ 0, & n \neq m \end{cases}$$

Same for cosines: $\langle \cos(\frac{2\pi n}{L}x), \cos(\frac{2\pi m}{L}x)\rangle$. Also, $\langle \sin(\frac{2\pi n}{L}x), \cos(\frac{2\pi m}{L}x)\rangle = 0$ for all n, m. We use these functions as an orthogonal basis to decompose other functions.

Trigonometric (Fourier) series: Given a function $f : [0, L] \to \mathbb{R}$ which is piecewise continuous, can we decompose:

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty}\left(a_n \cdot \cos\left(\frac{2\pi n}{L}x\right) + b_n \cdot \sin\left(\frac{2\pi n}{L}x\right)\right)?$$

(The same considerations hold in the case when $f : \mathbb{R} \to \mathbb{R}$ is periodic with period L; I can think of it as a function on $[0, L]$).

The answer is yes! Note that if such a decomposition exists, I can compute (formally) the coefficients because of the orthogonality conditions:

$$\int_0^L f(x)\mathrm{d}x = \frac{L}{2}a_0$$
$$\int_0^L f(x)\cos\left(\frac{2\pi n}{L}x\right)\mathrm{d}x = \frac{L}{2}a_n, \quad n = 1, 2, \ldots$$
$$\int_0^L f(x)\sin\left(\frac{2\pi n}{L}x\right)\mathrm{d}x = \frac{L}{2}b_n, \quad n = 1, 2, \ldots$$

So it makes sense to define:

$$a_n = \frac{2}{L}\int_0^L f(x)\cdot\cos\left(\frac{2\pi n}{L}x\right)\mathrm{d}x, \quad n = 0, 1, 2, \ldots$$
$$b_n = \frac{2}{L}\int_0^L f(x)\cdot\sin\left(\frac{2\pi n}{L}x\right)\mathrm{d}x, \quad n = 1, 2, \ldots$$

Dirichlet's Theorem

Assume that $f : [0, L] \to \mathbb{R}$ is piecewise continuous. Let $f_N(x)$ denote the partial sum:

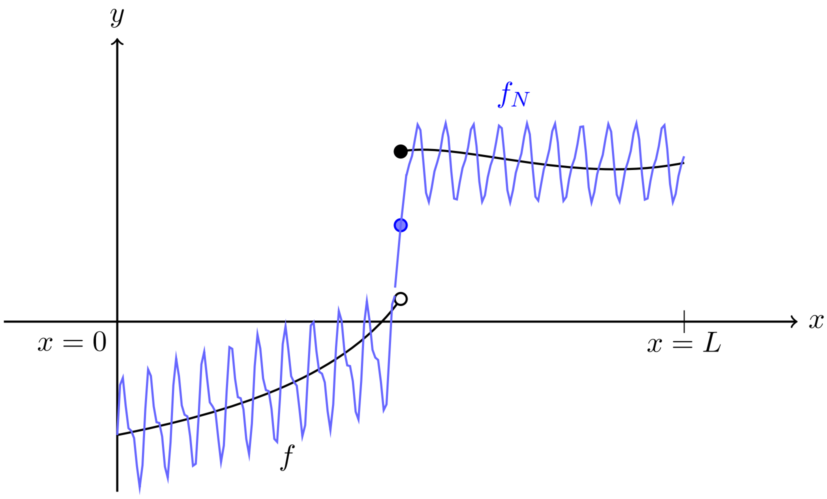
$$f_N(x) = \frac{a_0}{2} + \sum_{n=1}^{N} \left(a_n \cos\left(\frac{2\pi n}{L}x\right) + b_n \sin\left(\frac{2\pi n}{L}x\right)\right)$$

Then $\lim_{N\to+\infty} f_N(x) = f(x)$ at every point x where f is differentiable. Moreover:

$$\int_0^L |f_N(x) - f(x)| \mathrm{d}x \xrightarrow{N\to+\infty} 0$$

At points of jump discontinuities, $f_N(x)$ converges to the midpoint:

$$f_N(x) \longrightarrow \frac{f(x^+) + f(x^-)}{2}$$



Special cases:

- **When f is odd** ($f(-x) = -f(x)$): Then $a_n = 0$ for all n, and

$$f(x) = \sum_{n=1}^{\infty} b_n \cdot \sin\left(\frac{2\pi n}{L}x\right)$$

  with $b_n = \frac{2}{L}\int_0^L f(x) \cdot \sin\left(\frac{2\pi n}{L}x\right) \mathrm{d}x = \frac{4}{L}\int_0^{L/2} f(x) \sin\left(\frac{2\pi n}{L}x\right) \mathrm{d}x$. (Here, use the fact that the integral over any interval of length L is the same since the product is L-periodic).

- **When f is even** ($f(-x) = f(x)$): Then $b_n = 0$ for all n, and

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cdot \cos\left(\frac{2\pi n}{L}x\right)$$

  with $a_n = \frac{4}{L}\int_0^{L/2} f(x) \cos\left(\frac{2\pi n}{L}x\right) \mathrm{d}x$.

The functions $e_n(x) = \cos(\frac{2\pi n}{L}x)$ and $h_n(x) = \sin(\frac{2\pi n}{L}x)$ are sometimes called modes. Note that $e_n'(0) = 0 = e_n'(L)$ and $h_n(0) = 0 = h_n(L)$.

11.2.2 Solving the 1-Dimensional Heat Equation

Let's return to the 1-dimensional heat equation:

$$\begin{cases} \frac{\partial u}{\partial t} - \frac{\partial^2 u}{\partial x^2} = f & \text{for } t > 0, 0 < x < L \\ u|_{t=0} = u_0 & \text{(initial condition)} \\ u|_{x=0} = 0, \quad u|_{x=L} = 0 & \text{(boundary conditions)} \end{cases} \tag{25}$$

Overview of the method:

1. Decompose the heat equation into Fourier modes (in x).
2. The PDE becomes an ODE (in time) for the Fourier coefficients: Solve it.
3. Find a formula for the solution by summing the mode solutions.

Uniqueness

The initial-boundary value problem for the heat equation has a unique solution.

Proof

If u_1, u_2 are two solutions, then $v = u_1 - u_2$ solves the homogeneous problem: $\frac{\partial v}{\partial t} - \frac{\partial^2 v}{\partial x^2} = 0$, $v|_{t=0} = 0$, $v|_{x=0} = v|_{x=L} = 0$. Then using the energy method:

$$\begin{aligned} \frac{\mathrm{d}}{\mathrm{d}t}\left(\int_0^L v^2(x,t)\mathrm{d}x\right) &= \int_0^L 2v\frac{\partial v}{\partial t}\mathrm{d}x = 2\int_0^L v\frac{\partial^2 v}{\partial x^2}\mathrm{d}x \\ &= 2\left[v \cdot \frac{\partial v}{\partial x}\right]_{x=0}^{x=L} - 2\int_0^L \left(\frac{\partial v}{\partial x}\right)^2 \mathrm{d}x \\ &= -2\int_0^L \left(\frac{\partial v}{\partial x}\right)^2 \mathrm{d}x \leq 0 \end{aligned}$$

So $\int_0^L v^2(x,t)\mathrm{d}x$ is decreasing. Since $v(x,0) = 0$, we have $\int_0^L v^2(x,t)\mathrm{d}x \leq \int_0^L v^2(x,0)\mathrm{d}x = 0$. Thus $v(x,t) = 0$ for all $t \geq 0, 0 \leq x \leq L$.

So, in view of uniqueness, if we can find a solution in any way, then that is the unique solution.

Fourier Decomposition Trick: To decompose into Fourier modes, it helps to do the following trick (since we want $u|_{x=0} = 0, u|_{x=L} = 0$): Extend $u(t,x)$ in x to a $2L$-periodic function on $\mathbb{R}$ which is odd. Do the same for u_0, f.

Note: If u is a $2L$-periodic and odd function which is continuous, then necessarily $u|_{x=0} = 0$ and $u|_{x=L} = 0$.

Fourier series (with respect to the period $2L$):

$$u(x,t) = \sum_{n=1}^{\infty} b_n(t) \cdot \sin\left(\frac{2\pi n}{2L}x\right) = \sum_{n=1}^{\infty} b_n(t) \cdot \sin\left(\frac{\pi n}{L}x\right)$$

where $b_n(t) = \frac{2}{L}\int_0^L u(x,t)\cdot \sin\left(\frac{\pi n}{L}x\right) \mathrm{d}x$.

Similarly, decompose the known data:

$$u_0(x) = \sum_{n=1}^{\infty} b_{0n}\cdot \sin\left(\frac{\pi n}{L}x\right), \quad f(x,t) = \sum_{n=1}^{\infty} f_n(t)\cdot \sin\left(\frac{\pi n}{L}x\right)$$

Differentiating the series formally:

$$\frac{\partial u}{\partial x}(x,t) = \sum_{n=1}^{\infty} b_n(t)\cdot \frac{\pi n}{L}\cdot \cos\left(\frac{\pi n}{L}x\right)$$
$$\frac{\partial^2 u}{\partial x^2}(x,t) = \sum_{n=1}^{\infty} b_n(t)\cdot \left(-\frac{\pi^2 n^2}{L^2}\right)\cdot \sin\left(\frac{\pi n}{L}x\right)$$
$$\frac{\partial u}{\partial t}(x,t) = \sum_{n=1}^{\infty} b_n'(t)\cdot \sin\left(\frac{\pi n}{L}x\right)$$

Injecting into the PDE $\frac{\partial u}{\partial t} - \frac{\partial^2 u}{\partial x^2} = f$ gives:

$$\sum_{n=1}^{\infty}\left(b_n'(t) + \frac{\pi^2 n^2}{L^2}b_n(t)\right)\cdot \sin\left(\frac{\pi n}{L}x\right) = \sum_{n=1}^{\infty} f_n(t)\cdot \sin\left(\frac{\pi n}{L}x\right)$$

Correction of the ODE sign

In the raw notes, the equation was written as $b_n' - \frac{\pi^2 n^2}{L^2}b_n = f_n$. However, taking $\frac{\partial u}{\partial t} - \frac{\partial^2 u}{\partial x^2}$, the double spatial derivative already introduces a negative sign ($-\frac{\pi^2 n^2}{L^2}$). Subtracting it yields a positive term: $b_n' - (-\dots) = b_n' + \frac{\pi^2 n^2}{L^2}b_n$. This yields the correct stable decay expected for the heat equation.

So, by equating coefficients, we obtain a family of ODEs:

$$b_n'(t) + \frac{\pi^2 n^2}{L^2}b_n(t) = f_n(t) \quad \text{for } n = 1,2,3,\dots \text{ and } t > 0$$

The initial conditions $u(x,0) = u_0(x)$ imply $b_n(0) = b_{0n}$. The above is an initial value problem of the standard form $y' + ay = g(t)$, $y(0) = y_0$, which we can solve using the integrating factor method or the Laplace transform.

Solution:

$$b_n(t) = b_{0n}\cdot e^{-\frac{\pi^2 n^2}{L^2}t} + \int_0^t e^{-\frac{\pi^2 n^2}{L^2}(t-s)} f_n(s)\mathrm{d}s$$

So we have found u:

$$u(x,t) = \sum_{n=1}^{\infty} b_n(t)\sin\left(\frac{\pi n}{L}x\right)$$

Remarks:

- **If** $f = 0$**:** $u(x,t) = \sum_{n=1}^{\infty} b_{0n} e^{-\frac{\pi^2 n^2}{L^2} t} \cdot \sin\left(\frac{\pi n}{L} x\right)$. The solution decays exponentially fast to 0 as $t \to +\infty$ (and faster decay occurs in higher modes).

- **Special solutions:** If $u_0 = 0$ and $f(x,t) = \sin(\frac{\pi x}{L})$: Then $f_n(t) = 1$ for $n = 1$ and 0 for $n \geq 2$. $\implies b_1(t) = \int_0^t e^{-\frac{\pi^2}{L^2}(t-s)} \cdot 1 \mathrm{d}s = \frac{L^2}{\pi^2}\left(1 - e^{-\frac{\pi^2}{L^2}t}\right)$ and $b_n(t) = 0$ for $n \geq 2$.

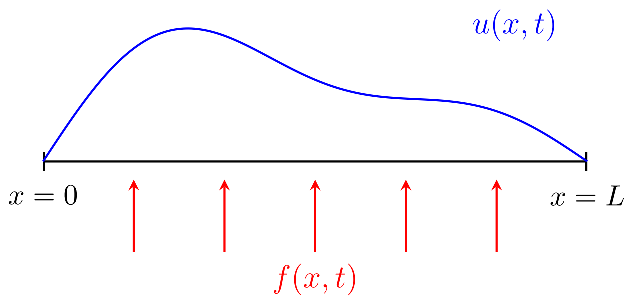
$$u(x,t) = \frac{L^2}{\pi^2}\left(1 - e^{-\frac{\pi^2}{L^2}t}\right) \cdot \sin\left(\frac{\pi x}{L}\right)$$

- **Neumann Boundary Conditions:** If we wanted Neumann boundary conditions (namely $\frac{\partial u}{\partial x}\big|_{x=0} = 0 = \frac{\partial u}{\partial x}\big|_{x=L}$), we would extend u as an *even* $2L$-periodic function. In that case:

$$u(x,t) = \frac{a_0(t)}{2} + \sum_{n=1}^{\infty} a_n(t) \cdot \cos\left(\frac{\pi n x}{L}\right)$$

12 Week-12 - 11.05.2026 - The 1-Dimensional Heat Equation

Last time: We began studying the 1-dimensional heat equation.



Initial-Boundary Value Problem for the Heat Equation

The problem is defined for $t > 0$ and $0 < x < L$ as:

$$\frac{\partial u}{\partial t} - \frac{\partial^2 u}{\partial x^2} = f$$

$$u|_{t=0} = u_0 \quad \text{(initial condition)}$$

$$u|_{x=0} = 0, \quad u|_{x=L} = 0 \quad \text{(boundary conditions)}$$

Overview of the method:

- Decompose the heat equation into Fourier modes (in x).
- The PDE becomes an ODE (in time) for the Fourier coefficients. Solve it.
- Find a formula for the solution by summing the mode solutions.

Uniqueness

The initial value problem for the heat equation has a unique solution.

Proof

If u_1 and u_2 are two solutions, then their difference $v = u_1 - u_2$ solves the homogeneous problem:

$$\frac{\partial v}{\partial t} - \frac{\partial^2 v}{\partial x^2} = 0 \quad \text{for } t > 0,\ 0 < x < L$$
$$v|_{t=0} = 0$$
$$v|_{x=0} = 0 = v|_{x=L}$$

Consider the "energy" or L^2 norm of v:

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(\int_0^L v^2(x,t)\,\mathrm{d}x\right) = \int_0^L 2v \cdot \frac{\partial v}{\partial t}\,\mathrm{d}x$$

Using the PDE $\frac{\partial v}{\partial t} = \frac{\partial^2 v}{\partial x^2}$, we substitute this in:

$$= 2\int_0^L v \cdot \frac{\partial^2 v}{\partial x^2}\,\mathrm{d}x$$

Integrating by parts:

$$= 2\left(\left[v \cdot \frac{\partial v}{\partial x}\right]_{x=0}^{x=L} - \int_0^L \left(\frac{\partial v}{\partial x}\right)^2 \mathrm{d}x\right)$$

Since $v|_{x=0} = 0 = v|_{x=L}$, the boundary term evaluates to zero. Therefore,

$$= -2\int_0^L \left(\frac{\partial v}{\partial x}\right)^2 \mathrm{d}x \le 0$$

This implies that $\int_0^L v^2(t,x)\,\mathrm{d}x$ is a decreasing function of t. Given the initial condition $v|_{t=0} = 0$, we have:

$$\int_0^L v^2(t,x)\,\mathrm{d}x \le \int_0^L v^2(0,x)\,\mathrm{d}x = 0$$

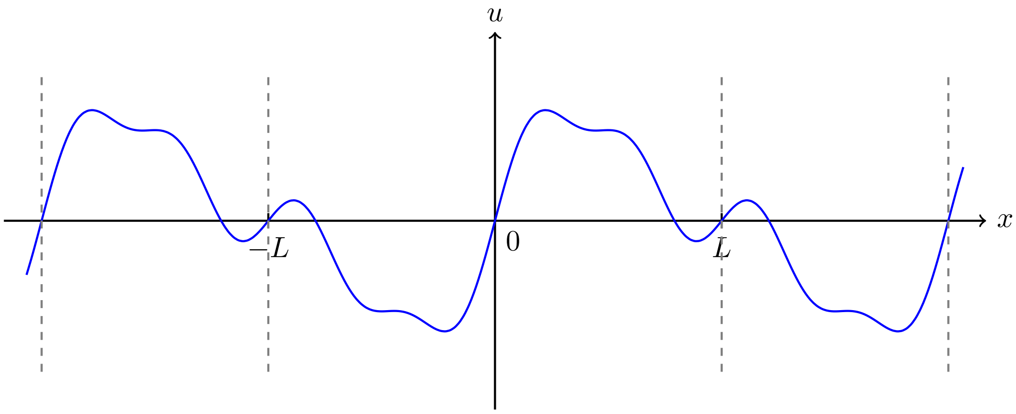
Since the integral of a non-negative function is strictly zero, we must have $v(t,x) = 0$ for all $t \ge 0$ and $0 \le x \le L$. Thus, $u_1 = u_2$.

Implication of Uniqueness

In view of uniqueness: If we can find a solution *in any way*, then that is **the** unique solution.

12.1 Decomposing into Fourier Modes

It helps to do the following trick: since we want $u|_{x=0} = 0$ and $u|_{x=L} = 0$, we extend $u(t,x)$ in x to a $2L$-periodic function on $\mathbb{R}$ which is **odd**.



We do the same for u_0 and f. *Note:* If u is a $2L$-periodic and odd function which is continuous, then necessarily $u|_{x=0} = 0$ and $u|_{x=L} = 0$.

For the solution of the heat equation, it can be shown that if f is piecewise continuous and the same is true for u_0, then for $t > 0$, $u(t,x)$ and $\frac{\partial u}{\partial x}(t,x)$ are continuous in x.

Fourier series (with respect to the $2L$-period): Because of the odd extension, only sine terms appear.

$$u(x,t) = \sum_{n=1}^{\infty} b_n(t) \sin\left(\frac{\pi n}{L} x\right), \quad \text{where } b_n(t) = \frac{2}{L}\int_0^L u(x,t) \sin\left(\frac{\pi n}{L} x\right) \mathrm{d}x$$

$$u_0(x) = \sum_{n=1}^{\infty} b_{0n} \sin\left(\frac{\pi n}{L} x\right)$$

$$f(x,t) = \sum_{n=1}^{\infty} f_n(t) \sin\left(\frac{\pi n}{L} x\right)$$

Computing the derivatives:

$$\frac{\partial u}{\partial x}(x,t) = \sum_{n=1}^{\infty} b_n(t) \left(\frac{\pi n}{L}\right) \cos\left(\frac{\pi n}{L} x\right)$$

$$\frac{\partial^2 u}{\partial x^2}(x,t) = \sum_{n=1}^{\infty} b_n(t) \left(-\frac{\pi^2 n^2}{L^2}\right) \sin\left(\frac{\pi n}{L} x\right)$$

$$\frac{\partial u}{\partial t}(x,t) = \sum_{n=1}^{\infty} b_n'(t) \sin\left(\frac{\pi n}{L} x\right)$$

Substituting these into the heat equation $\frac{\partial u}{\partial t} - \frac{\partial^2 u}{\partial x^2} = f$ gives:

$$\sum_{n=1}^{\infty} \left(b_n'(t) + \frac{\pi^2 n^2}{L^2} b_n(t)\right) \sin\left(\frac{\pi n}{L} x\right) = \sum_{n=1}^{\infty} f_n(t) \sin\left(\frac{\pi n}{L} x\right)$$

Mathematical Correction

In the original handwritten notes, a minus sign was written erroneously as $b_n'(t) - \frac{\pi^2 n^2}{L^2} b_n(t)$. This has been corrected to a **plus** sign here. This ensures mathematical accuracy because $\frac{\partial^2 u}{\partial x^2}$ already produces a negative term, so $-\frac{\partial^2 u}{\partial x^2}$ produces a strictly positive term.

By matching the coefficients, for $n = 1, 2, 3, \dots$ and for $t > 0$, we get the ODE:

$$b_n'(t) + \frac{\pi^2 n^2}{L^2} b_n(t) = f_n(t)$$

With the initial conditions matching $u(x,0) = u_0(x)$, we have $b_n(0) = b_{0n}$.

The above is an initial value problem of the form:

$$\begin{cases} y' + ay = F \\ y(0) = y_0 \end{cases}$$

We have seen how to solve it (about 2 weeks ago) using the Laplace transform.

Solution for the Fourier Coefficients

$$b_n(t) = b_{0n} e^{-\frac{\pi^2 n^2}{L^2} t} + \int_0^t e^{-\frac{\pi^2 n^2}{L^2}(t-s)} f_n(s)\, \mathrm{d}s$$

So we have found u:

$$u(x,t) = \sum_{n=1}^{\infty} \left(b_{0n} e^{-\frac{\pi^2 n^2}{L^2} t} + \int_0^t e^{-\frac{\pi^2 n^2}{L^2}(t-s)} f_n(s)\, \mathrm{d}s \right) \sin\left(\frac{\pi n}{L} x\right)$$

Homogeneous Case

If $f \equiv 0$, then $u(x,t) = \sum_{n=1}^{\infty} b_{0n} e^{-\frac{\pi^2 n^2}{L^2} t} \sin\left(\frac{\pi n}{L} x\right)$. This decays exponentially fast to 0 as $t \to +\infty$ (with an even faster decay in the higher modes).

Special Solution 1

If $u_0 = 0$ and $f(x,t) = \sin\left(\frac{\pi x}{L}\right)$:

$$f_n(t) = \begin{cases} 1, & n = 1 \\ 0, & n \geq 2 \end{cases}$$

This yields:

$$b_n(t) = \begin{cases} \frac{L^2}{\pi^2} \left(1 - e^{-\frac{\pi^2}{L^2} t}\right), & n = 1 \\ 0, & n \geq 2 \end{cases}$$

So the specific solution is:

$$u(x,t) = \frac{L^2}{\pi^2} \left(1 - e^{-\frac{\pi^2}{L^2} t}\right) \sin\left(\frac{\pi x}{L}\right)$$

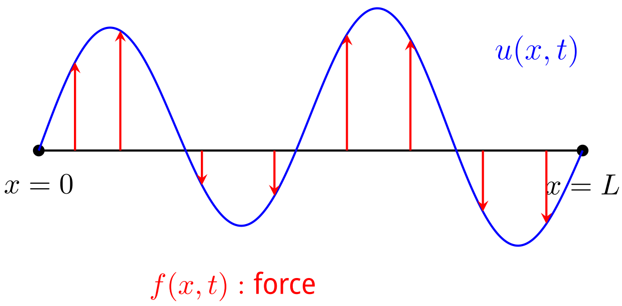
Neumann Boundary Conditions

If we wanted Neumann boundary conditions, namely $\frac{\partial u}{\partial x}\big|_{x=0} = 0 = \frac{\partial u}{\partial x}\big|_{x=L}$, we would extend u as an **even** $2L$-periodic function. In that case, the series expansion uses cosines:

$$u(x,t) = \frac{a_0(t)}{2} + \sum_{n=1}^{\infty} a_n(t) \cos\left(\frac{\pi n}{L} x\right)$$

12.2 The Wave Equation

Elastic cord:



Initial-Boundary Value Problem for the Wave Equation

Defined for $t > 0$ and $0 < x < L$:

$$\frac{\partial^2 u}{\partial t^2} - c^2 \frac{\partial^2 u}{\partial x^2} = f$$

$$u|_{t=0} = u_0 \quad \text{(initial position)}$$

$$\left.\frac{\partial u}{\partial t}\right|_{t=0} = u_1 \quad \text{(initial velocity)}$$

$$u|_{x=0} = 0, \quad u|_{x=L} = 0 \quad \text{(Dirichlet boundary conditions: fixed endpoints)}$$

Here, $c > 0$ is the speed of propagation.

Energy Conservation

We define the energy $E[u](t)$ as the sum of kinetic and potential energy:

$$E[u](t) = \frac{1}{2}\int_0^L \underbrace{\left(\frac{\partial u}{\partial t}\right)^2}_{\text{Kinetic}} + \underbrace{c^2\left(\frac{\partial u}{\partial x}\right)^2}_{\text{Potential}} \,\mathrm{d}x$$

Then, the rate of change of energy equals the power exerted by the external force:

$$\frac{\mathrm{d}}{\mathrm{d}t}E[u](t) = \int_0^L \frac{\partial u}{\partial t}(x,t) f(x,t)\,\mathrm{d}x$$

Proof

Differentiating the energy with respect to time:

$$\frac{\mathrm{d}}{\mathrm{d}t}E[u](t) = \frac{1}{2}\int_0^L 2\frac{\partial u}{\partial t}\cdot\frac{\partial^2 u}{\partial t^2} + 2c^2\frac{\partial u}{\partial x}\cdot\frac{\partial^2 u}{\partial x\partial t}\,\mathrm{d}x$$

Substitute $\frac{\partial^2 u}{\partial t^2} = c^2\frac{\partial^2 u}{\partial x^2} + f$ into the first term:

$$= \int_0^L \frac{\partial u}{\partial t}\left(f + c^2\frac{\partial^2 u}{\partial x^2}\right) + c^2\frac{\partial u}{\partial x}\frac{\partial^2 u}{\partial x\partial t}\,\mathrm{d}x$$

Separate the integral and apply integration by parts to $c^2\frac{\partial u}{\partial t}\frac{\partial^2 u}{\partial x^2}$:

$$\begin{aligned} &= \int_0^L \frac{\partial u}{\partial t} f\,\mathrm{d}x + \left[c^2\frac{\partial u}{\partial t}\frac{\partial u}{\partial x}\right]_{x=0}^{x=L} \\ &\qquad - \int_0^L c^2\frac{\partial^2 u}{\partial t\partial x}\frac{\partial u}{\partial x}\,\mathrm{d}x + \int_0^L c^2\frac{\partial u}{\partial x}\frac{\partial^2 u}{\partial x\partial t}\,\mathrm{d}x \end{aligned}$$

The last two integrals perfectly cancel each other out. Since $u(0,t) = 0 = u(L,t)$, their time derivatives also vanish, $\frac{\partial u}{\partial t}(0,t) = 0 = \frac{\partial u}{\partial t}(L,t)$. Therefore, the boundary term vanishes.

$$\frac{\mathrm{d}}{\mathrm{d}t}E[u](t) = \int_0^L \frac{\partial u}{\partial t}(x,t)\cdot f(x,t)\,\mathrm{d}x$$

Uniqueness

The conservation of energy implies the uniqueness of solutions for the wave equation.

Proof

If u_1 and u_2 are two solutions, their difference $v = u_1 - u_2$ solves the homogeneous wave equation:

$$\begin{aligned} \frac{\partial^2 v}{\partial t^2} - c^2 \frac{\partial^2 v}{\partial x^2} &= 0 \\ v|_{t=0} &= 0 \\ \left.\frac{\partial v}{\partial t}\right|_{t=0} &= 0 \\ v|_{x=0} &= 0 = v|_{x=L} \end{aligned}$$

Conservation of energy for v (since $f = 0$) implies $\frac{\mathrm{d}}{\mathrm{d}t} E[v](t) = 0$. Thus, for all $t \geq 0$, $E[v](t) = E[v](0)$. But $v|_{t=0} = 0$ and $\frac{\partial v}{\partial t}|_{t=0} = 0$ both vanish, so $E[v](0) = 0$. So $E[v](t) = 0$. Since $E[v](t) = \frac{1}{2} \int_0^L \left(\left(\frac{\partial v}{\partial t}\right)^2 + c^2 \left(\frac{\partial v}{\partial x}\right)^2 \right) \mathrm{d}x = 0$, and the integrand is strictly non-negative, we infer that:

$$\frac{\partial v}{\partial t} = 0 \quad \text{and} \quad \frac{\partial v}{\partial x} = 0 \quad \forall t, x$$

Hence $v = \text{const}$. Since $v|_{t=0} = 0$, it follows that $v = 0 \implies u_1 = u_2$.

12.2.1 Construction of the Solution for the Wave Equation

We extend $u(x,t)$, $f(x,t)$, $u_0(x)$, $u_1(x)$ as odd, $2L$-periodic functions of $x \in \mathbb{R}$. The Fourier series then only contains sine terms:

$$\begin{aligned} u(x,t) &= \sum_{n=1}^{\infty} b_n(t) \sin\left(\frac{\pi n}{L} x\right) \\ u_0(x) &= \sum_{n=1}^{\infty} b_{0n} \sin\left(\frac{\pi n}{L} x\right) \\ u_1(x) &= \sum_{n=1}^{\infty} b_{1n} \sin\left(\frac{\pi n}{L} x\right) \\ f(x,t) &= \sum_{n=1}^{\infty} f_n(t) \sin\left(\frac{\pi n}{L} x\right) \end{aligned}$$

Differentiating yields:

$$\begin{aligned} \frac{\partial^2 u}{\partial t^2}(x,t) &= \sum_{n=1}^{\infty} b_n''(t) \sin\left(\frac{\pi n}{L} x\right) \\ \frac{\partial^2 u}{\partial x^2}(x,t) &= \sum_{n=1}^{\infty} b_n(t) \left(-\frac{\pi^2 n^2}{L^2}\right) \sin\left(\frac{\pi n}{L} x\right) \end{aligned}$$

Substituting into $\frac{\partial^2 u}{\partial t^2} - c^2 \frac{\partial^2 u}{\partial x^2} = f$ gives:

$$\sum_{n=1}^{\infty} \left(b_n''(t) + c^2 \frac{\pi^2 n^2}{L^2} b_n(t) \right) \sin\left(\frac{\pi n}{L} x\right) = \sum_{n=1}^{\infty} f_n(t) \sin\left(\frac{\pi n}{L} x\right)$$

This generates a system of ODEs for the coefficients:

$$\begin{cases} b_n''(t) + c^2 \frac{\pi^2 n^2}{L^2} b_n(t) = f_n(t) \\ b_n(0) = b_{0n} \\ b_n'(0) = b_{1n} \end{cases}$$

We can solve this using the Laplace transform:

$$z^2 \mathcal{L}[b_n](z) - z b_n(0) - b_n'(0) + \frac{c^2 \pi^2 n^2}{L^2} \mathcal{L}[b_n](z) = \mathcal{L}[f_n](z)$$

$$\implies \mathcal{L}[b_n](z) = \frac{\mathcal{L}[f_n](z) + z \cdot b_{0n} + b_{1n}}{z^2 + \frac{c^2 \pi^2 n^2}{L^2}}$$

Taking the inverse Laplace transform for $n = 1, 2, 3, \ldots$:

$$b_n(t) = b_{0n} \cos\left(\frac{c\pi n}{L} t\right) + \frac{b_{1n}}{c\frac{\pi n}{L}} \sin\left(\frac{c\pi n}{L} t\right) + \int_0^t \frac{1}{c\frac{\pi n}{L}} \sin\left(\frac{c\pi n}{L}(t-s)\right) f_n(s) \, \mathrm{d}s$$

Special Cases for the Wave Equation

Case 1: If $f \equiv 0$, $u_0 = \sin\left(\frac{\pi x}{L}\right)$, and $u_1 = 0$:

$$\begin{aligned} f_n(t) &= 0 \\ b_{0n} &= \begin{cases} 1, & n = 1 \\ 0, & n \geq 2 \end{cases} \\ b_{1n} &= 0 \end{aligned}$$

This implies $b_n(t) = \cos\left(\frac{c\pi}{L}t\right)$ for $n = 1$, and 0 otherwise. The final solution is:

$$u(x,t) = \cos\left(\frac{c\pi}{L}t\right)\sin\left(\frac{\pi x}{L}\right)$$

Case 2: If $u_0 = 0$, $u_1 = 0$, and $f(x,t) = \sin\left(\frac{\pi x}{L}\right)$:

$$\begin{aligned} b_{0n} &= 0 \\ b_{1n} &= 0 \\ f_n(t) &= \begin{cases} 1, & n = 1 \\ 0, & n \geq 2 \end{cases} \end{aligned}$$

By explicitly evaluating the integral for $n = 1$, we get:

$$b_1(t) = \frac{1 - \cos\left(\frac{c\pi}{L}t\right)}{\left(c\frac{\pi}{L}\right)^2}$$

Mathematical Correction

The handwritten notes omitted the square in the denominator: $c\frac{\pi}{L}$. Standard integration of $\int \frac{1}{\omega}\sin(\omega(t-s))\,ds$ yields $\frac{1-\cos(\omega t)}{\omega^2}$. Hence it is explicitly corrected here to $\left(c\frac{\pi}{L}\right)^2$.

The specific solution is:

$$u(x,t) = \frac{1 - \cos\left(\frac{c\pi}{L}t\right)}{\left(c\frac{\pi}{L}\right)^2}\sin\left(\frac{\pi x}{L}\right)$$

12.3 Generalizing the Method

Extensions of the Fourier Method

The above method applies to more general initial-boundary value problems ($x \in [0, L], t \geq 0$) for linear PDEs, subject to the following principles:

1. **Constant Coefficients:** We typically need the coefficients of the spatial operator in the equation to be constant with respect to x. For example, if we try the same approach for:
$$\frac{\partial^2 u}{\partial t^2} - c^2(t)\frac{\partial^2 u}{\partial x^2} = f$$
It yields the ODE:
$$b_n''(t) + c^2(t)\frac{\pi^2 n^2}{L^2} b_n(t) = f_n(t)$$
In principle, if $c(t)$ is known explicitly, the above ODE can be solved, but it is much more complicated than the case of a constant c.

2. **General Boundary Conditions:** One can also address more general boundary conditions, e.g., periodic boundary conditions:
$$u(0,t) = u(L,t)$$
$$\frac{\partial u}{\partial x}(0,t) = \frac{\partial u}{\partial x}(L,t)$$
In this case, decomposing relies on the full Fourier series:
$$u(x,t) = \frac{a_0(t)}{2} + \sum_{n=1}^{\infty} \left(a_n(t) \cos\left(\frac{2\pi n}{L}x\right) + b_n(t) \sin\left(\frac{2\pi n}{L}x\right) \right)$$
And we solve coupled ODEs for both $a_n(t)$ and $b_n(t)$.

3. **Inhomogeneous Boundary Conditions:** (Covered in exercise sets).

4. **General Linear Operators:** Solving initial-boundary value problems of the form:
$$\frac{\partial u}{\partial t} + Pu = 0$$
$$u|_{t=0} = u_0$$
$$u|_{x=0} = 0, \quad u|_{x=L} = 0$$
Where P is a linear differential operator:
$$P = c_k(x)\frac{\partial^k}{\partial x^k} + \cdots + c_1(x)\frac{\partial}{\partial x} + c_0(x)$$
This requires decomposing $u(x,t)$ into a series *not necessarily trigonometric*, but rather with respect to a basis of **"eigenfunctions"** of P (with Dirichlet boundary conditions), known as a **Sturm-Liouville problem**.

   In this class, we will not deal with this more general case; we will strictly consider problems where the trigonometric series is the appropriate expansion.

12.4 Fourier Transform and PDEs

While on $x \in [0, L]$ we utilize the Fourier series decomposition, on the whole real line $x \in \mathbb{R}$, we shift to the **Fourier transform** in x.

Fourier Transform on $\mathbb{R}$

For an absolutely integrable function $f : \mathbb{R} \to \mathbb{C}$ (provided $\int_{-\infty}^{+\infty} |f(x)| \, \mathrm{d}x < +\infty$), the Fourier transform is defined as:

$$\mathcal{F}[f](\xi) = \hat{f}(\xi) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} e^{-i\xi x} f(x) \, \mathrm{d}x$$

13 Week - 13-18.05.2026 - Fourier Transform and PDEs

- If $x \in [0, L]$: We use the **Fourier series decomposition**.
- If $x \in \mathbb{R}$: We use the **Fourier transform** in x.

Recall

If $f : \mathbb{R} \to \mathbb{C}$, the Fourier transform is defined as:

$$\mathcal{F}[f](a) = \hat{f}(a) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} e^{-iax} f(x) \mathrm{d}x$$

(provided $\int_{-\infty}^{+\infty} |f(x)| \mathrm{d}x < +\infty$).

Important properties:

$$\mathcal{F}\left[\frac{\partial f}{\partial x}\right](a) = ia\mathcal{F}[f](a),$$

$$\mathcal{F}[f](a) \cdot \mathcal{F}[g](a) = \frac{1}{\sqrt{2\pi}} \mathcal{F}[f * g](a)$$

13.1 The Heat Equation

Consider the inhomogeneous heat equation:

$$\begin{cases} \dfrac{\partial u}{\partial t} - \dfrac{\partial^2 u}{\partial x^2} = f(x,t), \quad x \in \mathbb{R}, t > 0, \\ u(x,0) = u_0(x) \end{cases} \tag{26}$$

(We assume that $\int_{-\infty}^{+\infty} |u_0(x)| \mathrm{d}x < +\infty$ and $\int_{-\infty}^{+\infty} |f(x,t)| \mathrm{d}x < +\infty$ so that we can appropriately apply the Fourier transform).

Applying the Fourier transform in x:

$$\hat{u}(a,t) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} e^{-iax} u(x,t) \mathrm{d}x$$
$$\hat{u}_0(a) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} e^{-iax} u_0(x) \mathrm{d}x$$
$$\hat{f}(a,t) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} e^{-iax} f(x,t) \mathrm{d}x$$

Then the PDE becomes:

$$\frac{\partial u}{\partial t} - \frac{\partial^2 u}{\partial x^2} = f \implies \mathcal{F}\left[\frac{\partial u}{\partial t} - \frac{\partial^2 u}{\partial x^2}\right](a,t) = \hat{f}(a,t)$$

This yields the system:

$$\begin{cases} \dfrac{\partial \hat{u}}{\partial t}(a,t) + a^2 \hat{u}(a,t) = \hat{f}(a,t) \\ \hat{u}(a,0) = \hat{u}_0(a) \end{cases}$$

For each frequency $a \in \mathbb{R}$, this is a first-order **ODE in time**! We can solve it (e.g., using the Laplace transform in t, or an integrating factor).

Solution:

$$\hat{u}(a,t) = e^{-a^2 t}\hat{u}_0(a) + \int_0^t e^{-a^2(t-s)}\hat{f}(a,s)\mathrm{d}s \tag{27}$$

We need to apply the *inverse Fourier transform* to get an expression for $u(x,t)$.

Recall

The Fourier transform of a Gaussian is a Gaussian:

$$\mathcal{F}\left[e^{-\frac{x^2}{2}}\right](a) = e^{-a^2/2}$$

Also, recall the rescaling property:

$$\mathcal{F}[f(\lambda x)](a) = \frac{1}{|\lambda|}\mathcal{F}[f]\left(\frac{a}{\lambda}\right)$$

Using this, we can invert $e^{-a^2 t}$:

$$e^{-a^2 t} = e^{-\frac{(\sqrt{2t}\cdot a)^2}{2}} = \frac{1}{\sqrt{2t}}\mathcal{F}\left[e^{-\frac{1}{2}\left(\frac{x}{\sqrt{2t}}\right)^2}\right](a) = \mathcal{F}\left[\frac{1}{\sqrt{2t}}e^{-\frac{x^2}{4t}}\right](a)$$

Going back to our ODE solution (2):

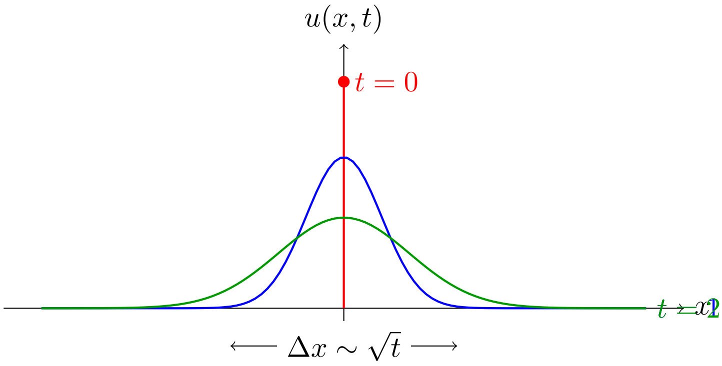
$$\hat{u}(a,t) = \mathcal{F}\left[\frac{1}{\sqrt{2t}}e^{-\frac{x^2}{4t}}\right](a)\cdot\mathcal{F}[u_0](a) + \int_0^t \mathcal{F}\left[\frac{1}{\sqrt{2(t-s)}}e^{-\frac{x^2}{4(t-s)}}\right](a)\cdot\mathcal{F}[f(\cdot,s)](a)\mathrm{d}s$$

Inverting (and using the property about convolutions):

$$\begin{aligned} u(x,t) = {} & \frac{1}{\sqrt{4\pi t}}\int_{-\infty}^{+\infty} e^{-\frac{(x-y)^2}{4t}}u_0(y)\mathrm{d}y \\ & + \int_0^t \int_{-\infty}^{+\infty} \frac{1}{\sqrt{4\pi(t-s)}}e^{-\frac{(x-y)^2}{4(t-s)}}f(y,s)\mathrm{d}y\,\mathrm{d}s \end{aligned} \tag{28}$$

Special case: When $f = 0$, we just have the first term:

$$u(x,t) = \frac{1}{\sqrt{4\pi t}}\int_{-\infty}^{+\infty} e^{-\frac{(x-y)^2}{4t}}u_0(y)\mathrm{d}y$$



Duhamel's Principle

If $P[u_0](x,t)$ is the above integral expression for the homogeneous case, then the solution in the case when $f \neq 0$ is given by:

$$u(x,t) = P[u_0](x,t) + \int_0^t P[f(\cdot,s)](x,t-s)\mathrm{d}s$$

13.2 Another Example: Wave Equation

$$\begin{cases} \dfrac{\partial^2 u}{\partial t^2} - \dfrac{\partial^2 u}{\partial x^2} = f(x,t) & \text{for } x \in \mathbb{R}, t > 0 \\ u(x,0) = u_0(x) \\ \dfrac{\partial u}{\partial t}(x,0) = u_1(x) \end{cases} \tag{29}$$

(We set $c = 1$ for simplicity).

Apply a Fourier transform in x:

$$\hat{u}(a,t) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} e^{-iax} u(x,t)\mathrm{d}x$$

and similarly for $\hat{f}(a,t)$, $\hat{u}_0(a)$, and $\hat{u}_1(a)$.

Then, noting that $\mathcal{F}\left[\frac{\partial^2 u}{\partial x^2}\right] = (ia)^2\hat{u} = -a^2\hat{u}$, we get:

$$\begin{cases} \dfrac{\partial^2 \hat{u}}{\partial t^2}(a,t) + a^2\hat{u}(a,t) = \hat{f}(a,t) \\ \hat{u}(a,0) = \hat{u}_0(a) \\ \dfrac{\partial \hat{u}}{\partial t}(a,0) = \hat{u}_1(a) \end{cases} \tag{30}$$

Solution (using Laplace transform in t, or variation of parameters):

$$\hat{u}(a,t) = \cos(at)\hat{u}_0(a) + \frac{1}{a}\sin(at)\hat{u}_1(a) + \int_0^t \frac{1}{a}\sin(a(t-s))\hat{f}(a,s)\mathrm{d}s$$

We need to invert the Fourier transform; this is a bit trickier than before.

- For the $\cos(at)\hat{u}_0(a)$ term, recall the translation property:

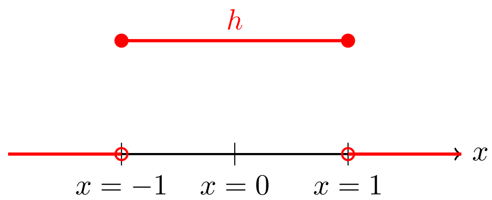
$$\mathcal{F}[h(x+c)](a) = e^{iac}\mathcal{F}[h(x)](a)$$

So:

$$\begin{aligned}\hat{u}_0(a) \cdot \cos(at) &= \hat{u}_0(a)\frac{e^{iat}+e^{-iat}}{2} \\ &= \frac{1}{2}\mathcal{F}[u_0(x+t)+u_0(x-t)](a)\end{aligned}$$

- For the $\frac{1}{a}\sin(at)\hat{u}_1(a)$ term, we use the fact that if:

$$h(x) = \begin{cases} 1, & -1 \le x \le 1 \\ 0, & \text{else} \end{cases}$$



Then $\hat{h}(a) = \sqrt{\frac{2}{\pi}} \cdot \frac{\sin(a)}{a}$ (Exercise).

Using the rescaling property $\mathcal{F}[h(\lambda x)](a) = \frac{1}{|\lambda|}\mathcal{F}[h]\left(\frac{a}{\lambda}\right)$ and the fact that

$$h_t(x) = \begin{cases} 1, & -t \le x \le t \\ 0, & \text{else} \end{cases}$$

is just $h_t(x) = h(x/t)$, we compute:

$$\hat{h}_t(a) = \sqrt{\frac{2}{\pi}} \cdot \frac{\sin(at)}{a}$$

Therefore:

$$\begin{aligned}\hat{u}_1(a) \cdot \frac{\sin(at)}{a} &= \mathcal{F}[u_1](a) \cdot \sqrt{\frac{\pi}{2}} \cdot \mathcal{F}[h_t](a) \\ &= \frac{1}{2}\mathcal{F}[u_1 * h_t](a)\end{aligned}$$

Evaluating the convolution:

$$(u_1 * h_t)(x) = \int_{-\infty}^{+\infty} h_t(x-y)u_1(y)\mathrm{d}y = \int_{x-t}^{x+t} u_1(y)\mathrm{d}y$$

Correction of D'Alembert's Formula

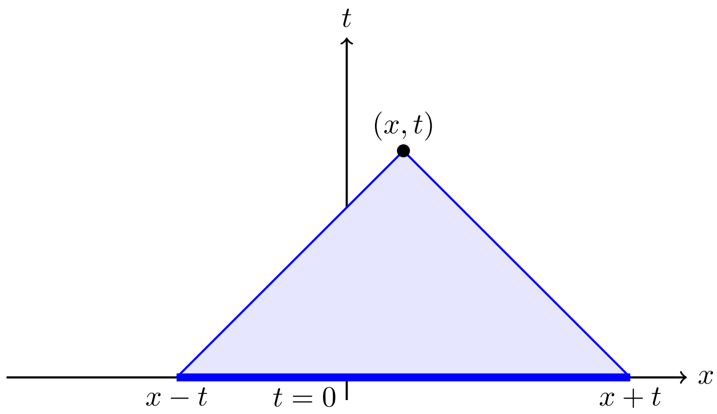
In the original handwritten notes, the integral over the source term $f(y,s)$ is incorrectly written as $\int_{x-t}^{x+t} f(y,s)\mathrm{d}y\mathrm{d}s$.
By Duhamel's Principle, the source term is treated as a series of initial velocity profiles $f(y,s)$ injected at time s. The domain of dependence for the perturbation at (x,t) originating from time s is governed by the time elapsed, which is $(t-s)$, not t. Therefore, the spatial bounds for the integral at time s must be $[x-(t-s), x+(t-s)]$. We correct this below.

So, overall (D'Alembert's Formula):

D'Alembert's Formula

$$u(x,t) = \frac{1}{2}\Big(u_0(x+t) + u_0(x-t)\Big) + \frac{1}{2}\int_{x-t}^{x+t} u_1(y)\,\mathrm{d}y + \frac{1}{2}\int_0^t \int_{x-(t-s)}^{x+(t-s)} f(y,s)\,\mathrm{d}y\,\mathrm{d}s \tag{31}$$

Finite speed of propagation: The value of $u(x,t)$ depends only on the value of f in the shaded light-cone region and the value of (u_0, u_1) on the base segment.



13.3 The Schrödinger Equation

Suppose that $u : \mathbb{R} \times (0, +\infty) \to \mathbb{C}$ solves the following equation:

$$i\frac{\partial u}{\partial t}(x,t) + \frac{\partial^2 u}{\partial x^2}(x,t) - V(x,t)u(x,t) = 0 \quad \text{for } t > 0, x \in \mathbb{R} \tag{32}$$

This is the Schrödinger equation with potential V.

- The potential is always assumed to be real-valued (i.e. $V(x,t) \in \mathbb{R}\ \forall x, t$).
- The solution u is necessarily complex-valued in view of the presence of i in the equation.
- **Physical interpretation:** The above equation describes the evolution of a quantum particle moving under the influence of a (classical) potential $V(x,t)$.

The quantity $\int_a^b |u(x,t)|^2 \mathrm{d}x$ is the probability that the particle at time t is in the interval $a \leq x \leq b$. In order for the above "probabilistic" interpretation to make sense, we must have:

$$\int_{-\infty}^{+\infty} |u(x,t)|^2 \mathrm{d}x = 1 \quad \text{for all times.}$$

In particular, $\int_{-\infty}^{+\infty} |u(x,t)|^2 \mathrm{d}x$ must be **constant** in t for solutions u to (1).

Proof that $\int_{-\infty}^{+\infty} |u(x,t)|^2 \mathrm{d}x$ is conserved if $u, \frac{\partial u}{\partial x} \to 0$ as $x \to \pm\infty$

$$\begin{aligned}
\frac{\partial}{\partial t}\int_{-\infty}^{+\infty} |u(x,t)|^2 \mathrm{d}x &= \frac{\partial}{\partial t}\int_{-\infty}^{+\infty} u(x,t)\cdot\overline{u(x,t)}\mathrm{d}x \\
&= \int_{-\infty}^{+\infty}\left(\frac{\partial u}{\partial t}\overline{u} + u\overline{\frac{\partial u}{\partial t}}\right)\mathrm{d}x \\
&= \int_{-\infty}^{+\infty} 2\,\mathrm{Re}\left\{\frac{\partial u}{\partial t}\overline{u}\right\}\mathrm{d}x \qquad \text{(A)}
\end{aligned}$$

From the equation $i\frac{\partial u}{\partial t} + \frac{\partial^2 u}{\partial x^2} - Vu = 0$, we have $\frac{\partial u}{\partial t} = i\frac{\partial^2 u}{\partial x^2} - iVu$. Substituting this into (A):

$$\begin{aligned}
\text{(A)} &= \int_{-\infty}^{+\infty} 2\,\mathrm{Re}\left\{\left(i\frac{\partial^2 u}{\partial x^2} - iVu\right)\overline{u}\right\}\mathrm{d}x \\
&= \int_{-\infty}^{+\infty} 2\,\mathrm{Re}\left\{i\frac{\partial^2 u}{\partial x^2}\overline{u} - iV|u|^2\right\}\mathrm{d}x
\end{aligned}$$

Since $V(x,t)$ and $|u(x,t)|^2$ are real-valued, the term $-iV|u|^2$ is purely imaginary. Therefore, its real part is zero: $\mathrm{Re}\{-iV|u|^2\} = 0$. Returning to the integral:

$$= 2\int_{-\infty}^{+\infty} \mathrm{Re}\left\{i\frac{\partial^2 u}{\partial x^2}\overline{u}\right\}\mathrm{d}x$$

Integrating by parts:

$$= 2\,\mathrm{Re}\left\{\left[i\frac{\partial u}{\partial x}\overline{u}\right]_{x=-\infty}^{+\infty} - \int_{-\infty}^{+\infty} i\frac{\partial u}{\partial x}\frac{\partial \overline{u}}{\partial x}\mathrm{d}x\right\}$$

If $u(x,t), \frac{\partial u}{\partial x}(x,t) \to 0$ as $x \to \pm\infty$ (a reasonable assumption for a "localized" particle), then the boundary terms vanish, and the above is:

$$= -2\int_{-\infty}^{+\infty} \mathrm{Re}\left\{i\frac{\partial u}{\partial x}\frac{\partial \overline{u}}{\partial x}\right\}\mathrm{d}x$$

But $i\frac{\partial u}{\partial x}\frac{\partial \overline{u}}{\partial x} = i\left|\frac{\partial u}{\partial x}\right|^2$ is purely imaginary, so its real part is 0. Thus, the derivative is 0, and probability is conserved.

Initial Value Problem for the (inhomogeneous) Schrödinger equation:

$$\begin{cases} i\frac{\partial u}{\partial t}(x,t) + \frac{\partial^2 u}{\partial x^2}(x,t) - V(x,t)u(x,t) = F(x,t) & \text{for } t>0, x\in\mathbb{R} \\ u(x,0) = u_0(x) \end{cases} \qquad (33)$$

Uniqueness of solutions: If u_1, u_2 are solutions, then $w = u_1 - u_2$ solves the homogeneous system:

$$\begin{cases} i\frac{\partial w}{\partial t} + \frac{\partial^2 w}{\partial x^2} - Vw = 0 \\ w(x,0) = 0 \end{cases}$$

By the conservation law proved above:

$$\forall t>0: \int_{-\infty}^{+\infty} |w(x,t)|^2 \mathrm{d}x = \cdots = \int_{-\infty}^{+\infty} |w(x,0)|^2 \mathrm{d}x = 0$$

So $w \equiv 0$ for $t > 0, x \in \mathbb{R}$, thus $u_1 = u_2$. (So we have uniqueness in the class of solutions that decay as $x \to \pm\infty$, which are the physically relevant ones).

Remark

Qualitatively, solutions to the Schrödinger equations behave in some sense like waves (but with unbounded speed of propagation). The Schrödinger equation belongs to the class of *dispersive* equations.

13.3.1 Construction of a solution when $V(x,t) \equiv 0$

Consider the free-particle inhomogeneous equation:

$$\begin{cases} i\frac{\partial u}{\partial t}(x,t) + \frac{\partial^2 u}{\partial x^2}(x,t) = F(x,t), \quad x \in \mathbb{R}, t > 0 \\ u(x,0) = u_0(x) \end{cases}$$

Applying the Fourier transform in x:

$$\hat{u}(a,t) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} e^{-iax} u(x,t) \mathrm{d}x$$
$$\hat{F}(a,t) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} e^{-iax} F(x,t) \mathrm{d}x$$
$$\hat{u}_0(a) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} e^{-iax} u_0(x) \mathrm{d}x$$

Then $\hat{u}$ satisfies the first-order ODE in time (note the i factor in front of $\frac{\partial}{\partial t}$):

$$\begin{cases} i\frac{\partial \hat{u}}{\partial t}(a,t) - a^2\hat{u}(a,t) = \hat{F}(a,t) \\ \hat{u}(a,0) = \hat{u}_0(a) \end{cases}$$

We solve this using integrating factors or Laplace transform:

$$\hat{u}(a,t) = e^{-ia^2t}\hat{u}_0(a) - i\int_0^t e^{-ia^2(t-s)}\hat{F}(a,s)\mathrm{d}s$$

Now we need to invert the Fourier transform. Note that e^{-ita^2} looks like a Gaussian with a complex rescaling. In particular, the above looks like the solution of the heat equation, but with it in place of t.

For the Heat Solution, we had:

$$\mathcal{F}^{-1}[e^{-a^2t}\hat{u}_0(a)] = \frac{1}{\sqrt{4\pi t}} \int_{-\infty}^{+\infty} e^{-\frac{(x-y)^2}{4t}} u_0(y)\mathrm{d}y$$

In the Schrödinger case, analytically continuing $t \to it$:

$$\mathcal{F}^{-1}[e^{-ia^2t}\hat{u}_0(a)] = \frac{1}{\sqrt{4\pi it}} \int_{-\infty}^{+\infty} e^{-\frac{(x-y)^2}{4it}} u_0(y)\mathrm{d}y$$

In particular, the total solution takes the form:

$$u(x,t) = \frac{1}{\sqrt{4\pi it}} \int_{-\infty}^{+\infty} e^{-\frac{(x-y)^2}{4it}} u_0(y)\mathrm{d}y - i\int_0^t \int_{-\infty}^{+\infty} \frac{1}{\sqrt{4\pi i(t-s)}} e^{-\frac{(x-y)^2}{4i(t-s)}} F(y,s)\mathrm{d}y\mathrm{d}s \tag{34}$$

You can verify that the above is indeed a solution (and, thus, due to uniqueness, the unique solution), even though the analytic continuation method used to derive it is formal. We can explicitly compute the inverse Fourier transform rigorously in special cases using complex contour integration.

13.3.2 Example

If $F = 0$ and $u_0(x) = e^{-\frac{x^2}{2}}$. Then $\hat{u}_0(a) = e^{-\frac{a^2}{2}}$ (Fourier transform of Gaussian is Gaussian), then:

$$\hat{u}(a,t) = e^{-ia^2t}\hat{u}_0(a) = e^{-ia^2t}e^{-a^2/2} = e^{-(it+\frac{1}{2})a^2}$$

So:

$$u(x,t) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} e^{ixa} \cdot e^{-(it+\frac{1}{2})a^2}\mathrm{d}a$$

$$= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} e^{ixa-(it+\frac{1}{2})a^2}\mathrm{d}a \quad (*)$$

The above integral can be computed using the methods of complex integration. By completing the square in the exponent, we set:

$$Z = \left(a - \frac{ix}{2it+1}\right)\sqrt{2it+1}$$

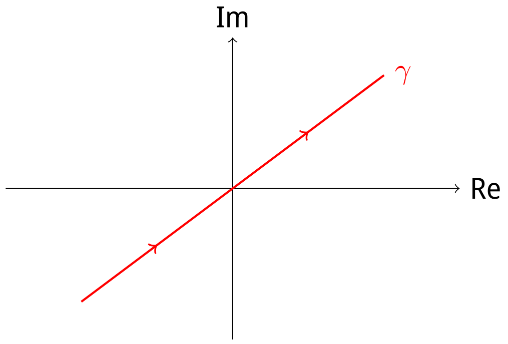
and

$$e^{ixa-(it+\frac{1}{2})a^2} = e^{-\frac{x^2}{4it+2}} \cdot e^{-\frac{Z^2}{2}}$$

$$\mathrm{d}a = \frac{\mathrm{d}Z}{\sqrt{2it+1}}$$

Note that when a goes from $-\infty$ to $+\infty$, $Z = Z(a)$ parametrizes the straight line in the complex plane:

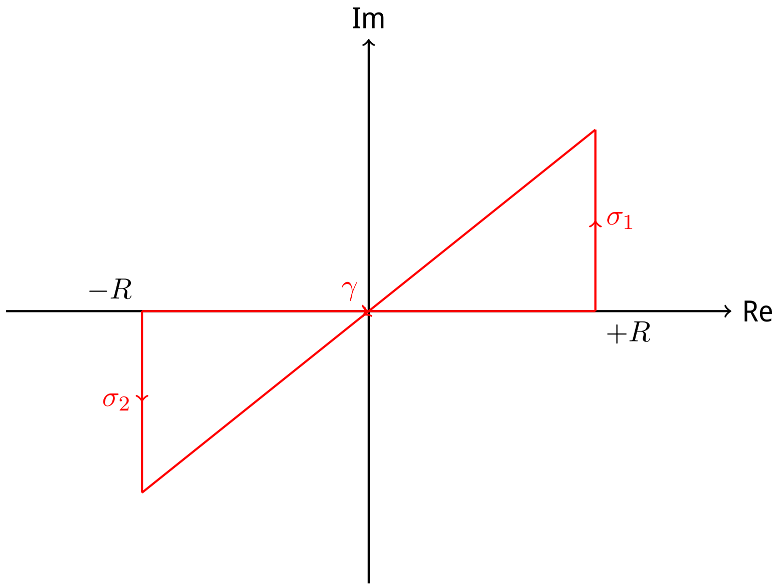
$$\gamma = \left\{\left(a - \frac{ix}{2it+1}\right)\sqrt{2it+1} : a \in \mathbb{R}\right\}$$



So, from (*):

$$\begin{aligned} u(x,t) &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} e^{ixa-(it+\frac{1}{2})a^2} \mathrm{d}a \\ &= \frac{1}{\sqrt{2\pi}} \int_\gamma e^{-\frac{x^2}{4it+2}} \cdot e^{-Z^2/2} \cdot \frac{\mathrm{d}Z}{\sqrt{2it+1}} \\ &= \frac{1}{\sqrt{2\pi}} \frac{e^{-\frac{x^2}{4it+2}}}{\sqrt{2it+1}} \int_\gamma e^{-Z^2/2} \mathrm{d}Z \end{aligned}$$

Using Cauchy's Theorem, we can move the integration curve γ back to the real axis $\mathbb{R}$:



The contributions from the vertical segments σ_1 and σ_2 vanish:

$$\int_{\sigma_1} e^{-Z^2/2} \mathrm{d}Z, \int_{\sigma_2} e^{-Z^2/2} \mathrm{d}Z \to 0 \quad \text{as } R \to +\infty$$

because the integrand $e^{-Z^2/2}$ decays exponentially fast for $\mathrm{Re}(Z) \to \pm\infty$ within the sector bounded by $|\arg(Z)| < \pi/4$.

But $\int_{-\infty}^{+\infty} e^{-Z^2/2} \mathrm{d}Z = \sqrt{2\pi}$. So:

Free Schrödinger Equation Solution with Gaussian Initial Data

$$u(x,t) = \frac{e^{-\frac{x^2}{4it+2}}}{\sqrt{2it+1}}$$

13.4 More Remarks

If we wanted to solve the equation on $0 < x < L$ with (say) Dirichlet boundary conditions $u|_{x=0} = 0, u|_{x=L} = 0$ (the infinite-depth well problem), we would proceed as in the case of the heat or the wave equation.

Namely, we set:

$$u(x,t) = \sum_{n=1}^{\infty} b_n(t) \cdot \sin\left(\frac{\pi n}{L} x\right)$$

and solve the (1st order) ODE for $b_n(t)$.

But now: $b_n(t) \in \mathbb{C}$.

Thank you!
(and please complete the in-depth evaluations for the course)